

## Paper B: corrections to the manuscript (master plan)

Against the author’s draft committed at c2ae782f, 1 October 2026 (C.1 to C.4 rebased on the author’s applied text)

Each row sets the current manuscript text (left) against the proposed replacement (right), both typeset as they would appear. BEFORE text is cut from the committed manuscript by script, so it matches the manuscript exactly. Under each entry, **Why** gives the error and **Evidence** the Lean or sympy record or page that settles it (audit items refer to `notes/citation_audit.md`). Entry C.15 (inside C.11.9) reconciles Section 3 with `notes/positioning_economics.tex`; the abstract (C.1) leads with the identification point for the same reason. Cross-references print as their label (e.g. PROP:IMM); footnotes print inline. **Nothing here is applied.** The author approves entries by number; approved entries are applied to the manuscript, compiled and committed. Every AFTER text passes the mechanical checks of `notes/writing_discipline.md` (no colons, no dashes doing a sentence’s work, “sequence” for reading order, the fixed adoption terminology, length within 80–120% of the draft except for deletions, citation fixes and the removed duplicate); the generator refuses to build otherwise.

### Section C: corrections to the current draft

| Part                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | BEFORE (current manuscript)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | AFTER (proposed)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
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| <b>C.1</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Abstract, sentences 1, 2-4 and 6 (rebased on the author’s draft c2ae782f)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 1/3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | The effect of the order in which evidence arrives on an individual judgment has sparked much interest in both experimental and theoretical studies.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | The effect of the sequence in which evidence arrives on an individual judgment has sparked much interest in both experimental and theoretical studies.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 2/3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Considering a case where evaluators make decisions using two correlated soft cues, the current papers asks which of their beliefs and decision statistics register a resulting effect of arrival sequence with respect to a sequence-free benchmark. With a view of implications for empirical measurement, the paper shows that an arbitrary statistic is an order of magnitude closer to the benchmark when it is insensitive to small shifts in the marginal probabilities. The paper shows how the collective marginal probabilities and the share of evaluators whose decision the sequence changes depart from the sequence-free benchmark at first order, whereas the believed association departs only at second order. | Considering a case where evaluators make decisions using two correlated soft cues, the current paper asks which statistics of their beliefs and decisions register a resulting effect of arrival sequence with respect to a sequence-free benchmark. With a view to implications for empirical measurement, the paper shows that a smooth statistic of the belief is an order of magnitude closer to the benchmark than the marginal probabilities when it is insensitive to small shifts in those marginals. The average marginal probabilities and the share of evaluators whose decision the sequence changes depart from the sequence-free benchmark at first order, whereas the believed association departs only at second order. |
| 3/3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Second, whether the effect of order is detectable or not depends on the kind of question being asked rather than on the number of evaluators considered.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Second, at a given precision of measurement, whether the sequence effect is detectable depends on the kind of question being asked.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Why.</b> The author applied C.1 in their own wording at c2ae782f; this entry keeps only what still needs correcting. Sentence 1 and the last sentence use “order” for reading sequence, which the paper reserves for order in <i>c</i> (writing discipline 6). Sentence 2: “the current papers asks”, “with respect to”, and “which of their beliefs and decision statistics” (a belief is not a statistic). Sentence 3: “With a view of” is “With a view to”; “an arbitrary statistic” is false as stated, since only statistics insensitive to the marginals are closer (Proposition PRO), so “smooth ... of the belief”, the comparator “than the marginal probabilities” and “those marginals” are restored from the approved C.1. Sentence 4: two consecutive sentences open “The paper shows”, and “shows how” states a result as a method; “collective” becomes “average” (Proposition DRF reads the mean belief). Leaving the loss out of sentence 4 is accurate, since the loss is also second order (Theorem LOS). Last sentence: “rather than on the number of evaluators” is the identification overclaim (todo, 2026-10-01). Only the odds ratio is blind at every sample size (Lemma SEP); the association is second order and visible once precision is finer than $c^2$ (MS precision passage, $c^2 \ll \varepsilon \ll c$ ), and precision improves with the number of evaluators. Fixing the precision makes the claim true. |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |

| Part                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | BEFORE (current manuscript)                                                                                                                                                                                                                                                                                                 | AFTER (proposed)                                                                                                                                                                                                                                                                                                        |
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| <b>Evidence.</b> PRO (propPRO_protection, propPRO_uniqueness); DEC and LOS (second order); DRF and SHR (first order); LemmaSEP.lean (the odds ratio). All in the Lean and sympy suites.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                         |
| <b>Grounds.</b><br><b>Theorem.</b> PropPRO.lean, propPRO_protection<br>For all real $x, y$ and all $(\alpha, \beta, \lambda, q_0, r_0)$ , $\langle xJ + y \nabla \text{assoc}(q \otimes r), M_\lambda \rangle = 0$ , where $M_\lambda = \lambda \kappa R_1 + (1 - \lambda) \kappa' R_2$ , $R_1 = q \otimes (1, -1)$ , $R_2 = (1, -1) \otimes r$ , $\kappa = (\alpha - q_0)r_0(1 - r_0)/Z$ , $\kappa' = (\beta - r_0)q_0(1 - q_0)/Z$ and $Z = \alpha\beta(1 - \alpha)(1 - \beta)$ .<br>Proposition PRO (i): a statistic whose gradient lies in the plane $\text{span}\{J, \nabla \text{assoc}\}$ has zero first-order coefficient, identically in the prior and in the mix.<br><b>Theorem.</b> PropPRO.lean, propPRO_uniqueness and annihilator_eq_span<br>Let $\lambda \in (0, 1)$ , $q_0, r_0 \in (0, 1)$ , $\beta \neq r_0$ and $x = \langle \nabla F, R_1 \rangle$ , $y = \langle \nabla F, R_2 \rangle$ . If $\lambda(\alpha - q_0)r_0(1 - r_0)x + (1 - \lambda)(\beta - r_0)q_0(1 - q_0)y = 0$ at two priors $\alpha_1 \neq \alpha_2$ , then $x = y = 0$ ; and any $V$ with $\langle V, R_1 \rangle = \langle V, R_2 \rangle = 0$ equals $(V_{01} + r_0(V_{00} - V_{01}))J + \frac{V_{00} - V_{01}}{1 - q_0} \nabla \text{assoc}$ .<br>Proposition PRO (ii): protection on an open set of priors forces $\nabla F$ into the plane, so the classification is an “if and only if”.<br><b>Theorem.</b> PropPRO.lean, margA_not_in_span; Aggregate.lean, propDRF<br>For $0 < r_0 < 1$ the $A$ -marginal read-out $(0, 0, 1, 1)$ is not of the form $xJ + y \nabla \text{assoc}(q_0, r_0)$ . For $\alpha, \beta \notin \{0, 1\}$ and $Z \neq 0$ , $\frac{d}{dc} [\lambda(P_{AB}^J - P^B)(A=1) + (1 - \lambda)(P_{BA}^J - P^B)(A=1)]_{c=0} = (1 - \lambda)q_0(1 - q_0)(r_0 - \beta)/Z$ .<br>Proposition DRF: a marginal departs at first order for every $\lambda < 1$ , so “an arbitrary statistic” is not an order closer; only the protected ones are.<br><b>Theorem.</b> PropPRO.lean, lemmaASC_first_order_vanishes (and_BA)<br>For all $q_0, r_0, t, \kappa$ : $\text{assoc}(q \otimes r + t\kappa R_1) = \text{assoc}(q \otimes r) + t^2 \text{assoc}(\kappa R_1)$ , and likewise along $R_2$ : the first-order term $t \langle \nabla \text{assoc}, \kappa R_\sigma \rangle$ is identically zero.<br>Proposition DEC: the believed association is in the protected class, second order along either route.<br><b>Theorem.</b> Decision.lean, volume_flipSet and lintegral_stake_le_volume<br>With $u$ the benchmark surplus, $\delta$ the first-order score coefficient and $\text{Flips}(u, c, \delta) := \neg(0 \leq u \iff 0 \leq u + c\delta)$ : $\text{Leb}\{u : \text{Flips}(u, c, \delta)\} =  c\delta $ and $\int_{\{\text{Flips}\}}  u  du \leq  c\delta ^2$ , for all real $c, \delta$ .<br>Proposition SHR and Theorem LOS: the affected share is first order, the surplus-weighted loss second order. |                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                         |
| <b>C.2</b> Introduction, paragraph 1 (rebased on the author’s draft c2ae782f)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                         |
| 1/5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | That arrival sequence of cues does effect judgment is rarely under doubt – since if individual judgments did not depend on the arrival sequence, then each cue must move belief by a factor fixed before it is received and lead to the unlikely situation where experience does not change how later cues are read at all. | That the arrival sequence of cues affects judgment is rarely in doubt, since if individual judgments did not depend on the arrival sequence, then each cue must move belief by a factor fixed before it is received, leading to the unlikely situation where experience does not change how later cues are read at all. |
| 2/5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | The sequence-dependence of Jeffrey conditioning for soft evidence in particular has attracted much debate (Diaconis and Zabell, 1982; Hawthorne, 2004). However, but the question of how that dependence registers in the statistics an observer reads has received little attention.                                       | The sequence dependence of Jeffrey conditioning for soft evidence in particular has attracted much debate (Diaconis and Zabell, 1982; Hawthorne, 2004), but the question of how that dependence registers in the statistics an observer reads has received little attention.                                            |
| 3/5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | It finds that the share of population affected by evidence sequence deviates from a sequence-invariant benchmark – but the believed association between attributes and the total loss due to sequence effects do not.                                                                                                       | It finds that the share of the population whose decision the sequence changes departs from a sequence-invariant benchmark at first order, while the believed association between attributes and the surplus-weighted loss depart from it only at second order.                                                          |
| 4/5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Given that a difference from the benchmark below the observer’s precision is invisible in the statistic read, the paper discusses the implications for identification of sequence effects.                                                                                                                                  | (sentence deleted)                                                                                                                                                                                                                                                                                                      |
| 5/5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Despite its simplicity the two-cue setting demonstrates the central mechanism of how an attribute-local updating leaves the interactions among attributes untouched remains valid for any number of cues and attributes.                                                                                                    | Despite its simplicity, the two-cue setting demonstrates the central mechanism, that attribute-local updating leaves the interactions among attributes untouched, and the mechanism remains valid for any number of cues and attributes (Lemma LEM:SEP).                                                                |

| Part | BEFORE (current manuscript)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | AFTER (proposed) |
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|      | <p><b>Why.</b> The author applied C.2 in their own wording at c2ae782f; this entry keeps what still needs correcting. Modus tollens sentence: “does effect” is “affects”, “under doubt” is “in doubt”, the double hyphen is a dash doing a sentence’s work, and “and lead to” breaks the parallel with “move”. Caution kept from the earlier entry: “must” is stronger than the literature. Fixed factors imply commutation (Wagner 2003 Theorem 3.2, Hawthorne factorUpdate_comm, Cripps order_invariance); the converse is exact in this paper’s two-cue model, where the sequence effect vanishes iff <math>q_0 = \alpha</math> and <math>r_0 = \beta</math>, but is not a theorem for arbitrary updating rules. Debate sentence: “However, but” and the hyphen in the noun “sequence dependence”. ZhaoOsherson2010 is no longer cited anywhere in the manuscript, which is the author’s choice; its bibliography entry is now unused. Sentence “It finds that”: the loss is not free of the sequence effect, it carries it at second order (LOS), and “– but” is a dash doing a sentence’s work. Sentence “Given that”: cut (rule 4 of the writing discipline); the precision point is made in the C.4 paragraph, and “identification of sequence effects” is the identification overclaim C.1 removes from the abstract. Last sentence: “demonstrates the central mechanism of how ... remains valid” has two verbs for one subject; Lemma SEP is the general-N result it refers to.</p> <p><b>Evidence.</b> verify_jeffrey.md J1; verify_kinematics.md; verify_hawthorne_weisberg.md; Theorem LOS, Proposition SHR; LemmaSEP.lean (general N). Modus tollens: Cripps.lean order_invariance, Hawthorne.lean factorUpdate_comm, Wagner2003.lean thm32; sympy/verify_soft_vs_hard.py (exact condition for the sequence effect).</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                  |
|      | <p><b>Grounds.</b></p> <p><b>Quote.</b> Zhao and Osherson 2010, p. 290 (opening part, before Experiment 1)<br/> “The normative status of Jeffrey’s rule has nonetheless been questioned because successive uses produce distinct distributions depending on the order in which events are considered (Döring, 1999). In our view, such doubts disappear on closer inspection of the evidential weight of probability judgements (Osherson, 2002; Wagner, 2002).”<br/> They report the doubt as settled, not the mechanism as “unclear”.</p> <p><b>Quote.</b> Jeffrey 1983, pp. 182-183 (Section 11.11)<br/> “It is straightforward to verify that the present kinematical scheme is not generally commutative ... the resulting probability measure may be one thing or another depending on the order of the two applications. That is as it should be, for just after the <math>A</math>-application, one’s degrees of belief in the <math>A_i</math> are <math>a_i</math>, no matter what they were before ... The notion that the kinematical scheme ought to be generally commutative (Domotor, <i>Philosophy of Science</i> 47 [1980]: 395) stems from a conflation of two attitudes toward The Given”<br/> Jeffrey holds non-commutativity to be correct and the demand for commutativity to rest on a conflation.</p> <p><b>Quote.</b> Diaconis and Zabell 1982, p. 827, Section 4.2, Remark 2<br/> “Thus noncommutativity is not a real problem for successive Jeffrey updating.”<br/> D-Z are on the side of the sequence effect, not of the doubt.</p> <p><b>Quote.</b> Hawthorne 2004, p. 99 (Section 5)<br/> “The second concern is that, as a result of this amnesia, the order in which states or experiences are acquired will almost always have a very significant influence on an agent’s belief strengths. This order effect is very troubling.”<br/> The one cited author who treats the order effect as a defect; the three sources disagree, which is a debate.</p> <p><b>Theorem.</b> Decision.lean, lintegral_stake_le_volume_with_volume_flipSet<br/> For all real <math>c, \delta</math>: <math>\int_{\{u: \text{Flips}(u, c, \delta)\}}  u  du \leq  c\delta  \cdot  c\delta </math>, whereas <math>\text{Leb}\{u : \text{Flips}(u, c, \delta)\} =  c\delta </math>.<br/> Theorem LOS: the loss carries the sequence effect at second order; it is not free of it, and the share (Proposition SHR) is first order.</p> <p><b>Theorem.</b> Hawthorne.lean, factorUpdate_comm; Cripps.lean, order_invariance<br/> With <math>\text{factorUpdate}(p, u, w)(x) = w(u(x))p(x) / \sum_y w(u(y))p(y)</math>, a belief multiplied by fixed factors <math>w</math> on one partition and <math>z</math> on another, both normalisers nonzero:<br/> <math>\text{factorUpdate}(\text{factorUpdate}(p, u, w), v, z) = \text{factorUpdate}(\text{factorUpdate}(p, v, z), u, w)</math>. Likewise, under Cripps’s Symmetry and Divisibility, two signals with fixed likelihoods give the same posterior in either sequence.<br/> (sentence 1) The first half of the modus tollens. Read as Bayes factors, fixed before either cue arrives, the cues cannot leave a trace of the sequence.</p> <p><b>Computation.</b> sympy/verify_soft_vs_hard.py, symbolic checks (15/15)<br/> Symbolic in <math>\alpha, \beta, c, q_0, r_0</math>: a credential read first moves the <math>B</math>-marginal the letter meets to <math>\beta + c(q_0 - \alpha)/(\alpha(1 - \alpha))</math>, so the factor the letter implies changes unless <math>c = 0</math> or <math>q_0 = \alpha</math>, and symmetrically for the credential. Every cell of <math>P_{AB}^J - P_{BA}^J</math> carries the factor <math>c</math>, and with <math>c \neq 0</math> the difference vanishes if and only if <math>q_0 = \alpha</math> and <math>r_0 = \beta</math>.<br/> (sentence 1) Under the delivered-credence reading the sequence leaves a trace exactly when an earlier cue changes the factor a later one implies, which is the lived-experience point in the model’s terms. The sentence itself claims only the first half; this row shows that the delivered-credence reading, one of the readings it leaves open, is one in which what came earlier changes how later cues are read.</p> |                  |
| C.3  | Introduction, paragraph 2 and its footnote (rebased on the author’s draft c2ae782f)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                  |

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The belief-adjustment literature often models an impression as a single score (Hawthorne, 2004) or considers traits one attribute at a time (Asch, 1946). The characterisation of when two sequences lead to the same belief does consider a joint belief over attributes Diaconis and Zabell (1982), but the order dependence is more more treated as a defect to repair [Footnote: Hawthorne (2004) repairs this by letting a cue multiply the old belief instead of replacing it, and under his order-free variants the sequence effect disappears.] rather than as something to measure. While changing the input from a credence that replaces the prior to a factor that multiplies it – does settle whether beliefs depend on sequence, it hardly addresses the empirical issue that an observer cannot tell whether impressions replace prior belief or adjust it.</p> | <p>The discussions on how Jeffrey conditioning (Jeffrey, 1983) depends on arrival sequence have focused more often on whether the commutation criterion is exactly satisfied or not but not enough on how measurable its failure is in observed data. The belief-adjustment literature often models an impression as a single score (Hogarth and Einhorn, 1992), and the impression-formation literature considers traits one attribute at a time (Asch, 1946). The characterisation of when two sequences lead to the same belief does consider a joint belief over attributes (Diaconis and Zabell, 1982), but elsewhere sequence dependence is more often treated as a defect to repair [Footnote: Hawthorne (2004) offers factor-based alternatives in which a cue supplies a normed-likelihood or likelihood-ratio factor instead of a new probability. Under these, updates on distinct attributes commute, and in his Basis-Commuting Version all updates do.] rather than as something to measure. While changing the input from a credence that replaces the prior to a factor that multiplies it does settle whether beliefs depend on sequence, it hardly addresses the empirical issue that an observer cannot tell whether impressions replace prior belief or adjust it.</p> |
| <p><b>Why.</b> The author applied C.3 in their own wording at c2ae782f; this entry keeps what still needs correcting. (i) “models an impression as a single score \citep{Hawthorne2004}”: the single-score model is Hogarth-Einhorn’s belief-adjustment model, and Hawthorne is not part of that literature. (ii) Asch is impression formation, not belief adjustment, and reports eighteen traits one by one (audit A28). (iii) “\cit{DiaconisZabell1982}” is glued to “attributes” and should be \citep. (iv) “the order dependence is more more treated as a defect”: doubled “more”, “order” for reading sequence, and the clause still groups Diaconis-Zabell with the defect view, though they hold that non-commutativity “is not a real problem” (M24); “elsewhere” detaches it from them. (v) The footnote still says Hawthorne “repairs” the defect and that the effect disappears; he offers alternatives, and freedom from sequence holds across distinct bases, fully only in his Basis-Commuting Version (M19). (vi) “Jeffrey\citep” lacks a space and the citation sits inside “Jeffrey conditioning”. (vii) The double hyphen before “does settle” is a dash doing a comma’s work.</p> <p><b>Evidence.</b> Asch1946 record (Asch.lean, Table 7); HogarthEinhorn.lean; DiaconisZabell.lean and README; Hawthorne.lean (extUpdate_basisCommuting, factorUpdate_comm), Sections 6-8.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <p><b>Grounds.</b></p> <p><b>Theorem.</b> <i>Asch.lean, seriesA_length, seriesB_eq_reverse, checkListI_length, allTraits_length, table7_rows_are_checklist_pairs, d_ne_zero, restrained_64_9</i><br/> <math> \text{Series A}  = 6</math> and <math>\text{Series B} = \text{reverse}(\text{Series A})</math>; <math> \text{Check List I}  = 18</math> and Table 7 has 18 rows, row <math>i</math> naming a member of pair <math>i</math>; for every one of the 18 traits <math>\text{ie}(t) - \text{ei}(t) \neq 0</math>; e.g. restrained 64 vs 9, good-looking 74 vs 35, humorous 52 vs 21, good-natured 18 vs 0.<br/> <i>Experiment VI reports eighteen per-trait percentages under two orders, i.e. eighteen marginals, not a single score.</i></p> <p><b>Quote.</b> <i>Asch 1946, pp. 270-271, Experiment VI and Table 7</i><br/> p. 270: “The two series are identical with regard to their members, differing only in the order of succession of the latter.” p. 271: “The check-list data appearing in Table 7 furnish quantitative support for the conclusions drawn from the written sketches.”<br/> <i>The check list is Asch’s own quantitative instrument for the order comparison.</i></p> <p><b>Quote.</b> <i>Diaconis and Zabell 1982, p. 827, Section 4.2, Remarks 1-2</i><br/> “Remark 1. There is no reason to require <math>P_{\mathcal{EF}} = P_{\mathcal{FE}}</math> for successive updating to be useful and valid. ... Remark 2. ... Thus noncommutativity is not a real problem for successive Jeffrey updating.”<br/> <i>D-Z do not hold the defect view the manuscript groups them with.</i></p> <p><b>Quote.</b> <i>Hawthorne 2004, pp. 96 and 108 (Sections 4 and 8)</i><br/> p. 96: “Indeed, it may turn out that there is no one true theory of uncertain updating – that each theory has its uses, its domain of applicability.” p. 108: “Let us call this the Basis-Commuting Version of the Likelihood-Ratio Update Model. Sequential updating is completely order-independent on this model.”<br/> <i>He offers alternatives rather than a repair, and full order-freedom is claimed only for the Basis-Commuting Version.</i></p> <p><b>Theorem.</b> <i>Hawthorne.lean, factorUpdate_comm and extUpdate_basisCommuting</i><br/> For bases <math>u : \Omega \rightarrow \iota</math>, <math>v : \Omega \rightarrow \kappa</math> and factor vectors <math>w, z</math> with <math>\sum_y w(u(y))p(y) \neq 0</math> and <math>\sum_y z(v(y))p(y) \neq 0</math>: <math>\text{fU}(\text{fU}(p, u, w), v, z) = \text{fU}(\text{fU}(p, v, z), u, w)</math>. If the factor <math>\Lambda_k(\gamma)</math> of every basis-homogeneous subsequence is invariant under permutation of <math>\gamma</math>, then <math>\text{extUpdate}(p, \delta) = \text{extUpdate}(p, \delta')</math> for every permutation <math>\delta'</math> of <math>\delta</math>.<br/> <i>Factor updates commute across distinct bases; commutation within a basis, hence full order-freedom, needs the extra Basis-Commuting hypothesis.</i></p> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |

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| C.4  | Introduction, the two-channel paragraph (rebased on the author’s draft c2ae782f)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                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|      | <p>Whether impressions replace prior belief or adjust is a question which Hawthorne (2004), writing as a logician rather than a psychologist, leave open. To set the scope of its conclusions, the paper considers that an arrival-sequence dependence can enter through two channels, an <b>association</b> channel, in which one cue changes what the other implies and which exists only when the attributes are believed related, and a <b>position</b> channel, in which the observer weights the later cue less whatever the attributes are (Hogarth and Einhorn, 1992; Asch, 1946). In the position channel, the later cue is adopted only in part and every marginal registers the sequence even when the attributes are believed unrelated. On the other hand, when each impression is adopted in full, as in the successive updating of Diaconis and Zabell (1982), the position channel is disabled with only the association channel remaining. In the two-attribute setting with full-adoption, the paper thus considers three coordinates of a belief – two marginal probabilities and the cross-attribute association – and explains how a difference from the sequence-free benchmark below the observer’s precision is invisible in the statistic read by the observer. The question is therefore not simply whether the two sequences agree but by how much they disagree in an arbitrary statistic. The paper finds that the marginal probabilities and decisions from them carry the difference at first order in the prior covariance, while the believed association and statistics carry it only at second order. This carries clear implications for what an audit can or cannot measure about sequence dependence.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <p>Whether impressions replace prior belief or adjust it is a question which Hawthorne (2004, pp. 115–116), writing as a logician rather than a psychologist, leaves open. To set the scope of its conclusions, the paper considers that arrival-sequence dependence can enter through two channels, an <b>association</b> channel, in which one cue changes what the other implies and which exists only when the attributes are believed related, and a <b>position</b> channel, in which the weight a cue receives depends on where in the sequence it arrives, whatever the attributes are (Hogarth and Einhorn, 1992). The position channel is a property of the evaluator rather than of the evidence. When the later cue is adopted only in part, every marginal registers the sequence even when the attributes are believed unrelated, and an audit that pools evaluators who read in different sequences records an association that none of them holds. The paper therefore sets the position channel aside until Section SEC:ROBUST and asks whether, when each impression is adopted in full, as in the successive updating of Diaconis and Zabell (1982), and the attributes are believed to go together, an audit detects the sequence dependence that the believed link alone produces. In the two-attribute setting with full adoption, the paper thus considers three coordinates of a belief, the two marginal probabilities and the cross-attribute association, and explains how a difference from the sequence-free benchmark below the observer’s precision is invisible in the statistic read by the observer. The question is therefore not simply whether the two sequences agree but by how much they disagree in an arbitrary statistic. The paper finds that the marginal probabilities and the share of decisions they change carry the difference at first order in the prior covariance, while the believed association and the statistics that move with it carry it only at second order. This carries clear implications for what an audit can or cannot measure about sequence dependence.</p> |
|      | <p><b>Why.</b> The author applied C.4 in their own wording at c2ae782f; this entry keeps what still needs correcting and adds one sentence. Corrections: “adjust” to “adjust it”; “leave open” to “leaves open” (Hawthorne is one author; he gives a tentative view and says his interest is normative, pp. 115-116); “an arrival-sequence dependence” loses its article; the position channel was defined as “the observer weights the later cue less” and cited to Hogarth-Einhorn and Asch, but Hogarth-Einhorn’s step-by-step adjustment predicts recency (Appendix B) and Asch denies that “sheer temporal position” matters (p. 272), so the definition is made neutral and Asch is dropped (audit P10-P12); the two dashes become commas; “decisions from them” becomes “the share of decisions they change”, since the loss is a decision statistic and is second order (LOS); “and statistics” becomes “and the statistics that move with it”. Scope defence (author, 2026-10-01): the position channel is a property of the evaluator, not of the evidence, and it leaves a signature any marginal audit detects, so the paper sets it aside and asks the question that remains when evaluators adopt a cue alike whichever position it arrives in. Two sentences are replaced rather than added (discipline rule 4): the position channel sentence and the “On the other hand” sentence, whose content the new sentences carry; the earlier one-line full-adoption sentence is cut, its pointer to Section 6 now in the scope sentence. Checked against rule 3: every contrast names both sides (evaluator rather than evidence; adopted only in part against adopted in full; the believed link against the position), no bare “it”, no assertion of importance. Evidence for the claims: at <math>c = 0</math> the fully adopting evaluator shows no sequence effect (IMM) and the partially adopting one shows <math>(1 - \omega)(\alpha - q_0)</math> (ADJ); the pooled association at <math>c = 0</math> is <math>-\lambda(1 - \lambda)(1 - \omega)^2(\alpha - q_0)(\beta - r_0)</math>, nonzero for interior weights.</p> <p><b>Evidence.</b> HogarthEinhorn.lean (appB_recency, eq8_estimation_first_dominates); Hawthorne 2004 pp. 115-116 (grounds_E_literature); Ladder.lean (ladder_gap, ladder_assoc_coeff, ladder_oddsShadow_seqEffect, oddsRatio_rescale); sympy/verify_ladder.py (37/37).</p> <p><b>Grounds.</b></p> <p><b>Theorem.</b> <i>HogarthEinhorn.lean, appB_recency, appB_recency_pos, oneSided_eq_eq8, oneSided_primacy_iff</i><br/> With <math>\text{est}(w, S, s) = S + w(s - S)</math> (HE Eq. 3/4): for all reals, <math>\text{est}(w_b, \text{est}(w_a, S_0, a), b) - \text{est}(w_a, \text{est}(w_b, S_0, b), a) = w_a w_b (b - a)</math>, which is <math>&gt; 0</math> when <math>w_a, w_b &gt; 0</math> and <math>a &lt; b</math> (recency). The one-sided rule <math>\text{est}(w, \text{est}(1, S_0, a), b)</math> equals HE’s Eq. 8 with <math>k = 2</math>, and for <math>a &lt; b</math> it gives <math>S_{ab} - S_{ba} &lt; 0</math> iff <math>w &lt; 1/2</math>.<br/> <i>Damping every cue gives recency; primacy comes from adopting the first cue in full (Eq. 8), and then only for <math>w &lt; 1/2</math>.</i></p> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |

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B.5) and T-p. 12 (Eq. 8)</i></p> <p>T-p. 35: “<math>D = w_a w_b [s(x_b) - s(x_a)]</math>. Because <math>s(x_b) &gt; s(x_a)</math>, <math>D &gt; 0</math> and recency always obtains.” T-p. 12: “<math>S_k = s(x_1) + w_k [s(x_2, \dots, x_k) - R]</math> ... the effective weight accorded to <math>s(x_1)</math> must be greater than the weight attached to any of the other pieces of evidence.”</p> <p><i>Their step-by-step partial adjustment predicts recency; their primacy is the End-of-Sequence anchoring on the first item.</i></p> <p><b>Quote.</b> <i>Asch 1946, p. 272</i></p> <p>“It is not the sheer temporal position of the item which is important as much as the functional relation of its content to the content of the items following it.”</p> <p><i>Asch denies a position channel, so he cannot be cited for “weights the later cue less”.</i></p> <p><b>Theorem.</b> <i>BenjaminBodohCreedRabin.lean, margOddsA_orders_ne and example_marginal_gap</i></p> <p>On the <math>2 \times 2</math> joint with prior <math>u \otimes v</math> (independent, <math>c = 0</math>), <math>\alpha \neq 1</math>, positive cue likelihoods <math>a</math> on <math>A</math> and <math>b</math> on <math>B</math>, the <math>A</math> cue informative (<math>a_0 \neq a_1</math>): the odds <math>P(A=0)/P(A=1)</math> after base-rate-neglect updating differ between the orders <math>AB</math> and <math>BA</math>. Instance: uniform prior, <math>\alpha = 1/2</math>, <math>a = b = (49/50, 1/50)</math> gives <math>P(A=0) = 7/8</math> reading <math>A</math> first and <math>49/50</math> reading <math>A</math> last.</p> <p><i>A position channel operates with likelihood inputs adopted in full, so “which channels operate is fixed by adoption” holds for this model only.</i></p> <p><b>Quote.</b> <i>Diaconis and Zabell 1982, p. 825 (Section 3.1) and p. 827 (Section 4.2)</i></p> <p>p. 825: “To use Jeffrey’s rule at the second stage we must, of course, accept the J-condition ... Clearly, the order of updating matters, since the second opinion dominates.” p. 827: “2. When is successive updating reasonable?”</p> <p><i>Successive updating with each impression adopted in full is D-Z’s setting, examined, not a premise they lay down.</i></p> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>C.5</b> Introduction, hiring-panel paragraph, last sentence                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         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                                                                                         | In demonstrating how the sequence-dependence of certain statistics can be invisible to an observer, the panel example also clarifies that the invisibility does not depend on the extent to which a later impression erases the earlier one (the amnesic updating concern in the literature).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Apart from showing how the sequence dependence of some statistics can be invisible to an observer, the panel example also shows why that invisibility needs each impression to be adopted in full. The letter then fixes the belief about trustworthiness whatever the credential implied, so the sequence acts only through the believed link between the traits, and the believed association differs between the two sequences only at second order, whereas a letter adopted only in part moves that belief from wherever the credential left it and the association differs at first order. What remains untouched at every degree of adoption is the odds ratio, since each impression rescales rows or columns of the belief and never the pairing within them (Lemma LEM:SEP, Section SEC:ROBUST). |
| <p><b>Why.</b> The BEFORE sentence is false for the cross-product association, which is the paper’s <b>assoc</b>: the invisibility at second order does depend on how far a later impression erases the earlier one. The author asked (2026-10-01) that C.5 say whether omega touches the main result and why, in economic terms. It does. Full adoption: a Jeffrey step sets its own marginal to the delivered credence whatever the other cue had implied, so at independence the two sequences agree (IMM) and the sequence acts only through the prior association; each step rescales rows or columns, so the association is <math>c</math> times a product of factors that agree between sequences at <math>c = 0</math>, and the difference is second order (ORD, DEC). Partial adoption: the damped step moves the marginal only part of the way from where the first cue left it, so which cue is discounted depends on the sequence; both marginals differ at order zero even at <math>c = 0</math>, by <math>(1 - \omega)(\alpha - q_0)</math> and <math>-(1 - \omega)(\beta - r_0)</math>; the rescaling factors then differ at order zero, so the association, <math>c</math> times those factors, differs at first order with coefficient <math>(1 - \omega)H/Z</math> (LAD); and the mean belief of a population mixing the sequences acquires a cross-product association <math>-\lambda(1 - \lambda)(1 - \omega)^2(\alpha - q_0)(\beta - r_0)</math> at <math>c = 0</math> that no member holds (the “average association” clause). Odds ratio: invariant under row and column rescaling, so the same in both sequences for every <math>c</math> and every omega (Lemma SEP, which covers damped steps). So the main result, the second-order invisibility of the association and of the statistics that move with it, holds under full adoption and not otherwise; what holds at every omega is the odds ratio. “Amnesic” is dropped here since Hawthorne is not cited in this paragraph (writing discipline 6). The commented-out line below the paragraph states the odds-ratio version. Trimmed after the scope defence went into C.4 (2026-10-01): the clauses on ratings of unrelated traits and on the pooled association now live there, and rule 5 allows each point once; what stays is the mechanism in the vignette’s terms, the contrast between full and partial adoption, and the odds ratio. Before that, AFTER was the author’s rephrasing of 2026-10-01 with four changes: “the panel” to “the panel example”, “or not” dropped after “whether”, the double hyphen before “since” replaced by a comma (writing discipline 9), and “in both full and partial adoption” to “at every degree of adoption”, since no adoption is a case too; the references are added.</p> <p><b>Evidence.</b> PropIMM.lean; PropORD.lean; PropDEC.lean; Ladder.lean (ladder_gap, ladder_gap_mA1, ladder_assoc_coeff, ladder_assoc_coeff_witness, oddsRatio_rescale, jeffreyA_eq_rescale, jeffreyB_eq_rescale); LemmaSEP.lean (general N, any attribute-local rescaling); sympy/verify_ladder.py (53/53); sympy/verify_interior_omega.py row 4 (the pooled association at <math>c = 0</math>).</p> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |

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| <div>Grounds.</div> <div>Computation. <i>sympy/verify_interior_omega.py, check (3)</i></div> <div>Between-sequence effect on assoc with the second cue adopted with weight <math>\delta</math> (the manuscript's <math>\omega</math>): the <math>c^0</math> term is 0 for every <math>\delta</math>; the <math>c^1</math> coefficient is <math>(1 - \delta) G</math> with <math>G</math> affine in <math>\delta</math>, <math>G \neq 0</math> at the generic prior for <math>\delta = 1/2</math> and for <math>\delta = 0</math>; at <math>\delta = 1</math> the factor <math>(1 - \delta)</math> makes the effect second order in <math>c</math>.</div> <div><i>Under partial adoption the cross-product association carries the sequence at first order in <math>c</math>.</i></div> <div>Computation. <i>sympy/verify_interior_omega.py, check (4)</i></div> <div>Pooled belief <math>\lambda P_{AB} + (1 - \lambda)P_{BA}</math> at <math>c = 0</math>: <math>\text{assoc}(\bar{P}) - \text{assoc}(P^B) = -\lambda(1 - \lambda)(1 - \delta)^2(\alpha - q_0)(\beta - r_0)</math>, nonzero for interior <math>\lambda, \delta</math> whenever both cues differ from the prior marginals, and zero at <math>\delta = 1</math>.</div> <div><i>The pooled association is distorted already at <math>c = 0</math> unless the later cue is adopted in full.</i></div> <div>Computation. <i>sympy/verify_interior_omega.py, check (2)</i></div> <div>Each damped step multiplies the table it meets by a factor constant along rows (resp. columns), so both routes are separable reweightings of the prior and <math>\text{OR}(P_{AB}) = \text{OR}(P_{BA}) = \text{OR}(P)</math> for every <math>\delta</math> (witnessed at <math>\delta = 1/3</math>, <math>c = 1/40</math>, generic prior).</div> <div><i>Only the odds ratio is identical across sequences for every adoption weight.</i></div> <div>Theorem. <i>LemmaSEP.lean, isSeparable_applySteps and sep_rescales_association</i></div> <div>For <math>N</math> binary attributes, any finite sequence of attribute-local updates gives <math>P'(x) = P(x)\prod_a g_a(x_a)</math>; and for <math>i \neq j</math> such an <math>F</math> satisfies <math>F_{11}F_{00} - F_{10}F_{01} = g_i(1)g_i(0)g_j(1)g_j(0) \left(\prod_{a \neq i,j} g_a(x_a)\right)^2 (P_{11}P_{00} - P_{10}P_{01})</math>.</div> <div><i>Lemma SEP: a route rescales the association and cannot shift it, whence the odds ratio is invariant; the cross-product itself is not.</i></div> <div>Theorem. <i>PropDEC.lean, oddsRatio_gap_eq_zero</i></div> <div>For <math>\alpha, \beta, q_0, r_0 \notin \{0, 1\}</math>, nonzero intermediate <math>B</math>-marginals, nonzero off-diagonal prior cells and nonzero benchmark normaliser and off-diagonal cells: <math>\text{OR}(P_{AB}^J) - \text{OR}(P^B) = 0</math>, identically in <math>c</math>.</div> <div><i>The odds-ratio form of the panel sentence is exact; the cross-product form is protected only at <math>\omega = 1</math> (Proposition DEC).</i></div>                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <div>C.6 Introduction, premise paragraph, from “Read as a Bayes factor”</div> <table><tr><td></td><td>Read as a Bayes factor, the credential carries a likelihood ratio stating how much more probable it is under competence than in its absence, and that ratio multiplies whatever belief it meets. A Bayes factor, however, requires the probability of the same credential for a candidate who is not competent, and an impression does not supply that counterfactual. As a mechanism that holds conditionals fixed, Jeffrey updating is the unique coherent revision for credence delivered on the cue's partition and is equivalent to making the minimal change of the prior consistent with the delivered marginal (Diaconis and Zabell, 1982)</td><td>Read as a Bayes factor, the credential carries the ratio of new to old odds on competence, which multiplies whatever belief it meets. The two readings agree on a single cue (Proposition PROP:IMM) and differ in what stays fixed when the cue meets another prior. The panelist's own impression is naturally a credence, whereas a factor suits evidence reported by someone else, whose report mixes the evidence with that person's prior (Jeffrey, 2004, ch. 3). Jeffrey updating holds fixed the conditionals given the cue's partition, and its posterior is the unique belief with the delivered marginal closest to the prior in Kullback–Leibler divergence (Diaconis and Zabell, 1982, Theorem 5.1)</td></tr></table> <div>Why. (i) “A Bayes factor requires the probability of the same credential for a candidate who is not competent” is contradicted by Jeffrey's own definition, new odds over old odds, which needs no likelihood (2002 draft of Jeffrey 2004, ch. 3; audit J2). Jeffrey draws the line between the readings by provenance instead (own experience gives credences, others' reports should be converted to factors), which supports the paper's modelling choice. (ii) “the unique coherent revision” is not in Diaconis-Zabell; what they prove is the unique Kullback-Leibler (and Hellinger) minimiser (M22). “equivalent to making the minimal change” without naming the distance is also loose (not unique in variation distance). The footnote after the citation is unchanged. Needs the Jeffrey2004 bib entry (C.12); the Drive copy is the 2002 draft, so cite the chapter, not a page.</div> <div>Evidence. <i>verify_jeffrey.md J2; DiaconisZabell.lean (thm51_KL_eq_iff, jcond_iff_jeffrey); PropIMM.lean (propIMM_single_cue).</i></div> <div>Grounds.</div> <div>Quote. <i>Jeffrey (2002 draft of Jeffrey 2004), ch. 3, §3.3, p. 61, eq. (4)</i></div> <div><math display="block">\beta(D_i : D_1) = \frac{\text{new } D_i}{\text{new}(D_1)} \bigg/ \frac{\text{old}(D_i)}{\text{old}(D_1)} = \frac{\pi(D_i)}{\pi(D_1)} \quad (D_i : D_1 \text{ Bayes factor}).</math><p>This is your Bayes factor for <math>D_i</math> against <math>D_1</math>; ... it is what remains of your new odds when the old odds have been factored out.” And: “probability factors are a little easier to compute than Bayes factors, starting from old and new probabilities.”</p><div><i>Jeffrey's Bayes factor is new odds over old odds, computed from the prior and the delivered credence; no likelihood under a counterfactual enters, so “requires the probability of the same credential for a candidate who is not competent” is false of it.</i></div></div> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |  | Read as a Bayes factor, the credential carries a likelihood ratio stating how much more probable it is under competence than in its absence, and that ratio multiplies whatever belief it meets. 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|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <p><b>Quote.</b> <i>Jeffrey (2002 draft), ch. 3, §3.2 p. 59 and §3.3 p. 59</i></p> <p>“But the results <math>new'(D_i)</math> of that interaction are <i>data</i> for the kinematical formula for updating <math>old'(H)</math>; the formula itself does not compute those results.” “You are unlikely to simply adopt such an observer’s updated probabilities as your own, for they are necessarily a confusion of what the other person has gathered from the observation itself . . . with that person’s prior judgmental state, for which you may prefer to substitute your own.”</p> <p><i>Jeffrey draws the line by provenance: one’s own experience yields credences taken as data, another’s report is to be converted to factors (§3.3), which is what the corrected sentence says.</i></p> <p><b>Quote.</b> <i>Jeffrey (2002 draft), ch. 2, p. 42</i></p> <p>“Assuming rigidity relative to <math>D</math>, the odds factor for a theory <math>T</math> against an alternative theory <math>S</math> that is due to learning that <math>D</math> is true will be the left-hand side of the following equation, the right-hand side of which is called ‘the likelihood ratio’: Bayes Factor = Likelihood Ratio: <math>\frac{old(T D)/old(S D)}{old(T)/old(S)} = \frac{old(D T)}{old(D S)}</math>,”</p> <p><i>The Bayes factor equals a likelihood ratio only when <math>D</math> is learned for certain under rigidity; the counterfactual-likelihood reading the manuscript objects to is this special case.</i></p> <p><b>Theorem.</b> <i>DiaconisZabell.lean, thm51_KL_eq_iff (with thm51_KL_le); D-Z Theorem 5.1, (5.6)</i></p> <p>Let <math>\Omega</math> be finite, <math>P(\omega) &gt; 0</math> for all <math>\omega</math>, <math>e : \Omega \rightarrow I</math> surjective onto a finite <math>I</math> with cells <math>E_i = e^{-1}(i)</math>, and <math>Q \geq 0</math> with <math>Q(E_i) = p_i</math> for every <math>i</math>. Then <math>\text{KL}(Q \parallel P) = \sum_{\omega} Q(\omega) \log \frac{Q(\omega)}{P(\omega)} \geq \sum_i p_i \log \frac{p_i}{P(E_i)}</math>, with equality if and only if <math>Q(\omega) = p_{e(\omega)} P(\omega)/P(E_{e(\omega)})</math>, Jeffrey’s update.</p> <p><i>What D-Z prove is that Jeffrey’s posterior is the unique Kullback–Leibler minimiser among beliefs with the delivered marginal; “the unique coherent revision” is not in the paper.</i></p> <p><b>Theorem.</b> <i>DiaconisZabell.lean, remark_a_tv_not_unique; D-Z Remark (a), p. 828</i></p> <p>On <math>\Omega = \{0, 1, 2\}</math> with cells <math>\{0, 1\}</math>, <math>\{2\}</math>, prior <math>P = (1/4, 1/4, 1/2)</math> and new cell probabilities <math>(3/4, 1/4)</math>: Jeffrey’s update is <math>J = (3/8, 3/8, 1/4)</math>; <math>Q = (1/2, 1/4, 1/4)</math> has the same cell probabilities, <math>Q \neq J</math>, and <math>\ Q - P\  = \ J - P\  = \frac{1}{2} \sum_i  p_i - P(E_i)  = 1/4</math>, the minimum of (5.4). D-Z: “Although the probability measure given by Jeffrey’s rule minimizes the variation distance, it does not do so uniquely”.</p> <p><i>“Equivalent to making the minimal change” without naming the distance is loose: in variation distance the minimiser is not unique.</i></p> <p><b>Computation.</b> <i>sympy/verify_soft_vs_hard.py (15/15 checks pass)</i></p> <p>Worked example <math>\alpha = \beta = 1/2</math>, <math>c = 1/20</math>, <math>q_0 = 1/5</math>, <math>r_0 = 7/10</math>. The letter’s implied factor (new odds over old odds on <math>B</math>) is <math>\frac{7/3}{1} = 7/3</math> read first, against <math>\beta = 1/2</math>, and <math>\frac{7/3}{11/14} = 98/33</math> read second, against the post-credential <math>B</math>-marginal <math>11/25</math>. Hard cues with fixed ratios <math>1/4</math> and <math>7/3</math> commute cell by cell and reproduce <math>P^B</math>, with <math>A</math>-marginal <math>27/119</math> in either sequence; the soft sequences end at <math>18/77</math> (credential first) and <math>1/5</math> (letter first).</p> <p><i>The two readings differ in what is held fixed when the cue meets a moved marginal, not in whether a counterfactual likelihood exists.</i></p> <p><b>Theorem.</b> <i>JeffreyOrder/PropIMM.lean, propIMM_single_cue (with bayesAW_total)</i></p> <p>For <math>\alpha \neq 0</math>, <math>1 - \alpha \neq 0</math> and every <math>\beta</math>, <math>c</math>, <math>q_0</math>: <math>\text{bayesA}(P(\alpha, \beta, c), \alpha, q_0) = \text{jeffreyA}(P(\alpha, \beta, c), q_0)</math>, i.e. reweighting <math>P</math> by <math>q_0/\alpha</math> on row <math>A=0</math> and <math>(1 - q_0)/(1 - \alpha)</math> on row <math>A=1</math> and normalising equals the Jeffrey step to <math>A</math>-marginal <math>q_0</math>, identically; the normaliser is 1.</p> <p><i>On a single cue the two readings agree, as the corrected sentence says (Proposition IMM).</i></p> |                                                                                                                                                                                                                      |
| <b>C.7</b> Introduction, terminology paragraph, Domotor sentence and two typos                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                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| 1/2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | The sequence-dependence of marginals follows easily – since the likelihood a delivered credence implies must be read against the marginal in force when the cue arrives (Domotor, 1980).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | The sequence dependence of the marginals follows directly, since a delivered credence is taken against the marginal in force when the cue arrives, which the first cue moves when $c \neq 0$ (Proposition PROP:DIV). |
| 2/2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | To avoid a clash in terminology we use “order” to describe proximity with a sequence-free Bayesian bechmark and “sequence“ to denote the sequence in which evidence is read.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | To avoid a clash in terminology we use “order” to describe proximity with a sequence-free Bayesian benchmark and “sequence” to denote the sequence in which evidence is read.                                        |
| <p><b>Why.</b> Domotor never mentions likelihoods or marginals; he supports only bare non-commutativity, and treats it as a defect of Jeffrey machines (p. 395), the opposite lean (audit M1). The mechanism is the paper’s own (Proposition DIV). If a Domotor citation is wanted, it fits the literature paragraph as a source of the commutativity demand that Jeffrey (1983, pp. 182-183) rejects. Typos: “bechmark”, sequence closed with backquotes.</p> <p><b>Evidence.</b> Domotor.lean (field_eq_jeffrey_embed, embed_depends_on_state); Cripps.lean (composite_AB_eq_bayes: the B-marginal moves by <math>c(q_0 - \alpha)/(\alpha(1 - \alpha))</math>).</p> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         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|      | <p><b>Grounds.</b></p> <p><b>Quote.</b> <i>Domotor (1980), p. 395</i></p> <p>“Along with Field (1978) we may argue that Jeffrey machines are inadequate because, among other things, the commutativity of evidence application: <math>[P_{(U,p)}]_{(V,q)} = [P_{(V,q)}]_{(U,p)}</math> fails in general.”</p> <p><i>Domotor supports bare non-commutativity and counts it against Jeffrey machines; the manuscript cites him for a mechanism and for the opposite lean.</i></p> <p><b>Quote.</b> <i>Domotor (1980), p. 397</i></p> <p>“A moment’s thought shows that Field’s input space is embeddable into Jeffrey’s input space by the map: <math>h_P : \mathbf{F}_X \rightarrow \mathbf{E}_X</math>, defined by <math>h_P(U, \alpha) = (U, p)</math>, where: <math>p(A) = e^{\alpha A} P(A) : \alpha_X</math>.”</p> <p>...“So, whatever advantage may have been gained in commutativity is lost in probabilistic independence.”</p> <p><i>The passage nearest a mechanism: a fixed factor <math>\alpha</math> yields a <math>p</math> that depends on the state <math>P</math>, used to argue that Field’s space is smaller, not to explain non-commutativity. Neither “likelihood” nor “marginal” occurs on the rendered pp. 394–397, the pages around the quoted passages.</i></p> <p><b>Theorem.</b> <i>Domotor.lean, field_eq_jeffrey_embed; embed_depends_on_state</i></p> <p>For <math>P \geq 0</math> on finite <math>\Omega</math>, a partition <math>u : \Omega \rightarrow I</math> and <math>\alpha : I \rightarrow \mathbb{R}</math>, put <math>\alpha_X = \sum_y e^{\alpha u(y)} P(y)</math> and <math>h_P(\alpha)_i = e^{\alpha_i} P(E_i) / \alpha_X</math>. Then Field’s conditional <math>x \mapsto e^{\alpha u(x)} P(x) / \alpha_X</math> equals Jeffrey’s rule on <math>(u, h_P(\alpha))</math>. The image depends on the state: <math>\alpha = (\log 2, 0)</math> on two atoms gives <math>p = (2/3, 1/3)</math> at <math>P = (1/2, 1/2)</math> but <math>p = (2/5, 3/5)</math> at <math>P = (1/4, 3/4)</math>.</p> <p><i>Read in reverse, a fixed delivered credence corresponds to a factor that depends on the belief it meets; Domotor states the embedding but draws neither reading.</i></p> <p><b>Theorem.</b> <i>Cripps.lean, composite_AB_eq_bayes</i></p> <p>Let <math>\mu</math> on <math>A \times B</math> be strictly positive with <math>\sum \mu = 1</math>, <math>q</math> on <math>A</math> strictly positive with <math>\sum_a q_a = 1</math>, and <math>r</math> on <math>B</math> with <math>\sum_b r_b = 1</math>. Then <math>J_B(J_A(\mu, q), r) = \text{bayes}(\mu, x \mapsto \frac{q_{x_1}}{\mu(A=x_1)} \cdot \frac{r_{x_2}}{(J_A(\mu, q))(B=x_2)})</math>, where <math>\text{bayes}(\mu, \ell)(\theta) = \mu(\theta)\ell(\theta) / \sum_{\theta'} \mu(\theta')\ell(\theta')</math> and <math>J_A(\mu, q)(x) = q_{x_1} \mu(x) / \mu(A=x_1)</math>. In the other sequence the <math>B</math>-likelihood is <math>r_{x_2} / \mu(B=x_2)</math>.</p> <p><i>The second cue’s likelihood is matched to the intermediate belief, which the first cue has moved; this is the corrected sentence’s mechanism, and it is the paper’s own.</i></p> <p><b>Computation.</b> <i>sympy, recomputed symbolically (the instance is check 3 of verify_soft_vs_hard.py)</i></p> <p>After the <math>A</math>-step to <math>q_0</math> from <math>P(\alpha, \beta, c)</math>, the <math>B</math>-marginal is <math>\beta + c \frac{q_0 - \alpha}{\alpha(1 - \alpha)}</math>, so the first cue moves it whenever <math>c \neq 0</math> and <math>q_0 \neq \alpha</math>. At <math>\alpha = \beta = 1/2</math>, <math>c = 1/20</math>, <math>q_0 = 1/5</math> it is <math>11/25</math>.</p> <p><i>The delivered credence <math>r_0</math> is then read against <math>11/25</math> instead of <math>1/2</math>, which is the whole of the sequence effect.</i></p> <p><b>Quote.</b> <i>PAPER B MANUSCRIPT.tex, Proposition DIV (micro divergence), (i)</i></p> <p>Let <math>Z := \alpha\beta(1 - \alpha)(1 - \beta) &gt; 0</math>. “At <math>c = 0</math> both reading sequences coincide with the benchmark, <math>P_{AB}^J = P_{BA}^J = P^B = q \otimes r</math>.” (i) For <math>c \neq 0</math>, <math>P_{AB}^J - P^B = c \kappa R_1 + O(c^2)</math>, <math>R_1 = q \otimes (1, -1)</math>, <math>\kappa = \frac{(\alpha - q_0) r_0 (1 - r_0)}{Z}</math>, “with <math>\kappa = 0 \iff q_0 = \alpha</math> or <math>r_0 \in \{0, 1\}</math>”.</p> <p><i>The manuscript’s own statement of the mechanism the sentence should cite.</i></p> |                                                                                                                                                                                                                                                                                                                                                                           |
|      | <h2>C.8 Introduction, key-finding paragraph, last two sentences</h2>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                           |
|      | <p>The implication for evaluation is that and audit’s answer depends on which statistic is aggregated rather than on how many evaluators are averaged over. One consequence, is that the instruments most naturally trusted for detecting stereotype—belief audits asking how attributes are thought to go together—may not even be detectable.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <p>The implication for evaluation is that an audit’s answer depends on which statistic is aggregated rather than on how many evaluators are averaged over. One consequence is that belief audits asking how attributes are thought to go together, the instruments most naturally trusted for detecting stereotype, cannot detect the sequence effect at first order.</p> |
|      | <p><b>Why.</b> Typos (“and audit’s”, “One consequence, is”), and “the instruments ... may not even be detectable” says the instruments are undetectable; it is the sequence effect that belief audits cannot detect at first order.</p> <p><b>Evidence.</b> Propositions DEC, PRO.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                           |
|      | <p><b>Grounds.</b></p> <p><b>Theorem.</b> <i>PropDEC.lean, assoc_PJab; PropPRO.lean, lemmaASC_first_order_vanishes</i></p> <p><math>\text{assoc}(P_{AB}^J) = \frac{q_0}{\alpha} \frac{1 - q_0}{1 - \alpha} \frac{r_0}{m_0} \frac{1 - r_0}{m_1} c</math> with <math>m_j</math> the <math>B</math>-marginal after the <math>A</math>-step; and along either route direction <math>\text{assoc}(q \otimes r + t \kappa R_\sigma) = \text{assoc}(q \otimes r) + t^2 \text{assoc}(\kappa R_\sigma)</math>.</p> <p><i>Proposition DEC: the believed association departs from the benchmark only at second order in <math>c</math>.</i></p> <p><b>Theorem.</b> <i>PropPRO.lean, propPRO_protection_and_margA_not_in_span</i></p> <p><math>\langle xJ + y \nabla \text{assoc}, \lambda \kappa R_1 + (1 - \lambda) \kappa' R_2 \rangle = 0</math> for all <math>x, y, \alpha, \beta, \lambda</math>; whereas for <math>0 &lt; r_0 &lt; 1</math> the marginal read-out <math>(0, 0, 1, 1)</math> is not in <math>\text{span}\{J, \nabla \text{assoc}\}</math>.</p> <p><i>Proposition PRO: which statistic is aggregated decides whether the sequence effect is first or second order.</i></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                           |

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| <b>Theorem.</b> <i>PropPRO.lean, propPRO_uniqueness</i><br>For $\lambda \in (0, 1)$ , $q_0, r_0 \in (0, 1)$ , $\beta \neq r_0$ : if the aggregate first-order coefficient $\lambda(\alpha - q_0)r_0(1 - r_0)x + (1 - \lambda)(\beta - r_0)q_0(1 - q_0)y$ vanishes at two priors $\alpha_1 \neq \alpha_2$ , then $x = y = 0$ .<br><i>The coefficient is a property of the population's belief, not of a sample; averaging over more evaluators does not change it.</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>C.9</b> Setup 2.1, the prior paragraph and the sequence/weight sentences                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 1/3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | The three numbers that fix $P$ are described as $\Delta^3$ – specified using the marginals $P(A=0) = \alpha$ , $P(B=0) = \beta$ and the prior covariance $c$ between the attributes:                                                                                                                                                         | The prior is a point of $\Delta^3$ fixed by three numbers, the marginals $P(A=0) = \alpha$ and $P(B=0) = \beta$ and the prior covariance $c$ between the attributes,                                                                                                                                                                                                                                                                                        |
| 2/3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Simply put, the marginals $\alpha = P(A=0)$ and $\beta = P(B=0)$ denotes the shares of applicants the panelist believes to lack competence and to lack trustworthiness and the prior covariance $c = \text{assoc}(P)$ denotes how strongly the panelist believes the                                                                         | Simply put, the marginals $\alpha = P(A=0)$ and $\beta = P(B=0)$ denote the shares of applicants the panelist believes to lack competence and to lack trustworthiness, and the prior covariance $c = \text{assoc}(P)$ denotes how strongly the panelist believes the                                                                                                                                                                                        |
| 3/3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | To address the issues in amnestic updating debate, we also consider an adoption weight $\omega \in [0, 1]$ – which denotes how far the letter displaces what the credential had already implied about trustworthiness. At $\omega = 1$ the letter sets the rating outright, and at $\omega = 0$ the credential's implication is never moved. | To compare full adoption with the belief-adjustment model of Hogarth and Einhorn (1992), in which a later impression is adopted only in part, we also consider an adoption weight $\omega \in [0, 1]$ (Section SEC:ROBUST), which denotes how far the letter displaces what the credential had already implied about trustworthiness. At $\omega = 1$ the letter sets the rating outright, and at $\omega = 0$ the credential's implication is never moved. |
| <b>Why.</b> Purpose of omega (author's decision, 2026-10-01): the weight nests Hogarth-Einhorn's averaging rule (their Eq. 4, applied to the second cue; E.10), so full adoption is one corner of a descriptive model rather than an assumption taken from the normative literature. The Hawthorne debate is now taken up in the introduction (C.4). Grammar ("described as $\Delta^3$ –", "denotes" for two subjects). "Amnestic" is used only where Hawthorne is quoted (writing_discipline.md 6). <b>Dependency:</b> omega is defined here but used nowhere else in the manuscript until plan 5.B (Proposition ADJ) is applied; either apply 5.B or move this sentence into 5.B. The Frechet-Hoeffding bounds in the new text are correct (checked cell by cell).<br><b>Evidence.</b> Bounds: nonnegativity of the four cells of P; Hawthorne pp. 98-99.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Grounds.</b><br><b>Quote.</b> <i>Hawthorne (2004), §5, pp. 98–99</i><br>"The resulting Amnestic Model raises two related concerns. First, it seems implausible that the most recent experience or non-propositional state should completely dictate belief strengths for basis sentences, with no regard for the import of previous experiences or states. There may be some specialized systems for which this model is appropriate. But it seems wrong as a model of (idealized) human agents" ... "This order effect is very troubling."<br><i>The source of the debate the corrected sentence names; Hawthorne's alternatives are factor models (§§6–8), not a partial weight, so the citation is for the objection only.</i><br><b>Quote.</b> <i>Hawthorne (2004), §5, p. 96</i><br>"AMNESTIC UPDATE-FACTOR THESIS. For any state $e$ that directly affects an evidence basis $\{E_i\}$ and for any other state $d$ and sequence of states $\alpha$ , $Q_{\alpha de}[E_i] = Q_{\alpha e}[E_i]$ ."<br>"Indeed Amnestic Updating is just Standard Sequential Updating – Jeffrey's original approach to sequential updating."<br><i>Full adoption, <math>\omega = 1</math>, is this thesis under Hawthorne's name for it.</i><br><b>Computation.</b> <i>Nonnegativity of the four cells of P (checked cell by cell in sympy)</i><br>$\alpha\beta + c \geq 0$ , $\alpha(1 - \beta) - c \geq 0$ , $(1 - \alpha)\beta - c \geq 0$ , $(1 - \alpha)(1 - \beta) + c \geq 0$ , hence $-\min\{\alpha\beta, (1 - \alpha)(1 - \beta)\} \leq c \leq \min\{\alpha(1 - \beta), (1 - \alpha)\beta\}$ , the Fréchet–Hoeffding bounds for a $2 \times 2$ table with marginals $\alpha, \beta$ ; at $\alpha = \beta = 1/2$ the range is $[-1/4, 1/4]$ .<br><i>The manuscript's bounds sentence is correct and is kept.</i> |                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>C.10</b> Setup 2.2, the invariance condition and the justification of the rule                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 1/2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | holding the conditionals fixed—the invariance condition (Jeffrey, 1983, ch. 11).                                                                                                                                                                                                                                                             | holding the conditionals given the cue's partition fixed, the invariance condition (Jeffrey, 2004, ch. 3), which Jeffrey (1983, ch. 11) expresses by saying that the change originates in the partition.                                                                                                                                                                                                                                                    |

| Part                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   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| The axiomatic grounding is twofold. The partition satisfying the invariance condition is the minimal sufficient statistic for revising the prior to any candidate posterior on that partition (Diaconis and Zabell, 1982), and the rule is the minimal revision of the prior consistent with the delivered marginal, in the $I$ -divergence sense (Diaconis and Zabell, 1982). | Two results of Diaconis and Zabell (1982) support the rule. The invariance condition makes the cue's partition sufficient for the prior and posterior, and minimal whenever the delivered credence differs from the prior marginal (their Theorem 2.2). The rule is also the unique minimiser of the $I$ -divergence from the prior among all beliefs with the delivered marginal (their Theorem 5.1). |
| <p><b>Why.</b> (i) Chapter 11 of Jeffrey (1983) never uses “invariance”; it says the change “originates in” the partition (pp. 168, 174). “Invariance” is the 2004 book’s term (audit J4). (ii) “The partition satisfying the invariance condition is the minimal sufficient statistic for revising the prior to any candidate posterior” misstates D-Z Theorem 2.2: sufficiency is for one pair <math>\{P, P^*\}</math>, every refinement of such a partition also qualifies, and the minimal one is the likelihood-ratio partition (M23; Jeffrey p. 174 also allows “a certain latitude in the choice” of partition, J3). (iii) D-Z give no axioms; they call this “mechanical updating”. The summation index is <math>x</math> because <math>\omega</math> is now the adoption weight.</p> <p><b>Evidence.</b> DiaconisZabell.lean (thm22_lr_minimal, jcond_of_refines, lr_jeffrey_iff, thm51_KL_eq_iff); verify_jeffrey.md J3-J4.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            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                                                                                           |
| <p><b>Grounds.</b></p> <p><b>Quote.</b> Jeffrey (1983), ch. 11, §11.3 p. 168 and §11.7 p. 174</p> <p>p. 168: “the values of <math>PROB</math> for all arguments ought to be deducible from a knowledge of (a) the values of <math>prob</math> for all arguments, (b) the value of <math>PROB</math> for the argument <math>B</math>, and the fact that (c) the change from <math>prob</math> to <math>PROB</math> originated in <math>B</math>”. p. 174: “(11-8) <math>PROB(A/A_i) = prob(A/A_i)</math> for each <math>i = 1, 2, \dots, m</math> which defines what we shall mean by saying that the change from <math>prob</math> to <math>PROB</math> originated in the set (11-6).”</p> <p>On the rendered pp. 168 and 174–175 Jeffrey’s term for the condition is that the change originated in the partition, not “invariance”.</p> <p><b>Quote.</b> Jeffrey (1983), §11.7, p. 174, and Example 7</p> <p>“Note that since distinct sets of propositions can have the same set of atoms, there is a certain latitude in the choice of a set (11-6) in which the change from <math>prob</math> to <math>PROB</math> is viewed as originating.” Example 7: the change of Example 6, originating in <math>G, B, V</math>, “might equally well be regarded as originating in the set <math>B_1 = G \vee B, B_2 = G</math>”.</p> <p>On Jeffrey’s own account the originating set is not unique, against “the partition satisfying the invariance condition”.</p> <p><b>Quote.</b> Jeffrey (2002 draft), ch. 3, §3.1 p. 56 and §3.2 pp. 57–58</p> <p>“Invariant conditional probabilities: (1) For all <math>H</math>, <math>new(H D) = old(H D)</math>”. “As long as invariance holds, updating is valid by a generalization of conditioning to which we now turn.”</p> <p>“If the invariance condition holds for each answer, <math>D_i</math>, we have the updating scheme <math>new(H) = old(H D_1) new(D_1) + old(H D_2) new(D_2) + \dots</math>.” “This is equivalent to invariance with respect to every answer: <math>new(H D_i) = old(H D_i)</math> for <math>i = 1, \dots, n</math>.”</p> <p>“Invariance” is the 2004 book’s term, introduced in §3.1 for one proposition and carried to partitions in §3.2.</p> <p><b>Quote.</b> Diaconis and Zabell (1982), §2.2, p. 824</p> <p>“To apply Jeffrey’s rule, it is required to find a partition <math>\{E_i\}</math> such that <math>P(A E_i) = P^*(A E_i)</math> for all <math>A</math> and <math>i</math>. This is simply the problem of finding a <i>sufficient partition</i> for the two-element family <math>\mathcal{F} = \{P, P^*\}</math>”. “A coarsest sufficient partition is said to be minimal sufficient. The following (well-known) theorem gives an alternative version of Jeffrey’s rule and states that there is always a coarsest partition for which Jeffrey’s rule is valid.”</p> <p>Sufficiency is relative to one pair <math>\{P, P^*\}</math>, not to “any candidate posterior”.</p> <p><b>Quote.</b> Diaconis and Zabell (1982), Theorem 2.2, p. 824</p> <p>“Theorem 2.2. Let <math>P, P^*</math> be probability measures with common support on the countable set <math>\Omega</math>. If <math>\{E_i\}</math> is a partition of <math>\Omega</math> such that <math>P(E_i) &gt; 0</math> and <math>P(A E_i) = P^*(A E_i)</math> for all subsets <math>A</math> and elements of the partition <math>E_i</math>, then for each <math>\omega \in \Omega</math>, <math>P^*(\omega) = \frac{P^*(E_i)}{P(E_i)} P(\omega)</math>, <math>\omega \in E_i</math>. (2.2) If <math>R = \{x : P^*(\omega)/P(\omega) = x, \omega \in \Omega\}</math>, and <math>E_x = \{\omega : P^*(\omega)/P(\omega) = x, \omega \in \Omega\}</math>, then <math>\{E_x : x \in R\}</math> is a minimal sufficient partition for <math>\{P, P^*\}</math>.”</p> <p>The minimal sufficient partition is the likelihood-ratio partition, which depends on the pair; the manuscript assigns minimality to the cue’s partition.</p> <p><b>Quote.</b> Diaconis and Zabell (1982), §5.2 and §5.3, p. 828</p> <p>“Mechanical updating allows the possibility of updating on collections of sets more general than partitions.” “Theorem 5.1 suggests that Jeffrey’s rule is an uncontroversial form of mechanical updating in the sense that it agrees with virtually every minimum-distance rule.”</p> <p>D-Z give no axioms; “axiomatic grounding” mislabels what they call mechanical updating.</p> <p><b>Theorem.</b> DiaconisZabell.lean, thm22_lr_minimal</p> <p>Let <math>P, P^* &gt; 0</math> on finite <math>\Omega</math> and let <math>e : \Omega \rightarrow \bar{I}</math> satisfy (J): <math>P^*(A E_i) = P(A E_i)</math> for every <math>A \subseteq \Omega</math> and <math>i</math>. Then <math>e(\omega) = e(\omega') \Rightarrow P^*(\omega)/P(\omega) = P^*(\omega')/P(\omega')</math>: every cell of a (J)-partition lies inside one level set of <math>P^*/P</math>; with the converse (lr_jeffrey_iff) the likelihood-ratio partition is the coarsest sufficient one, and it depends on the pair <math>\{P, P^*\}</math>.</p> <p>Minimality belongs to the likelihood-ratio partition, not to whichever partition carries the cue.</p> |                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                        |

| Part                                                                                      | BEFORE (current manuscript)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | AFTER (proposed)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
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|                                                                                           | <p><b>Theorem.</b> <i>DiaconisZabell.lean, jcond_of_refines (with jcond_atoms, jcond_trivial_self)</i><br/>Let <math>P, P^* &gt; 0</math> on finite <math>\Omega</math>, let <math>e : \Omega \rightarrow I</math> satisfy (J), and let <math>e' : \Omega \rightarrow K</math> refine <math>e</math>, i.e. <math>e'(\omega) = e'(\omega') \Rightarrow e(\omega) = e(\omega')</math>. Then <math>e'</math> satisfies (J). In particular the partition into atoms satisfies (J) for every pair, and if <math>P^* = P</math> the one-cell partition does.<br/>“The partition satisfying the invariance condition” is not unique: every refinement qualifies, and the cue’s partition is minimal only when the delivered credence differs from the prior marginal.</p> <p><b>Theorem.</b> <i>DiaconisZabell.lean, thm51_KL_eq_iff; D-Z Theorem 5.1, (5.6)</i><br/>For finite <math>\Omega</math>, <math>P &gt; 0</math>, <math>e : \Omega \rightarrow I</math> surjective and <math>Q \geq 0</math> with <math>Q(E_i) = p_i</math> for all <math>i</math>: <math>\text{KL}(Q \parallel P) = \sum_i p_i \log \frac{p_i}{P(E_i)}</math> if and only if <math>Q(\omega) = p_{e(\omega)}P(\omega)/P(E_{e(\omega)})</math>; with <code>thm51_KL_le</code> the right-hand side is the minimum over the feasible set.<br/><i>Supports the corrected second sentence: the rule is the unique I-divergence minimiser among beliefs with the delivered marginal.</i></p> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>C.11</b> Related literature, the paragraph from BHW/Banerjee to Becker and Good-Mittal |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 1/9                                                                                       | In Banerjee (1992), the sequence persists because a coarse public action fails to be a sufficient statistic for private information. In Bikhchandani et al. (1992), on the other hand, a cascade is an action that conveys no private signal.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | In both, the equilibrium action is not a sufficient statistic for private information, being binary in Bikhchandani et al. (1992) and a non-invertible choice rule in Banerjee (1992), so once a cascade or herd forms, later actions convey no private signal.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 2/9                                                                                       | One similarity with the literature, however, is that the sequence effect in (Bikhchandani et al., 1992; Banerjee, 1992) lives on the analogue of the marginals—which remain so in the current paper as well.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | One analogy remains. Those models have a single state, so their sequence effect falls on a belief about one variable, the counterpart of a marginal, and here too the marginals carry the sequence effect.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 3/9                                                                                       | A related normative literature evaluates group belief formation by whether aggregation commutes with updating—a design criterion for the pooling rule (Dietrich, 2021). While geometric pooling (Dietrich, 2021, Def. 1) satisfies the criterion, the population mean studied in the current paper (i.e. the average posterior across evaluators who differ only in the sequence they meet the two cues) is the outcome of linear pooling and generically fails the criterion. As further sections elaborate, this is consistent with the first-order divergence of every statistic outside the protected class of Proposition <code>PROP:PRO</code> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | A related normative literature evaluates group belief formation by whether aggregation commutes with updating, a criterion that geometric pooling meets and linear pooling generally does not (Dietrich, 2021, Theorem 2). The criterion does not discriminate here. Evaluators who share a prior and differ only in reading sequence do not hold conditionalisations of that prior on a common event, and the linear and geometric averages of their posteriors differ only at $\mathcal{O}(c^2)$ . The first-order divergence of every statistic outside the protected class of Proposition <code>PROP:PRO</code> is therefore a property of Jeffrey updating, not of linear pooling.                                                                                                                                           |
| 4/9                                                                                       | <b>Several alternatives to Bayesian updating have been used in the decision theory literature to explain whether sequential conditioning is sequence-independent or not.</b> For example, Epstein (2006) makes the updating rule subjective, and Ortoleva (2012) axiomatises departures triggered by unexpected news. Cripps (2021) shows that symmetry and divisibility jointly force sequence-independence, so the correlated two-cue composite of Proposition <code>PROP:DIV</code> fails divisibility—and, of the four axioms, only divisibility. <i>[Footnote: Cripps’s axioms govern rules on fixed-likelihood experiments; the statement here is behavioural—the map from prior and delivered credences to final belief, held against the sequence-invariance that his Symmetry and Divisibility axioms jointly force.]</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Decision theory has axiomatised several alternatives to Bayesian updating, each for a single piece of news. Epstein (2006) makes the updating rule, and not only the prior, subjective, and Ortoleva (2012) axiomatises departures triggered by unexpected news. Cripps (2021) treats the sequence of signals, and his Symmetry and Divisibility axioms imply that two conditionally independent signals give the same posterior in either sequence. Read with each cue as its likelihood matched to the prior (Proposition <code>PROP:IMM</code> ), the composite of Proposition <code>PROP:DIV</code> is a sequence of Bayes updates satisfying all four of his axioms, and it depends on the sequence because the second likelihood is matched to the intermediate belief, so the two sequences process different experiments. |
| 5/9                                                                                       | While the literature restores sequence-invariance for sequential Jeffrey updating by changing how successive inputs are pooled (Pettigrew and Weisberg, 2025),                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | While Pettigrew and Weisberg (2025) restore sequence-invariance by pooling the prior with each new input multiplicatively before the Jeffrey step,                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 6/9                                                                                       | The paper borrows the evaluator-with-binary-attributes frame from the statistical-discrimination lineage (Phelps, 1972; Arrow, 1973).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | The paper borrows the evaluator-with-binary-attribute frame from the statistical-discrimination lineage (Arrow, 1973; Coate and Loury, 1993).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 7/9                                                                                       | instead of stereotypes as representativeness-distortion or selective recall (Bordalo et al., 2016), the sense of stereotype in the current paper is that of a believed cross-attribute association produced by coherent updating on impoverished input, not by a distortion of memory or sampling.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | instead of stereotypes as the selective recall of representative types (Bordalo et al., 2016), whose exaggerated associations arise across groups, the stereotype in the current paper is a believed cross-attribute association held by one evaluator and produced by coherent updating on impoverished input, not by a distortion of memory.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |

| Part                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | BEFORE (current manuscript)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | AFTER (proposed)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
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| 8/9                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | The lineage distinction the concluding section trades on—protection intrinsic to the arithmetic of aggregation, rather than imposed by an enforced constraint on behaviour—goes back at least as far as Becker (1962).                                                                                                                                                                                                                                                                                                                                                                                                         | <i>(sentence deleted)</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 9/9                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | It is worth contrasting the phenomenon explored in the current paper against the amalgamation paradox (Good and Mittal, 1987), where a real effect present in every subpopulation is erased or reversed by a confound in how the subpopulations are weighted together. Whereas the amalgamation paradox erases a real effect by combining subpopulations under confounded weights, our decision-space result shows the same kind of erasure can occur within a single population – where a first-order share of individuals who are individually affected contributes only a second-order loss in aggregate (Theorem THM:LOS). | It is worth contrasting the phenomenon explored here with the amalgamation paradox (Good and Mittal, 1987), in which a measure of association on a combined table lies outside the range of the same measure on its subtables, because treatment is allocated unevenly across the subpopulations, even when they are equally large. That paradox arises from combining subpopulations. The decision-space result here shows a comparable loss of visibility within a single population, where a first-order share of individuals who are individually affected contributes only a second-order loss in aggregate (Theorem THM:LOS). The audit perspective has a precedent in the economics of discrimination. Heckman (1998) showed that audit-pair estimates of discrimination rest on an assumption about unobserved productivity that “nothing guarantees” (p. 109), so that such an audit “can find discrimination when in fact none exists; it can also disguise discrimination when it is present” (p. 102), and Bohren et al. (2019) identify the source of discrimination by conditioning on the history of evaluations, since in static data, as they note, different sources generate the same patterns of observable behaviour. The obstruction here is of the same kind and arises on the belief side of an audit. The first-order signal in a protected statistic is zero in the population itself, so its silence is a failure of identification and not of estimation, which no sample size repairs, and the identifying variable is the reading sequence, which pooled data discard. |
| <p><b>Why.</b> One entry per sentence group, in manuscript order. BHW/Banerjee: “coarse” belongs to BHW’s binary action, not Banerjee’s continuum; the two share one mechanism, so “on the other hand” goes; the marginals point is the paper’s analogy (M2, M3). Dietrich: Def. 1 is preference aggregation, and with a common prior linear pooling passes the criterion wherever it applies (M4, M5). Epstein/Ortoleva/Cripps (bold sentence): the framing is unsupported for Epstein and Ortoleva; the Cripps claim is false, since under Proposition IMM’s reading the composite satisfies all four axioms, and the footnote goes with it (M6-M9). Pettigrew-Weisberg pool the prior with each input, not successive inputs (M25). Phelps has no binary attribute, so he is dropped from this sentence and can be re-cited in plan 1.C for statistical discrimination in general (M10). BCGS name one mechanism, not two, and “sampling” is not theirs (M12). Becker: the concluding section never draws the distinction, and Becker’s own mechanism is averaging plus a budget constraint, the opposite side of it (“Our statement goes beyond arithmetic”, p. 7; M13); the AFTER deletes the sentence, and an alternative that keeps Becker is in literature/becker1962/README.md. Good-Mittal: their paradox covers amplification and effects created from none, and its cause is uneven allocation, not weighting by population shares (M14).</p> <p><b>Evidence.</b> BHW.lean, Banerjee.lean, Dietrich.lean, Epstein.lean, Ortoleva.lean (via the 2024 survey), Cripps.lean (<code>order_invariance</code>, <code>composite_AB_eq_bayes</code>), PettigrewWeisberg.lean, Phelps.lean, Arrow.lean, CoateLoury.lean, BCGS.lean, Becker.lean, GoodMittal.lean (<code>piR_amalg_general</code>, <code>equalSize_reversal</code>). The appended paragraph (C.15) carries the identification point of notes/positioning_economics.tex and plan 3.C/3.D into Section 3, with audit P17a (no sample-size claim attributed to Heckman; that clause is the paper’s own consequence of Proposition PRO), H2 (“nothing guarantees” is p. 109), H3 (“can find discrimination” is p. 102) and B37 (BIR credit “different sources” to Fang-Moro, so it is paraphrased) applied. Needs Heckman1998 and Bohren2019 in bibliography.bib (plan B.A).</p> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <p><b>Grounds.</b></p> <p><b>Quote.</b> <i>Bikhchandani, Hirshleifer and Welch 1992, p. 1000</i></p> <p>“DEFINITION. An informational cascade occurs if an individual’s action does not depend on his private information signal. If an individual <math>i</math> is in a cascade, then his action conveys no information and individual <math>i + 1</math> draws the same inference from all previous actions.”</p> <p><i>(part 1, BHW) A cascade is defined by the action’s silence about the signal; the action is binary (adopt or reject, p. 996), which is where “coarse” belongs.</i></p> <p><b>Quote.</b> <i>Banerjee 1992, p. 809</i></p> <p>“The key reason why we get a different result is that in our model the choices made by agents are not always sufficient statistics for the information they have. ... It is evident that, for a wide range of signals, the agents in our model will always choose the same option; this lack of invertibility is what causes the sufficient property to fail.”</p> <p><i>(part 1, Banerjee) The same mechanism as BHW, on a continuum of options, so “on the other hand” has no contrast to mark.</i></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |

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|      | <p><b>Theorem.</b> <i>BHW.lean, lik_ratio (with cascade_uninformative)</i><br/> For every <math>p</math> and every history <math>h</math> with <math>d(h) = \# \text{adopt} - \# \text{reject}</math>, <math>L_1(h) c_0(d) = L_0(h) c_1(d)</math>, so the public likelihood ratio <math>L_1/L_0</math> is a function of <math>d</math> alone: 1 at <math>d = 0</math>, <math>p/(1-p)</math> at <math>d = 1</math>, <math>p(1+p)/((1-p)(2-p))</math> at <math>d \geq 2</math>, and the reciprocals at <math>d = -1</math>, <math>d \leq -2</math>.<br/> (part 1, BHW) Two adoptions carry less than two <math>H</math> signals, since <math>p(1+p)/((1-p)(2-p)) &lt; (p/(1-p))^2 \iff 1 &lt; 2p</math>; and the ratio is constant once <math> d  \geq 2</math>, so later actions convey no private signal.</p> <p><b>Quote.</b> <i>Bikhchandani, Hirshleifer and Welch 1992, p. 996</i><br/> “The gain to adopting, <math>V</math>, is also the same for all individuals and is either zero or one, with equal prior probability <math>1/2</math>.”<br/> (part 2, marginals analogy) One binary state, so the sequence effect in BHW falls on a belief about one variable, the counterpart of a marginal.</p> <p><b>Quote.</b> <i>Banerjee 1992, p. 802</i><br/> “Let us assume that there is a unique <math>i^*</math> such that <math>z(i) = 0</math> for all <math>i \neq i^*</math> and <math>z(i^*) = z</math>, where <math>z &gt; 0</math>. ... Of course, everybody, given these payoffs, would want to invest in <math>i^*</math>. The trouble is no one knows which one it is.”<br/> (part 2, marginals analogy) One unknown, the identity of <math>i^*</math>; there is no second attribute and no association for the sequence effect to fall on, which is the analogy the AFTER keeps.</p> <p><b>Quote.</b> <i>Dietrich 2021, Definition 1, p. 7</i><br/> “Definition 1 An EU preference aggregation rule ... is linear-geometric if there exist individual weights ... such that for each preference profile ... the group preference relation ... has ... probability function ... given on states by ... up to a multiplicative constant.”<br/> (part 3, Dietrich) Definition 1 defines a rule on preference profiles, so it is preference aggregation; the belief-pooling statement is Theorem 2, which the AFTER cites.</p> <p><b>Quote.</b> <i>Dietrich 2021, Theorem 2, p. 11</i><br/> “Theorem 2 A belief aggregation rule (on the domain of coherent belief profiles) is dynamically rational, unanimity-preserving, and continuous if and only if it is geometric.”<br/> (part 3, Dietrich) The criterion is Dynamic Rationality, defined on p. 16 as “When information is learnt by everyone, the new group beliefs equal the old ones conditional on the information”.</p> <p><b>Theorem.</b> <i>Dietrich.lean, linPool_dynRational_on_common_prior</i><br/> Let <math>w_i \geq 0</math> with <math>\sum_i w_i = 1</math>, let <math>p_i = \pi</math> for every <math>i</math>, and let <math>E</math> be any event. Then <math>\sum_i w_i p_i(\cdot   E) = (\sum_i w_i p_i)(\cdot   E)</math>.<br/> (part 3, Dietrich) With a common prior the premise of Dynamic Rationality forces unanimity, and linear pooling passes wherever the criterion applies; Paper B’s evaluators do not condition on a common event, so the criterion is silent on them.</p> <p><b>Quote.</b> <i>Epstein 2006, abstract, p. 413</i><br/> “The main result is a representation theorem that generalizes (the dynamic version of) Anscombe–Aumann’s theorem so that both the prior and the way in which it is updated are subjective.”<br/> (part 4, Epstein) What Epstein makes subjective is the updating rule, not only the prior; nothing in the abstract concerns the sequence of several updates.</p> <p><b>Quote.</b> <i>Epstein 2006, p. 416</i><br/> “There are three periods—an ex ante stage 0, an interim period 1 when a signal <math>s_1 \in S_1</math> is realized, and period 2 when the remaining uncertainty is resolved through realization of some <math>s_2 \in S_2</math>.”<br/> (part 4, Epstein) One interim signal, so the model has no sequence of updates whose invariance could be at issue.</p> <p><b>Quote.</b> <i>Ortoleva 2024, p. 546 and footnote 2</i><br/> “we do not discuss updating ... when subjects receive more general types of information, such as vague, ambiguous, or imprecise information or information of the form ‘Event A is more likely than event B’;” with footnote 2: “These cases have been extensively studied in several disciplines and include classical approaches such as Jeffrey’s rule or the AGM theory of belief revisions”.</p> <p>(part 4, Ortoleva) Jeffrey’s rule is outside the survey’s scope, so the Hypothesis Testing model is not a theory of Jeffrey-type sequence effects.</p> <p><b>Quote.</b> <i>Ortoleva 2024, Section 4.1, p. 559</i><br/> “When new information <math>A</math> arrives, individuals first test to determine if they were using the right prior, using the threshold <math>\epsilon</math>. If <math>A</math> is not unlikely given <math>\pi</math>, that is, <math>\pi(A) &gt; \epsilon</math>, then <math>\pi</math> is kept and updated following Bayes’ rule ... when the new information was unexpected given the prior, <math>\pi(A) \leq \epsilon</math>, then the prior is questioned”.</p> <p>(part 4, Ortoleva) The departure is triggered by one unexpected event, the AFTER’s wording; the survey’s restatement of Ortoleva (2012) is the only source in hand.</p> <p><b>Quote.</b> <i>Cripps 2021, p. 9 (axioms named on pp. 7–9)</i><br/> “Furthermore, symmetry implies that the order in which the signals are revealed can be changed without an effect on the ultimate beliefs. Hence, these axioms imply that reversing the order in which two signals arrive has no effect on the ultimate beliefs.”</p> <p>(part 4, Cripps) His four axioms are Axiom 1 (Uninformativeness) and Axiom 2 (Symmetry), p. 7, Axiom 3 (Divisibility), p. 8, and Axiom 4 (Non-Dogmatic), p. 9; order invariance follows from Symmetry and Divisibility.</p> |                  |

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|      | <p><b>Theorem.</b> <i>Cripps.lean, order_invariance</i><br/> Let <math>U</math> satisfy Symmetry and Divisibility, let <math>\mu</math> have full support, and let <math>x, y : \Theta \rightarrow (0, 1)</math>. Writing <math>u(\mu, x)</math> for the first-signal update of <math>\mu</math> on the binary experiment <math>(x, 1 - x)</math>, <math>u(u(\mu, x), y) = u(u(\mu, y), x)</math>.<br/> (part 4, Cripps) The p. 9 remark as a theorem, with no Axiom 1 or 4 needed: two conditionally independent signals with fixed likelihoods give the same posterior in either sequence.</p> <p><b>Theorem.</b> <i>Cripps.lean, composite_AB_eq_bayes</i><br/> Let <math>\mu</math> be a full-support belief on <math>A \times B</math>, <math>q &gt; 0</math> with <math>\sum_a q_a = 1</math> and <math>\sum_b r_b = 1</math>. Then <math>J_B(J_A(\mu, q), r) = \text{Bayes}(\mu, \ell)</math> with <math>\ell(a, b) = \frac{q_a}{\mu_A(a)} \cdot \frac{r_b}{(J_A(\mu, q))_B(b)}</math>.<br/> (part 4, Cripps) Under Proposition IMM's reading the composite is one Bayes update, a divisible rule meeting all four axioms, whose second likelihood is matched to the intermediate belief; the other sequence feeds a different experiment, so no axiom fails.</p> <p><b>Theorem.</b> <i>Cripps.lean, rigid_not_uninformative</i><br/> For every <math>\varphi</math>, the rigid rule <math>(\mu, p, s) \mapsto J_A(\mu, \varphi(p_s))</math> violates Axiom 1 (Uninformativeness): an experiment with <math>p_s(\theta) = p_s(\theta')</math> for all <math>\theta, \theta'</math> still resets the <math>A</math>-marginal.<br/> (part 4, Cripps) Under the other reading, delivered credence fixed whatever the prior, the composite fails Uninformativeness (and Non-Dogmatic), so "of the four axioms only divisibility" is wrong on either reading.</p> <p><b>Quote.</b> <i>Pettigrew and Weisberg 2025, p. 3, Equation (1)</i><br/> "But we'll see that commutativity instead favours upco, also known as multiplicative pooling: <math>P'(E) = \frac{P(E)Q(E)}{P(E)Q(E) + P(\bar{E})Q(\bar{E})} \cdot (1)</math>"<br/> (part 5, Pettigrew-Weisberg) The pooling step is multiplicative and takes the prior <math>P(E)</math> as one of its two arguments.</p> <p><b>Quote.</b> <i>Pettigrew and Weisberg 2025, pp. 3–4</i><br/> "Suppose we begin with <math>P</math>, fix some pooling rule <math>f</math>, and use the following two-step procedure for responding to <math>Q</math>'s opinion about <math>E</math>. ... Step 1. Apply pooling rule <math>f</math> to <math>P(E)</math> and <math>Q(E)</math> to obtain <math>P'(E)</math>" (p. 3); "Begin with the case where <math>P</math> pools with <math>Q</math> first. Step 1 of Jeffrey pooling combines <math>P(E) = 4/10</math> with <math>Q(E) = 8/10</math> via upco, to yield <math>P'(E) = 8/11</math>" (p. 4).<br/> (part 5, Pettigrew-Weisberg) The prior is pooled with each source's opinion; successive inputs are never pooled with each other.</p> <p><b>Theorem.</b> <i>PettigrewWeisberg.lean, upco_eq_self_iff</i><br/> For <math>0 &lt; p, q &lt; 1</math>, <math>\frac{pq}{pq + (1-p)(1-q)} = q \iff p = \frac{1}{2}</math>.<br/> (part 5, Pettigrew-Weisberg) Pooling the prior with a delivered credence returns that credence only from an even prior, so their operation is not the manuscript's marginal reset.</p> <p><b>Theorem.</b> <i>PettigrewWeisberg.lean, PB_is_upco_pooling</i><br/> Let <math>P</math> be regular on the cells <math>E_i \cap F_j</math> and <math>x_i, y_j &gt; 0</math>. With the matched opinions <math>m_E(x)_i \propto x_i/P(E_i)</math> and <math>m_F(y)_j \propto y_j/P(F_j)</math>, Jeffrey pooling <math>P</math> with upco on <math>E</math> using <math>m_E(x)</math> and then on <math>F</math> using <math>m_F(y)</math>, in either sequence, gives <math>P^B(\omega) \propto P(\omega) \frac{x_{E(\omega)}}{P(E_{E(\omega)})} \frac{y_{F(\omega)}}{P(F_{F(\omega)})}</math>.<br/> (part 5, Pettigrew-Weisberg) Their commuting procedure, fed the likelihoods matched to the prior, is Paper B's benchmark; sequence dependence belongs to the delivered credences.</p> <p><b>Quote.</b> <i>Phelps 1972, p. 659, Equation (1)</i><br/> "The employer is able to measure the performance of each applicant in some kind of test, <math>y_i</math>, which, after suitable scaling, may be said to measure the applicant's promise or degree of qualification, <math>q_i</math>, plus an error term, <math>\mu_i</math>. (1) <math>y_i = q_i + \mu_i</math> where <math>\mu</math> is normally distributed with mean zero."<br/> (part 6, Phelps) Qualification is a continuous variable read through a normal error and a least-squares predictor (pp. 659–660); the only binary variable is the race dummy, so Phelps supplies no binary-attribute frame.</p> <p><b>Quote.</b> <i>Arrow 1973, Section 4, working paper p. 26</i><br/> "Only some workers, however, are qualified to hold skilled jobs. ... The employer cannot know of any given worker whether or not he is qualified; however, he does believe that the probability that a random W worker is qualified is <math>p_W</math> and that a random B worker is qualified is <math>p_B</math>."<br/> (part 6, Arrow) Binary qualification and two groups, with a belief <math>p_W, p_B</math> per group; this is the binary-attribute frame, and Arrow stays in the AFTER (the local copy is the 1971 working paper, hence its pagination).</p> <p><b>Theorem.</b> <i>CoateLoury.lean, negativeStereotype_iff_cov_pos</i><br/> For <math>0 &lt; \lambda &lt; 1</math> (the share of W's), <math>\pi_b &lt; \pi_w \iff 0 &lt; \text{Cov}(W, q)</math>, where <math>\text{Cov}(W, q) = \lambda\pi_w - \lambda(\lambda\pi_w + (1-\lambda)\pi_b) = \lambda(1-\lambda)(\pi_w - \pi_b)</math>.<br/> (part 6, Coate-Loury) A negative stereotype in the lineage the AFTER cites is a believed positive association between group and qualification, the binary frame the manuscript borrows.</p> |                  |

| Part | BEFORE (current manuscript)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     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|      | <p><b>Quote.</b> <i>Bordalo, Coffman, Gennaioli and Shleifer 2016 (May 2015 WP), p. 12</i></p> <p>“The DM has stored in memory the full conditional distribution ... but he assesses this distribution by recalling only a limited and selected set of types. Selective recall is driven by representativeness, formalized below following GS (2010).”</p> <p><i>(part 7, BCGS) One mechanism, representativeness-driven selective recall, not “representativeness-distortion or selective recall”; nothing in it is a sampling distortion.</i></p> <p><b>Quote.</b> <i>Bordalo, Coffman, Gennaioli and Shleifer 2016 (May 2015 WP), Section 4.3, p. 27</i></p> <p>“Again, there is a kernel of truth in these stereotypes, but also an exaggeration of the correlation between education and being on welfare: people neglect that most elements of the less educated group are not on welfare”.</p> <p><i>(part 7, BCGS) BCGS also produce an exaggerated cross-attribute association, so the contrast rests on where it arises, across groups, and on the mechanism, not on the concept.</i></p> <p><b>Theorem.</b> <i>BCGS.lean, welfare_within_group</i></p> <p>With <math>\text{cov}(p_{00}, p_{01}, p_{10}, p_{11}) = p_{11} - (p_{10} + p_{11})(p_{01} + p_{11})</math>: <math>\text{cov}(0, \frac{9}{11}, 0, \frac{2}{11}) = 0</math>, while the true within-group law has <math>\text{cov}(\frac{21}{50}, \frac{9}{50}, \frac{9}{25}, \frac{1}{25}) = -\frac{6}{125}</math>.</p> <p><i>(part 7, BCGS) In the Lean instance of the Section 4.3 example (BCGS give no numbers there) the <math>d = 2</math> stereotype of a group recalls only welfare types, so the within-group education-welfare association is zero; the exaggerated correlation lives in the population pooled across groups.</i></p> <p><b>Quote.</b> <i>Becker 1962, p. 7</i></p> <p>“This analytical statement must be distinguished from the frequently encountered arithmetical statement that a market would behave rationally even if only a few households did ... Our statement goes beyond arithmetic and stems from an analysis of the responses of rational and irrational households.”</p> <p><i>(part 8, Becker) Becker sets his result against the arithmetic of aggregation, the opposite of “protection intrinsic to the arithmetic of aggregation”.</i></p> <p><b>Quote.</b> <i>Becker 1962, p. 10</i></p> <p>“In both cases the word ‘survive’ simply refers to a resource constraint on behavior and does not literally distinguish ‘life’ from ‘death,’ although some households and firms may actually die from trying to ‘live’ beyond their means.”</p> <p><i>(part 8, Becker) His mechanism is a constraint on behaviour, so “rather than imposed by an enforced constraint on behaviour” inverts him.</i></p> <p><b>Quote.</b> <i>Becker 1962, p. 12</i></p> <p>“Indeed, the most important substantive result of this paper is that irrational units would often be ‘forced’ by a change in opportunities to respond rationally.”</p> <p><i>(part 8, Becker) Becker’s own summary of the paper: rational market responses are forced by the opportunity set.</i></p> <p><b>Theorem.</b> <i>Becker.lean, constraint_without_averaging_and_impulsive_mean</i></p> <p>With <math>I = p_2 = 1</math> and <math>p_1</math> rising from 1 to 11/10, the theorem states that some <math>x \in [0, 1]</math> and <math>x' \in [0, 10/11]</math> have <math>x &lt; x'</math> (the proof takes <math>x = \frac{1}{10}</math>, <math>x' = \frac{9}{10}</math>): one impulsive household can buy more after the rise. If <math>X</math> is uniform on <math>[0, I/p_1]</math> then <math>\mathbb{E}X = I/(2p_1)</math>.</p> <p><i>(part 8, Becker) The downward market slope needs both ingredients, the budget constraint through the support <math>[0, I/p_1]</math> and averaging over many households; neither alone delivers it.</i></p> <p><b>Computation.</b> <i>grep on PAPER_B_MANUSCRIPT.tex, Concluding remarks (lines 873–939)</i></p> <p>Case-insensitive search of the concluding section for “Becker”, “constraint”, “arithmetic”, “enforced”, “imposed” and “intrinsic”: zero hits. The manuscript’s only mention of Becker is line 255, the sentence under correction.</p> <p><i>(part 8, Becker) The “lineage distinction the concluding section trades on” is not drawn there, so the forward reference dangles and the AFTER deletes the sentence.</i></p> <p><b>Quote.</b> <i>Good and Mittal 1987, Definition 1.1, p. 695</i></p> <p>“DEFINITION 1.1. We say that the amalgamation (or aggregation) paradox, or simply the paradox, occurs if <math>\max_i \alpha(\mathbf{a}_i) &lt; \alpha(\mathbf{A})</math> or <math>\alpha(\mathbf{A}) &lt; \min_i \alpha(\mathbf{a}_i)</math>.” ... “in Yule’s formulation the subpopulations each had zero association in which case there is no reversal of sign. Hence we prefer the name ‘amalgamation paradox.’”</p> <p><i>(part 9, Good-Mittal) The paradox covers amplification beyond the maximum and association created from none, not only “erased or reversed”.</i></p> <p><b>Quote.</b> <i>Good and Mittal 1987, pp. 695–696</i></p> <p>“a drug can be judged to be beneficial, as measured by <math>\alpha</math>, for both men and women considered separately, but can seem to be harmful for the population at large, by looking only at the amalgamated table. This can happen even though <math>N_i \propto p_i</math>. We claim that such a situation can arise only if not enough care is used in the design of the experiment.”</p> <p><i>(part 9, Good-Mittal) Population-share weights do not prevent the paradox; its cause is uneven treatment allocation across subpopulations.</i></p> <p><b>Theorem.</b> <i>GoodMittal.lean, equalSize_reversal</i></p> <p>For <math>\mathbf{a}_1 = [8, 2; 60, 30]</math> and <math>\mathbf{a}_2 = [20, 70; 1, 9]</math>, <math>N_1 = N_2 = 100</math>, <math>\pi_R(\mathbf{a}_1) = \frac{2}{15} &gt; 0</math>, <math>\pi_R(\mathbf{a}_2) = \frac{11}{90} &gt; 0</math>, but <math>\pi_R(\mathbf{a}_1 + \mathbf{a}_2) = -\frac{33}{100} &lt; 0</math>, where <math>\pi_R = \frac{a}{a+b} - \frac{c}{c+d}</math>.</p> <p><i>(part 9, Good-Mittal) Two equally large subpopulations, the treatment beneficial in each and harmful in the aggregate: the p. 696 remark as an instance.</i></p> |                  |

| Part | BEFORE (current manuscript)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | AFTER (proposed)                                                                                                                                                                                                                                                                           |
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|      | <p><b>Theorem.</b> <i>GoodMittal.lean, piR_ amalg_ general</i></p> <p>For positive tables <math>\mathbf{a}_i</math> with <math>\mathbf{A} = \sum_i \mathbf{a}_i</math>, <math>\pi_R(\mathbf{A}) = \frac{\sum_i (a_i + b_i) \frac{a_i}{a_i + b_i}}{\sum_i (a_i + b_i)} - \frac{\sum_i (c_i + d_i) \frac{c_i}{c_i + d_i}}{\sum_i (c_i + d_i)}</math>.</p> <p>(part 9, Good-Mittal) Each row averages the subpopulation rates with its own row totals, not with <math>N_i/N</math>; the paradox is driven by the difference between those two weightings, that is, by uneven treatment allocation.</p> <p><b>Quote.</b> Heckman 1998, p. 109</p> <p>“Assumption b is a strong one, and it is the key identifying assumption of the audit method as currently practiced. Nothing guarantees that it will be satisfied.”</p> <p>(part 9, C.15 Heckman) “nothing guarantees” is on p. 109, and assumption b is equal means of unobserved productivity across groups, as the AFTER glosses it.</p> <p><b>Quote.</b> Heckman 1998, p. 102</p> <p>“The audit method can find discrimination when in fact none exists; it can also disguise discrimination when it is present.”</p> <p>(part 9, C.15 Heckman) The p. 102 sentence is Heckman’s general summary; the AFTER attributes no sample-size claim to him.</p> <p><b>Quote.</b> Bohren, Imas and Rosenberg 2019, working paper p. 1</p> <p>“As prior work has noted, it is difficult to identify the underlying source of discrimination from such static settings, as different sources generate the same patterns of observable behavior (Fang and Moro 2011).”</p> <p>(part 9, C.15 BIR) BIR credit the sentence to Fang and Moro, so the AFTER paraphrases it (“as they note”) rather than quoting it as theirs.</p> <p><b>Theorem.</b> <i>PropPRO.lean, propPRO_ protection</i></p> <p>For all real <math>x, y</math> and all <math>(\alpha, \beta, \lambda, q_0, r_0)</math>, <math>\langle xJ + y \nabla \text{assoc}(q \otimes r), M_\lambda \rangle = 0</math>, where <math>M_\lambda = \lambda \kappa R_1 + (1 - \lambda) \kappa' R_2</math>, <math>R_1 = q \otimes (1, -1)</math> and <math>R_2 = (1, -1) \otimes r</math>.</p> <p>(part 9, C.15) The first-order coefficient of a protected statistic vanishes identically in the prior and in the mix, so it is zero in the population itself; its silence is a failure of identification, which no sample size repairs.</p> |                                                                                                                                                                                                                                                                                            |
| C.13 | Section 4, the Foster-Greer-Thorbecke sentence                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                            |
|      | <p>This is the standard incidence-versus-intensity pairing of the measurement literature, where a headcount (a count of those past a threshold) and a mean shortfall (average of their distances from it) are first two members of a single parametrised family of indices (Foster et al., 1984).</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <p>The two decision statistics pair incidence with intensity, as do the first two poverty measures of Foster et al. (1984), the headcount ratio, the share below a threshold, and the average shortfall below it normalised by the threshold, to which those above it contribute zero.</p> |
|      | <p><b>Why.</b> <math>P_0</math> is the headcount <i>ratio</i>, a share, and <math>P_1 = (1/n) \sum g_i / z</math> is normalised by the threshold and averaged over the whole population; “average of their distances” is the income-gap measure, which is not in the family (M15). FGT do not use “incidence/intensity”; the AFTER does not attribute it to them.</p> <p><b>Evidence.</b> FGT.lean (P0_eq_H, P1_eq_popMean_normGap, I_not_decomposable).</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                            |
|      | <p><b>Grounds.</b></p> <p><b>Quote.</b> Foster, Greer and Thorbecke 1984, p. 763, Equation (3)</p> <p>“For each <math>\alpha \geq 0</math>, let <math>P_\alpha</math> be defined by (3) <math>P_\alpha(y; z) = \frac{1}{n} \sum_{i=1}^q \left( \frac{g_i}{z} \right)^\alpha</math>.”</p> <p>Every member divides by the population size <math>n</math> and by the line <math>z</math>; the sum runs over the <math>q</math> poor only.</p> <p><b>Quote.</b> Foster, Greer and Thorbecke 1984, p. 763</p> <p>“The measure <math>P_0</math> is simply the headcount ratio <math>H</math>, while <math>P_1</math> is <math>H \cdot I</math>, a renormalization of the income-gap measure. The measure <math>P</math> is obtained by setting <math>\alpha = 2</math>.”</p> <p><math>P_0</math> is a ratio, not a count, and <math>P_1</math> is the income-gap measure renormalised, not the “average of their distances” among the poor; FGT never say “incidence” or “intensity”.</p> <p><b>Theorem.</b> FGT.lean, P1_eq_popMean_normGap</p> <p>For any population <math>s</math>, incomes <math>y</math> and line <math>z</math>, <math>P_1(y; z) = \frac{1}{ s } \sum_{i \in s} \frac{\max(z - y_i, 0)}{z}</math>.</p> <p><math>P_1</math> is the whole-population mean of the normalised shortfall, the non-poor contributing zero, which is the AFTER’s description.</p> <p><b>Theorem.</b> FGT.lean, I_not_decomposable</p> <p>With <math>z = 1</math> and incomes <math>(0, \frac{1}{2}, 2)</math>: <math>I = \frac{3}{4}</math> on the whole population, <math>I = 1</math> on <math>\{0\}</math> and <math>I = \frac{1}{2}</math> on <math>\{1, 2\}</math>, and <math>\frac{3}{4} \neq \frac{1}{3} \cdot 1 + \frac{2}{3} \cdot \frac{1}{2} = \frac{2}{3}</math>.</p> <p>The mean shortfall among the poor (FGT’s <math>I</math> times <math>z</math>) is not additively decomposable with population-share weights, so it is not a member of the decomposable family the manuscript cites.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                            |

| Part                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | BEFORE (current manuscript)                                        | AFTER (proposed)                                                                                                                                                             |
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| <b>C.14</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Appendix, proof of Theorem LOS, the two Tao citations              |                                                                                                                                                                              |
| 1/2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Tonelli's theorem (Tao, 2011, Corollary 1.7.23) gives              | Tonelli's theorem (Tao, 2011, Theorem 1.7.15), applicable since the integrand is non-negative, gives                                                                         |
| 2/2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | As $\mu$ is finite, continuity from above (Tao, 2011, §1.4) yields | As $\mu$ is finite and $B_c$ decreases as $c \downarrow 0$ , continuity from above (Tao, 2011, Exercise 1.4.23(iii)), applied along any sequence $c_n \downarrow 0$ , yields |
| <b>Why.</b> Corollary 1.7.23 is Fubini-Tonelli; for the nonnegative integrand Tonelli (Theorem 1.7.15, or 1.7.18 for complete measures) is the exact reference. Continuity from above is Exercise 1.4.23(iii), stated for sequences; the proof takes $c$ to 0 over a continuum (M16). Numbering is from the author's preprint of Tao; confirm against the printed book.<br><b>Evidence.</b> Tao.lean (thm_1_7_15, ex_1_4_23_iii, measure_band_tendsto_zero, los_step4).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                    |                                                                                                                                                                              |
| <b>Grounds.</b><br><b>Quote.</b> Tao 2011, Theorem 1.7.15, pre-p. 200<br>"Theorem 1.7.15 (Tonelli's theorem, incomplete version). Let $(X, \mathcal{B}_X, \mu_X)$ and $(Y, \mathcal{B}_Y, \mu_Y)$ be $\sigma$ -finite measure spaces, and let $f : X \times Y \rightarrow [0, +\infty]$ be measurable with respect to $\mathcal{B}_X \times \mathcal{B}_Y$ . Then:"<br><i>Tonelli is stated for nonnegative <math>f</math> with no integrability hypothesis, which is what Step 3 of the LOS proof uses.</i><br><b>Quote.</b> Tao 2011, Corollary 1.7.23, pre-p. 206<br>"Corollary 1.7.23 (Fubini-Tonelli theorem). Let $(X, \mathcal{B}_X, \mu_X)$ and $(Y, \mathcal{B}_Y, \mu_Y)$ be complete $\sigma$ -finite measure spaces, and let $f : X \times Y \rightarrow \mathbf{C}$ be measurable with respect to $\mathcal{B}_X \times \mathcal{B}_Y$ . If $\int_X (\int_Y  f(x, y)  d\mu_Y(y)) d\mu_X(x) < \infty \dots$ then $f$ is absolutely integrable".<br><i>The corollary the manuscript cites is Fubini-Tonelli for complex-valued <math>f</math> under an absolute-integrability hypothesis, not the nonnegative Tonelli statement.</i><br><b>Quote.</b> Tao 2011, Exercise 1.4.23(iii), pre-p. 93<br>"(iii) (Downwards monotone convergence) If $E_1 \supset E_2 \supset \dots$ are $\mathcal{B}$ -measurable, and $\mu(E_n) < \infty$ for at least one $n$ , then $\mu(\bigcap_{n=1}^{\infty} E_n) = \lim_{n \rightarrow \infty} \mu(E_n) = \inf_n \mu(E_n)$ ."<br><i>Continuity from above is an exercise, stated for sequences and with a finiteness hypothesis; the LOS proof lets <math>c \downarrow 0</math> over a continuum.</i><br><b>Theorem.</b> Tao.lean, measure_band_tendsto_zero<br>Let $\mu$ be a finite measure and $(B_c)_{c>0}$ measurable sets with $B_{c'} \subseteq B_c$ whenever $0 < c' \leq c$ and $\bigcap_{c>0} B_c = \emptyset$ . Then $\mu(B_c) \rightarrow 0$ as $c \downarrow 0$ .<br><i>The real-parameter passage Step 4 needs, proved from the sequential statement since the neighbourhood filter of <math>0^+</math> is countably generated; the AFTER applies 1.4.23(iii) along any <math>c_n \downarrow 0</math>.</i> |                                                                    |                                                                                                                                                                              |

## Section E: the archived plan's entries, re-issued and corrected

The entries of notes/manuscript\_change\_plan\_asof\_2026-09-30.md that the author's draft did not already apply, each corrected against the audit and passed through the discipline gate. Order of application: E.10 (Propositions ORD and ADJ) before anything that cites them; E.14 (bibliography) before E.1, E.5, E.7 and C.15; E.11 before E.12.

| Part       | BEFORE (current manuscript, or the anchor for an insert)               | AFTER (proposed) |
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| <b>E.1</b> | Introduction, new paragraph after the hiring-panel paragraph (was 1.C) |                  |

| Part                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   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The impression-formation literature disagrees on the direction. Asch (1946) reports that the earlier terms set the direction in which later ones are read, and the review of Hogarth and Einhorn (1992) finds primacy, recency or no effect depending on the characteristics of the task. Under either direction the sequence in which cues arrive is not neutral, and it need not be assigned by chance. A candidate who comes through a referral is met first through the letter and then through the credential, while a candidate who applies unsolicited is met first through the credential. If referral is more common in one group than in another, the two groups differ in the share $\lambda$ of evaluators who read the credential first. Two groups presenting the same evidence to evaluators with the same priors and the same preferences then receive different mean beliefs, and the difference is the sequence effect multiplied by the difference in their shares. A gap of that kind is not statistical discrimination in the sense of Phelps (1972) and Arrow (1973), which rests on different prior beliefs about the groups, and it is none of the three sources that Bohren et al. (2019) distinguish, correct beliefs, biased beliefs and preferences, since none of these differs between the groups here. Which group it favours depends on whether the later or the earlier impression prevails, which Proposition PROP:ADJ reads from the marginals. The gap also inherits the classification of this paper. It appears in the marginal probabilities and in the share of decisions changed, and an audit of the believed association does not register it. |
| <p><b>Why.</b> Plan 1.C with audit P4 (“prior beliefs about the groups”), D13 (BIR’s three sources named rather than “none of the sources”, since their impartial type discriminates with correct beliefs and no animus), A24 (Asch’s “early terms dominate” softened to his own “direction” account) and H6 (the primacy/recency/no-effect finding is HE’s review of other studies) applied. Presupposes E.10 (Proposition ADJ) and the Bohren2019 bib entry. Goes after C.5’s sentence.</p> <p><b>Evidence.</b> The identity <math>\text{mean}(\lambda) - \text{mean}(\lambda') = (\lambda - \lambda')(P_{AB} - P_{BA})</math> is exact, verify_ORD.py step 9; Asch.lean; HogarthEinhorn.lean; BohrenImas-Rosenberg.lean.</p> <p><b>Grounds.</b></p> <p><b>Computation.</b> <i>sympy/verify_ORD.py, step (9), with steps (1), (3)–(5)</i><br/> With <math>\bar{P}_\lambda = \lambda P_{AB}^J + (1 - \lambda) P_{BA}^J</math>, the identity <math>\bar{P}_\lambda - \bar{P}_{\lambda'} = (\lambda - \lambda') (P_{AB}^J - P_{BA}^J)</math> holds entrywise and exactly, at every order in <math>c</math> (symbolic in <math>\alpha, \beta, c, q_0, r_0, \lambda, \lambda'</math>). Since <math>P_{AB}^J - P_{BA}^J = c(\kappa R_1 - \kappa' R_2) + \mathcal{O}(c^2)</math>, the group gap on the <math>A</math>-marginal is <math>(\lambda - \lambda') \kappa' c + \mathcal{O}(c^2)</math> with <math>\kappa' = (\beta - r_0)q_0(1 - q_0)/Z</math>, while on the believed association it is <math>\mathcal{O}(c^2)</math> (checked at <math>\lambda = 2/5</math> against <math>\lambda' = 3/4</math>).</p> <p><i>Two groups with the same priors, preferences and evidence differ by the sequence effect multiplied by the difference in their shares, and the gap inherits Proposition ORD’s classification (marginals first order, association second order).</i></p> <p><b>Quote.</b> <i>Asch 1946, pp. 271–272 (Experiment VI)</i><br/> “the first terms set up in most subjects a direction which then exerts a continuous effect on the latter terms” (p. 271); “a broad, uncrystallized but directed impression is born. The next characteristic comes not as a separate item, but is related to the established direction. Quickly the view formed acquires a certain stability, so that later characteristics are fitted... to the given direction” (p. 272).<br/> <i>Asch’s account is one of direction rather than dominance, and he adds that it is “not the sheer temporal position of the item which is important” (p. 272), which is why the entry no longer says that early terms dominate (audit A24).</i></p> <p><b>Quote.</b> <i>Hogarth and Einhorn 1992, abstract, T-p. 1 (transcription)</i><br/> “Much literature attests to the existence of order effects in the updating of beliefs. However, under what conditions do primacy, recency, or no order effects occur? This paper presents a theory of belief updating that explicitly accounts for order-effect phenomena as arising from the interaction of information-processing strategies and task characteristics.”<br/> <i>The three outcomes are their review’s classification of earlier studies (Table 1, T-p. 4) and their model’s predictions (Table 2); their own five experiments found recency or no effect, so the entry attributes the finding to the review (audit H6).</i></p> |                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |

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|            | <p><b>Quote.</b> <i>Bohren, Imas and Rosenberg, January 2019 working paper, pp. 2, 13–14, 19–20</i></p> <p>“we allow for three potential sources: (i) belief-based with correct beliefs, (ii) belief-based with incorrect, biased beliefs, and (iii) preference-based” (p. 2). Of the first, “discrimination is due to imperfect information (Phelps 1972)” or “is an equilibrium effect (Arrow 1973)” (pp. 13–14), and “evaluators hold prior beliefs about workers’ abilities that differ by group identity” (p. 14). The impartial type “has no belief-based partiality” (p. 19), yet “this type discriminates against males in the second period” (p. 20).</p> <p><i>The three sources are named rather than treated as exhaustive, since BIR’s own impartial type discriminates with correct beliefs and no animus (audit D13).</i></p> <p><b>Quote.</b> <i>Phelps 1972, p. 659; Arrow 1973, p. 26 (working paper)</i></p> <p>Phelps: “the employer who seeks to maximize expected profit will discriminate against blacks or women if he believes them to be less qualified, reliable, long-term, etc. on the average than whites and men, respectively, and if the cost of gaining information about the individual applicants is excessive.” Arrow: “The employer cannot know of any given worker whether or not he is qualified; however, he does believe that the probability that a random W worker is qualified is <math>p_W</math> and that a random B worker is qualified is <math>p_B</math>.”</p> <p><i>Both rest on prior beliefs about the groups, the wording the entry uses (audit P4); Phelps’s Case 2 and Further Case rest on variance and test reliability rather than on mean, so “different beliefs” alone would be too broad.</i></p> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>E.2</b> | Setup 2.1, what an impression is (was 2.A)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|            | yields an impression (a target marginal) $q = (q_0, q_1)$ on $A$ ’s partition and a cue on $B$ yields an impression $r = (r_0, r_1)$ on $B$ ’s partition—with no cue bearing on a joint event.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | yields an impression (a target marginal) $q = (q_0, q_1)$ on $A$ ’s partition, with $q_i$ the credence that $A = i$ and $q_0 + q_1 = 1$ , and a cue on $B$ yields an impression $r = (r_0, r_1)$ on $B$ ’s partition, with $r_j$ the credence that $B = j$ and $r_0 + r_1 = 1$ , and no cue bears on a joint event. An impression is thus a probability distribution on one attribute, on the same scale as the prior marginal it replaces, $(\alpha, 1 - \alpha)$ for $q$ and $(\beta, 1 - \beta)$ for $r$ . |
|            | <p><b>Why.</b> Plan 2.A as written; the manuscript never says that <math>q_0</math> belongs to <math>A = 0</math>, that <math>q_0 + q_1 = 1</math>, or that an impression is on the prior marginal’s scale. The dash construction is replaced by a clause.</p> <p><b>Evidence.</b> Notation only; every sympy script and Lean file uses <math>q_1 = 1 - q_0</math>, <math>r_1 = 1 - r_0</math>.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|            | <p><b>Grounds.</b></p> <p><b>Theorem.</b> <i>DiaconisZabell.lean, marg_jeffrey and jeffrey_jcond</i></p> <p>On a finite <math>\Omega</math> with a partition <math>\{E_i\}</math> given by a surjective labelling <math>e</math> and a prior <math>P &gt; 0</math>, Jeffrey’s rule <math>P^*(\omega) = p_{e(\omega)} P(\omega)/P(E_{e(\omega)})</math> satisfies <math>P^*(E_i) = p_i</math> for every cell (marg_jeffrey) and, provided <math>p_{e(\omega)} \neq 0</math> for all <math>\omega</math>, <math>P^*(A   E_i) = P(A   E_i)</math> for every event <math>A</math> and every cell <math>i</math> (jeffrey_jcond).</p> <p><i>An impression is a probability distribution on one attribute’s partition; the step installs it as the new marginal and leaves every conditional on that partition unchanged.</i></p> <p><b>Theorem.</b> <i>Aggregate.lean, jeffreyA_pins_mA0, jeffreyA_pins_mA1, jeffreyB_pins_mB1</i></p> <p>For the <math>2 \times 2</math> table <math>Q</math>, <math>J_A(Q, q_0) = \begin{pmatrix} q_0 Q_{00}/Q(A=0) &amp; q_0 Q_{01}/Q(A=0) \\ (1-q_0)Q_{10}/Q(A=1) &amp; (1-q_0)Q_{11}/Q(A=1) \end{pmatrix}</math>; if <math>Q(A=0) \neq 0</math> then <math>J_A(Q, q_0)(A=0) = q_0</math>, and if <math>Q(A=1) \neq 0</math> then <math>J_A(Q, q_0)(A=1) = 1 - q_0</math>, identically in <math>c</math>; likewise <math>J_B(Q, r_0)(B=1) = 1 - r_0</math> when <math>Q(B=1) \neq 0</math>.</p> <p><i>The impression <math>(q_0, 1 - q_0)</math> is on the same scale as the prior marginal <math>(\alpha, 1 - \alpha)</math> it replaces, and <math>q_1 = 1 - q_0</math> is built into the step.</i></p>                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>E.3</b> | Setup 2.2, the worked example, and why a soft cue differs from a hard one (was 2.B, extended)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |

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The credential delivers <math>q_0 = \frac{1}{5}</math> and the letter <math>r_0 = \frac{7}{10}</math>. After the credential alone the table is <math>\begin{pmatrix} .12 &amp; .08 \\ .32 &amp; .48 \end{pmatrix}</math>. The <math>A</math>-marginal is <math>(.20, .80)</math>, as delivered, and the <math>B</math>-marginal has moved from <math>(.50, .50)</math> to <math>(.44, .56)</math> although no cue about <math>B</math> has been read. Reading the letter next resets the <math>B</math>-marginal to <math>(.70, .30)</math> and moves the <math>A</math>-marginal from <math>.20</math> to <math>P_{AB}^J(A=0) = .234</math>. In the reverse sequence the <math>A</math>-marginal ends at <math>.20</math>, as delivered, and the <math>B</math>-marginal at <math>P_{BA}^J(B=0) = .643</math> in place of <math>.70</math>. The benchmark gives <math>P^B(A=0) = .227</math> and <math>P^B(B=0) = .647</math>. The sequence effect is therefore <math>.034</math> on the <math>A</math>-marginal and <math>.057</math> on the <math>B</math>-marginal. On the believed association it is <math>.0002</math>, since <math>\text{assoc}(P_{AB}^J) = .0273</math> and <math>\text{assoc}(P_{BA}^J) = .0271</math>, smaller than either marginal effect by a factor above 150. The odds ratio equals <math>9/4</math> for the prior, for both sequences and for the benchmark. Against the benchmark, sequence <math>AB</math> departs by <math>.053</math> on the marginal read last, by <math>.007</math> on the marginal read first, and by <math>.002</math> on the association.</p> <p><b>Why a soft cue differs from a hard one.</b> Suppose instead that the letter were a hard cue, a proposition <math>F</math> the panel becomes certain of, with a likelihood fixed before any belief is formed, say <math>P(F   B=0)/P(F   B=1) = 7/3</math>, and the credential likewise a proposition <math>E</math> with ratio <math>1/4</math>. Conditioning on <math>E</math> and then on <math>F</math> is conditioning on <math>E \cap F</math>, so the two sequences give the same belief, which is the benchmark <math>P^B</math>, with <math>A</math>-marginal <math>.227</math> in either sequence. The soft letter delivers the credence <math>r_0 = \frac{7}{10}</math> instead. The factor that credence implies is the ratio of the odds it delivers to the odds it meets, <math>7/3</math> when the letter is read first against the prior marginal <math>\frac{1}{2}</math>, but <math>98/33</math> when it is read after the credential, against the marginal <math>.44</math> that the credential has already moved. The panel knows the credence the letter left, not what that credence would have been at another prior, and so cannot extract a portable factor from it. That dependence of the implied factor on the belief it meets is the whole of the sequence effect, and it vanishes at <math>c = 0</math>, where the credential leaves the <math>B</math>-marginal where it was and the two implied factors coincide.</p> |
| <p><b>Why.</b> Plan 2.B, with the colon removed, plus the passage the author asked for on 2026-09-30, saying exactly what is special about a soft cue. A hard cue is a proposition with a belief-independent likelihood, so conditioning is defined and commutes; a soft cue fixes a credence, and the factor that credence implies is matched to the marginal in force when it arrives (<math>7/3</math> read first, <math>98/33</math> read second). On one cue the two readings agree (Proposition IMM). This replaces the counterfactual-likelihood argument the audit found contradicted by Jeffrey’s definition of a Bayes factor (J2). With <math>c = 1/20</math> against a bound of <math>1/4</math> the example illustrates the ordering of magnitudes, not the asymptotic rates.</p> <p><b>Evidence.</b> Every number in both paragraphs is asserted in <code>sympy/verify_example.py</code> (18/18) and <code>sympy/verify_soft_vs_hard.py</code> (15/15, added 2026-09-30, registered in <code>run_all.py</code>; its symbolic checks give the exact condition, <math>c = 0</math> or each cue delivering its prior marginal); the mechanism is <code>Cripps.lean composite_AB_eq_bayes</code> and <code>DiaconisZabell.lean thm21</code>.</p> <p><b>Grounds.</b></p> <p><b>Computation.</b> <code>sympy/verify_example.py</code> (18/18 checks)</p> <p>At <math>\alpha = \beta = \frac{1}{2}</math>, <math>c = \frac{1}{20}</math>, <math>q_0 = \frac{1}{5}</math>, <math>r_0 = \frac{7}{10}</math>: prior <math>\begin{pmatrix} 3/10 &amp; 1/5 \\ 1/5 &amp; 3/10 \end{pmatrix}</math>; after the credential <math>\begin{pmatrix} 3/25 &amp; 2/25 \\ 8/25 &amp; 12/25 \end{pmatrix}</math> with <math>B</math>-marginal <math>11/25</math>; <math>P_{AB}^J = \begin{pmatrix} 21/110 &amp; 3/70 \\ 28/55 &amp; 9/35 \end{pmatrix}</math>, <math>P_{BA}^J = \begin{pmatrix} 7/45 &amp; 2/45 \\ 56/115 &amp; 36/115 \end{pmatrix}</math>, <math>P^B = \begin{pmatrix} 3/17 &amp; 6/119 \\ 8/17 &amp; 36/119 \end{pmatrix}</math>. Hence <math>P_{AB}^J(A=0) = 18/77 \approx .234</math>, <math>P_{AB}^J(B=0) = 7/10</math>, <math>P_{BA}^J(A=0) = 1/5</math>, <math>P_{BA}^J(B=0) = 133/207 \approx .643</math>, <math>P^B(A=0) = 27/119 \approx .227</math>, <math>P^B(B=0) = 11/17 \approx .647</math>; <math>\text{assoc}(P_{AB}^J) = 3/110 \approx .0273</math>, <math>\text{assoc}(P_{BA}^J) = 28/1035 \approx .0271</math>, <math>\text{assoc}(P^B) = 60/2023 \approx .0297</math>; sequence effects <math>13/385 \approx .034</math>, <math>119/2070 \approx .057</math>, <math>1/4554 \approx .0002</math> (ratio <math>5382/35 &gt; 150</math>); gaps of <math>AB</math> from <math>P^B</math>: <math>9/170 \approx .053</math>, <math>9/1309 \approx .007</math>, <math>531/222530 \approx .002</math>; odds ratio <math>9/4</math> for all four tables.</p> <p><i>Every number printed in the worked example, in exact rationals.</i></p> |                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |

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| <p><b>Computation.</b> <i>sympy/verify_soft_vs_hard.py (15/15 checks)</i></p> <p>The letter's implied factor, the odds it delivers over the odds it meets, is <math>\frac{7/3}{1} = 7/3</math> when read first (against <math>\beta = \frac{1}{2}</math>) and <math>\frac{7/3}{11/14} = 98/33</math> when read after the credential (against 11/25); at <math>c = 0</math> both equal 7/3. Hard cues with fixed likelihood ratios 1/4 (credential) and 7/3 (letter) commute cell by cell, <math>E</math> then <math>F</math> equals <math>F</math> then <math>E</math>, and reproduce <math>P^B</math>, with <math>A</math>-marginal <math>27/119 \approx .227</math> in either sequence. On one cue the hard and soft readings coincide, <math>\text{cond}(P, \ell_A) = J_A(P, q)</math> (Proposition IMM).</p> <p><i>Every number printed in the soft-versus-hard paragraph.</i></p> <p><b>Theorem.</b> <i>DiaconisZabell.lean, thm21 and thm21_finite (D-Z Theorem 2.1, p. 824)</i></p> <p>For finite <math>\Omega</math> with <math>\sum_{\omega} P(\omega) = 1</math>, <math>P^* \geq 0</math> and <math>\sum_{\omega} P^*(\omega) = 1</math>: <math>P^*</math> can be obtained from <math>P</math> by conditioning on an event of a finite enlarged space iff there is <math>B \geq 1</math> with <math>P^*(\omega) \leq B P(\omega)</math> for all <math>\omega</math>; when <math>P &gt; 0</math> this always holds.</p> <p><i>Any soft posterior, the Jeffrey step included, is a conditioning on some richer space; what a soft cue lacks is a proposition with a likelihood fixed before the belief is formed, not conditionability.</i></p> <p><b>Theorem.</b> <i>Cripps.lean, composite_AB_eq_bayes (with jeffreyA_eq_bayes_matched)</i></p> <p>For a full-support belief <math>\mu</math> on <math>A \times B</math> and credences <math>q &gt; 0</math>, <math>r</math> with <math>\sum_a q_a = \sum_b r_b = 1</math>: <math>J_B(J_A(\mu, q), r) = \text{bayes}(\mu, \ell)</math> with <math>\ell(a, b) = \frac{q_a}{\mu(A=a)} \cdot \frac{r_b}{J_A(\mu, q)(B=b)}</math>, a single Bayes update whose <math>B</math>-likelihood is matched to the intermediate belief <math>J_A(\mu, q)</math>, not to <math>\mu</math>.</p> <p><i>The factor a soft cue implies is matched to the marginal in force when it arrives; the two sequences feed different likelihoods into the same commuting rule, which is the whole of the sequence effect.</i></p> <p><b>Quote.</b> <i>Jeffrey, Subjective Probability (2002 draft), p. 59, Example 4</i></p> <p>"Certainly her <math>\text{new}'(D_i)</math>'s will have arisen through an interaction of features of her prior mental state with her new experiences at the microscope. But the results <math>\text{new}'(D_i)</math> of that interaction are <i>data</i> for the kinematical formula for updating <math>\text{old}'(H)</math>; the formula itself does not compute those results."</p> <p><i>One's own experience yields credences that are data for kinematics, while another's report is what Jeffrey converts to factors (Section 3.3, p. 59), so a panelist's own impression is modelled as a credence and the counterfactual-likelihood argument is dropped (audit J2).</i></p> |                                                                                                                         |                                                                                                                                                                |
| <b>E.4</b> Setup 2.3, Assumption 2 (was 2.C)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                         |                                                                                                                                                                |
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| <b>Why.</b> Plan 2.C, with the colon replaced by a clause.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                         |                                                                                                                                                                |
| <b>Evidence.</b> Notation only.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                         |                                                                                                                                                                |
| <p><b>Grounds.</b></p> <p><b>Theorem.</b> <i>PropDIV.lean, kappa_eq_zero_iff and kappa'_eq_zero_iff</i></p> <p>With <math>Z = \alpha\beta(1-\alpha)(1-\beta) \neq 0</math>, <math>\kappa = (\alpha - q_0)r_0(1-r_0)/Z</math> and <math>\kappa' = (\beta - r_0)q_0(1-q_0)/Z</math>: <math>\kappa = 0 \iff \alpha = q_0</math> or <math>r_0 = 0</math> or <math>r_0 = 1</math>, and <math>\kappa' = 0 \iff \beta = r_0</math> or <math>q_0 = 0</math> or <math>q_0 = 1</math>. A cue with <math>r_0 \in \{0, 1\}</math> is decisive and kills the leading gap, so "soft" means <math>0 &lt; q_0 &lt; 1</math> and <math>0 &lt; r_0 &lt; 1</math>, the interior credences Assumption 2 now states.</p> <p><b>Theorem.</b> <i>PropPRO.lean, propPRO_uniqueness (hypotheses)</i></p> <p>Proposition PRO (ii) is proved under <math>0 &lt; \lambda &lt; 1</math>, <math>0 &lt; q_0 &lt; 1</math>, <math>0 &lt; r_0 &lt; 1</math>, <math>\beta \neq r_0</math> and two priors <math>\alpha_1 \neq \alpha_2</math>; the interior bounds on <math>q_0, r_0</math> are what make the factors <math>r_0(1-r_0)</math> and <math>q_0(1-q_0)</math> nonzero in the leading coefficient <math>\lambda(\alpha - q_0)r_0(1-r_0)x + (1-\lambda)(\beta - r_0)q_0(1-q_0)y</math>.</p> <p><i>The same interior condition on the credences is a hypothesis of the paper's central characterisation.</i></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                         |                                                                                                                                                                |
| <b>E.5</b> Related literature, new opening paragraphs on the commutativity literature (was 3.A)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                         |                                                                                                                                                                |

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|      | <p>While sequence-dependence has been of interest to empirical psychology at least since Asch (1946), the problem seems to have received interest in economics primarily through the observational learning literature.</p> | <p>The closest literature concerns the commutativity of probability kinematics itself. Diaconis and Zabell (1982) give the condition under which two successive Jeffrey revisions lead to the same belief, and Hawthorne (2004) names the full-adoption premise, which the present paper adopts, the Amnestic Update-Factor Thesis, crediting that criterion to them. His survey also fixes the vocabulary. Extensions of Jeffrey updating to sequences differ in what an experience is taken to deliver, new probabilities for its own basis directly, a multiplicative factor applied to the belief it meets, or a ratio between basis sentences that the prior does not constrain. The model used here takes the impression to deliver a credence, with the most recent one fixing its attribute outright, and the benchmark <math>P^B</math> of Section SEC:JEFFREY has the form of his extended update formula built on multiplicative factors. The comparison drawn throughout is therefore between his amnestic model and his factor models, and the identification questions this paper raises are the ones the amnestic model brings with it and the factor models do not, since factor revisions on distinct attributes commute. What Section SEC:JEFFREY calls a Bayes factor is, up to normalisation, his normed-likelihood factor, and as a ratio of two such factors it is his likelihood-ratio factor, which he too calls a Bayes factor; on a two-element basis all induce the same revision.</p> <p>The premise is disputed on two different grounds. Döring (1999) objects on normative grounds, that a sequence dependence of this size is unjustified in an account of rational belief change. Hawthorne (2004, p. 115) objects on psychological grounds and declines to settle them, asking “Which extension of Basic Jeffrey Updating is the more plausible model of human agents? I’m a logician, not a psychologist.” He judges amnestic updating less plausible than the factor-based extensions, “because it seems unlikely that we dismiss previous experiences so completely”, and turns to the normative properties of the competing models, where his interest lies. The ground on which the paper adopts the premise is information-theoretic, since Jeffrey’s rule is the revision that moves the prior least in <math>I</math>-divergence while taking on the delivered credence (Diaconis and Zabell, 1982). An objection of Hawthorne’s form is a reason to measure the degree of adoption, not a reason to discard the premise. With two cues that degree has exactly two ends, the impression read last prevailing and the impression read first prevailing, and Proposition PROP:ADJ recovers where between those ends an evaluator sits, from the rating of one trait read before and after the second cue and against what that cue delivers.</p> <p>Two responses to the resulting sequence dependence have been made. One treats it as a defect of the framework. Döring (1999) argues that the sequence effect can be pronounced enough to call for an adjustment that successive Jeffrey steps cannot supply, though a single Jeffrey step from the original prior can. The other re-describes the input so that the effect disappears. Field (1978) reparametrises the update so that its input is a portable factor rather than a delivered credence, and Wagner (2002) shows that when identical learning is represented by identical Bayes factors, sequential revisions commute; Garber (1980) objects that a portable factor compounds implausibly under repetition. The present paper takes neither route. It accepts the sequence dependence that the delivered-credence reading entails, and asks how far the two sequences disagree in each statistic of the resulting belief. That the disagreement is uneven across statistics, first order in the marginals and the share of decisions changed and second order in the believed association and the surplus-weighted loss, is what makes the debate answerable by measurement rather than by introspection about how completely an evaluator dismisses an earlier impression.</p> <p>The insensitivity of the association is moreover not peculiar to the extension adopted here. Each of the three extensions revises a basis by multiplying its cells by a factor that</p> |

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|      | <p><b>Why.</b> Plan 3.A with the audit applied. D1/P3, Doring’s objection is normative and Hawthorne’s psychological, so the two are separated and “disputed on psychological grounds” goes. P2, Doring’s own remedy is a single Jeffrey update from the original prior. M21, “the two ends of his taxonomy” becomes “his amnestic model and his factor models” (NL is the middle model), and the Bayes-factor identification names the normalisation and his own use of “Bayes factor” for LR factors (his note 20). “Order” becomes “sequence” throughout (the manuscript’s convention). Two colons and one dash removed. The Hawthorne quotations are verbatim from pp. 115-116; the second drops “not because it is un-Bayesian, but” before “because”, which leaves his stated reason unchanged. Needs Doring1999 and Garber1980 in bibliography.bib (E.14).</p> <p><b>Evidence.</b> Doring.lean (fig2, gap_tends_to_one); Hawthorne.lean (amnestic_thesis, factorUpdate_comm, NL_jeffrey, med_NL_denominator); Field.lean (field_eq7_comm); Wagner2002.lean (thm31); Garber.lean (garber_table); DiaconisZabell.lean (thm51_KL_eq_iff); LemmaSEP.lean.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                  |
|      | <p><b>Grounds.</b></p> <p><b>Quote.</b> Hawthorne 2004, p. 96 (Section 4)</p> <p>“AMNESTIC UPDATE-FACTOR THESIS. For any state <math>e</math> that directly affects an evidence basis <math>\{E_i\}</math> and for any other state <math>d</math> and sequence of states <math>\alpha</math>, <math>Q_{\alpha de}[E_i] = Q_{\alpha e}[E_i]</math>.” And: “Indeed Amnestic Updating is just Standard Sequential Updating – Jeffrey’s original approach to sequential updating.”</p> <p>Para 1: the full-adoption premise is named by Hawthorne, and he identifies it with Jeffrey’s sequential rule.</p> <p><b>Quote.</b> Hawthorne 2004, p. 97 and note 12 (p. 121)</p> <p>“a pair of states <math>e</math> and <math>f</math> will commute for <math>Q_\beta \dots</math> just in case neither <math>e</math> nor <math>f</math> can, on its own, influence (even indirectly) the basis sentences of the other – i.e. just in case for each <math>E_i</math>, <math>Q_{\beta f}[E_i] = Q_\beta[E_i]</math>, and for each <math>F_j</math>, <math>Q_{\beta e}[F_j] = Q_\beta[F_j]</math>.” Note 12: “Theorem 3.2 in (Diaconis and Zabell, 1982).”</p> <p>Para 1: he credits the commutation criterion to Diaconis and Zabell.</p> <p><b>Quote.</b> Hawthorne 2004, abstract p. 89 and section headings pp. 96, 99, 103</p> <p>“I will explore three models of sequential updating, the usual extension and two alternatives.” The headings: “5. The Amnestic Update Model” (p. 96), “6. The Normed-Likelihood Factor Model” (p. 99), “7. The Likelihood-Ratio Factor Model” (p. 103).</p> <p>Para 1: the taxonomy has three members, one amnestic model and two factor models; the normed-likelihood model is the middle one, not an end.</p> <p><b>Quote.</b> Hawthorne 2004, note 20 (p. 121)</p> <p>“In his most recent work Jeffrey favors updating based on Likelihood-Ratio factors as well. He thinks of them as ratios of new to old odds and calls them ‘Bayes factors’, following Good (1950).”</p> <p>Para 1: the name Bayes factor is given by Hawthorne to likelihood-ratio factors, which is why the entry identifies the paper’s Bayes factor with his LR factor as a ratio of two NL factors.</p> <p><b>Theorem.</b> Hawthorne.lean, NL_jeffrey, factorUpdate_seq, med_NL_denominator</p> <p><math>NL[Q, e, E_i] = Q_e[E_i]/Q[E_i]</math> (p. 95); for a Basic Jeffrey update to targets <math>q</math>, <math>NL[Q, e, E_i] = q_i/Q[E_i]</math> with no side condition. Two factor updates on bases <math>u, v</math> with <math>\sum_y w(u_y)Q(y) \neq 0</math> compose to <math>Q_{ef}(x) = w(u_x)z(v_x)Q(x)/\sum_y w(u_y)z(v_y)Q(y)</math> (pp. 102, 105). In his medical example the NL factors against the initial prior give <math>Q[C] = 1/2</math> only after dividing by the normaliser <math>301/625 \neq 1</math>.</p> <p>Para 1: <math>P^B</math> has the form of his extended update formula up to normalisation, so the entry says “up to normalisation”.</p> <p><b>Quote.</b> Doring 1999, S379 (Introduction) and S383</p> <p>“The following is an exercise in Bayesian rational psychology. Its contention is that Jeffrey conditionalization, a generalization of classical conditionalization, cannot be a complete account of rational belief change.” On the reversed conditional probabilities of his example: “This seems wholly unjustified if there is nothing essential about the order in which experiences were made.”</p> <p>Para 2: Doring’s objection is normative, about rational belief change, not about how people update (audit D1/P3).</p> <p><b>Quote.</b> Hawthorne 2004, pp. 115-116</p> <p>“Which extension of Basic Jeffrey Updating is the more plausible model of human agents? I’m a logician, not a psychologist. But Amnestic Updating seems psychologically less plausible than the more Bayesian approaches, not because it is un-Bayesian, but because it seems unlikely that we dismiss previous experiences so completely. However, I am mainly interested in whether these models capture useful normative conceptions of belief updating”</p> <p>Para 2: the entry’s second fragment begins at “because it seems unlikely”, dropping “not because it is un-Bayesian, but”; the sentence runs onto p. 116 at “of belief updating”.</p> <p><b>Quote.</b> Diaconis and Zabell 1982, p. 828, Theorem 5.1 and (5.3)</p> <p>“Let <math>Q</math> be a probability on <math>\Omega</math> such that <math>Q(E_i) = P^*(E_i)</math>. Then <math>\dots I(Q, P) \geq \sum P^*(E_i) \log(P^*(E_i)/P(E_i))</math>. (5.6) In (5.5) and (5.6) equality holds if and only if <math>Q(A) = \sum P(A   E_i) P^*(E_i)</math>.” With (5.3): “<math>I(Q, P) = \sum_\omega Q(\omega) \log(Q(\omega)/P(\omega))</math>.”</p> <p>Para 2: Jeffrey’s rule is the unique minimiser of the <math>I</math>-divergence from the prior among revisions that take on the delivered credence; the Lean statement is DiaconisZabell.lean, thm51_KL_eq_iff.</p> |                  |

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|      | <p><b>Theorem.</b> <i>Anchoring.lean, dampedB_deviation</i><br/> Let <math>\text{dampedB}(Q, r_0, \delta)</math> be the Jeffrey step on <math>B</math> to the target <math>(1 - \delta) Q(B=1) + \delta(1 - r_0)</math>. If <math>Q(B=1) \neq 0</math> then <math>\text{dampedB}(Q, r_0, \delta)(B=1) - (1 - r_0) = (1 - \delta) (Q(B=1) - (1 - r_0))</math>, an identity in <math>Q, r_0</math> and <math>\delta</math>. The paper's <math>\omega</math> is <math>\delta</math>.</p> <p><i>Para 2: the degree of adoption is recovered from the rating of one marginal before and after the second cue against the delivered credence <math>1 - r_0</math>, namely <math>\omega = 1 - (\text{after} - (1 - r_0)) / (\text{before} - (1 - r_0))</math>.</i></p> <p><b>Quote.</b> <i>Doring 1999, S379 (abstract)</i><br/> “Jeffrey conditionalization is sensitive to the order in which the evidence arrives. This order effect can be so pronounced as to call for a belief adjustment that cannot be understood as an assimilation of incoming evidence by Jeffrey’s rule.”</p> <p><i>Para 3: Doring treats the sequence effect as a defect of the framework.</i></p> <p><b>Quote.</b> <i>Doring 1999, S384-S385</i><br/> “The most straightforward approach would be to revert to the original probabilities and try to assimilate the later evidence all at once.” (S384) “Jeffrey conditionalizing in one step on the new assignments yields a posterior distribution in which both pieces of evidence receive equal weight” (S385).</p> <p><i>Para 3: his remedy is a single Jeffrey step from the original prior, which successive steps cannot supply (audit P2).</i></p> <p><b>Quote.</b> <i>Field 1978, pp. 365-366 (eq. (7))</i><br/> “If we express the combined result of both changes in terms of the Jeffrey parameters <math>q</math> and <math>q'</math> of the individual changes, we get a very complicated law; moreover, it is an asymmetric law ... If however we express the combined result in terms of the parameters <math>\alpha</math> and <math>\alpha'</math> of the individual changes, we get a law that is both simple and symmetric”</p> <p><i>Para 3: Field’s input is the portable parameter alpha rather than the delivered credence <math>q</math>, and with it successive changes commute.</i></p> <p><b>Quote.</b> <i>Wagner 2002, abstract (preprint p. 1)</i><br/> “When identical learning is properly represented, namely, by identical Bayes factors rather than identical posterior probabilities, then sequential probability-kinematical revisions behave just as they should.”</p> <p><i>Para 3: identical Bayes factors make sequential revisions commute.</i></p> <p><b>Quote.</b> <i>Garber 1980, p. 144</i><br/> “after nine repetitions of the same rather uninformative experience, S will become virtually certain that the ball is blue.” And: “If stimulation behaves in the way he supposes in (3), then practical certainty in <math>E</math> is much too easily obtained.”</p> <p><i>Para 3: the objection is that a portable factor compounds implausibly under repetition; Garber never discusses order.</i></p> <p><b>Theorem.</b> <i>Field.lean, field_eq7_comm</i><br/> For <math>E, E' : \Omega \rightarrow \text{Bool}</math>, <math>\alpha, \alpha' \in \mathbb{R}</math> and <math>P \geq 0</math> with <math>\sum_{\omega} P(\omega) = 1</math>: <math>\text{tilt}_{E'}(\alpha', \text{tilt}_E(\alpha, P)) = \text{tilt}_E(\alpha, \text{tilt}_{E'}(\alpha', P))</math>, where <math>\text{tilt}_E(\alpha, P)(\omega) = e^{\pm\alpha} P(\omega) / \sum_{\omega'} e^{\pm\alpha} P(\omega')</math>, the sign <math>+</math> on <math>E</math> and <math>-</math> off it (Field’s eq. (6)).</p> <p><i>Para 3: Field’s eq. (7) is symmetric, so his two updates commute.</i></p> <p><b>Theorem.</b> <i>Wagner2002.lean, thm31</i><br/> Let <math>p</math> be a probability, <math>q</math> come from <math>p</math> by probability kinematics on <math>\mathcal{E}</math> and <math>r</math> from <math>q</math> on <math>\mathcal{F}</math>; let <math>q'</math> come from <math>p</math> on <math>\mathcal{F}</math> and <math>r'</math> from <math>q'</math> on <math>\mathcal{E}</math>. If <math>\beta_{r',q'}(E_{i_1}:E_{i_2}) = \beta_{q,p}(E_{i_1}:E_{i_2})</math> for all <math>i_1, i_2</math> and <math>\beta_{q',p}(F_{j_1}:F_{j_2}) = \beta_{r,q}(F_{j_1}:F_{j_2})</math> for all <math>j_1, j_2</math>, then <math>r' = r</math> (finite partitions).</p> <p><i>Para 3: Wagner’s Theorem 3.1, with the primes as printed on the rendered page.</i></p> <p><b>Theorem.</b> <i>Garber.lean, garber_nine</i><br/> With <math>\alpha_0</math> fixed by the change <math>.3 \rightarrow .4</math> (so <math>e^{2\alpha_0} = 14/9</math>) and Field’s step iterated from <math>P_0(E) = .3</math>, <math>P_n(E) = 3 \cdot 14^n / (3 \cdot 14^n + 7 \cdot 9^n)</math>; then <math>P_9(E) &gt; .95</math> and <math>P_8(E) &lt; .95</math>.</p> <p><i>Para 3: Garber’s table (p. 144) is reproduced exactly; nine repetitions are the first to pass .95.</i></p> <p><b>Theorem.</b> <i>Hawthorne.lean, factorUpdate_smul, NL_factorUpdate, LR_factorUpdate</i><br/> A factor update is <math>\text{fu}(p, u, w)(x) = w(u_x) p(x) / \sum_y w(u_y) p(y)</math>. For <math>c \neq 0</math>, <math>\text{fu}(p, u, cw) = \text{fu}(p, u, w)</math>. If <math>Q[E_i] \neq 0</math>, <math>\text{NL}[p, \text{fu}(p, u, w), E_i] = w_i / \sum_y w(u_y) p(y)</math>; if also <math>Q[E_j], Q[E_k]</math> and the normaliser are nonzero, <math>\text{LR}[p, \text{fu}(p, u, w), E_j, E_k] = w_j / w_k</math>, whatever the prior.</p> <p><i>Para 4: amnestic, normed-likelihood and likelihood-ratio updates each multiply the basis cells by a factor that depends on that basis alone; they differ only in what fixes the factor.</i></p> <p><b>Theorem.</b> <i>LemmaSEP.lean, isSeparable_applySteps, sep_rescales_association</i><br/> For any <math>N</math>, any prior <math>P</math> on <math>\{0, 1\}^N</math> and any finite list of attribute-local steps (each multiplying cell <math>x</math> by a factor of the single coordinate <math>x_a</math> it reads), the result is <math>P(x) \prod_a g_a(x_a)</math> for some <math>g</math>. For such <math>F</math> and <math>i \neq j</math>: <math>F_{11}F_{00} - F_{10}F_{01} = g_i(1)g_i(0)g_j(1)g_j(0) \left( \prod_{a \neq i,j} g_a(x_a) \right)^2 (P_{11}P_{00} - P_{10}P_{01})</math>.</p> <p><i>Para 4: under every extension the believed association is rescaled by a positive factor and never shifted.</i></p> |                  |

| Part                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   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| <b>E.6</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             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Related literature, Asch as evidence (was 3.B)                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  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What the paper highlights is that such experimental analyses must also factor in the cross-attribute association – thus requiring the elicitation of a full posterior table rather than a single rating produced by step-by-step or end-of-sequence response modes. | While Bayes-factor updating predicts no sequence effect anywhere (Wagner, 2002), the experimental tradition has measured the phenomenon entirely in marginals. Hogarth and Einhorn (1992) track a single evaluative anchor. Asch (1946) is more generous and more instructive. His Experiment VI reports, for each of eighteen traits, the proportion of subjects choosing it over its opposite for the person described, separately for the two reading sequences. That the sequence effect differs in size across statistics is already visible there, although the differences Asch reports are all among marginals, which the paper treats alike. Between the sequences, <i>restrained</i> moves from 64 to 9 per cent and <i>good-looking</i> from 74 to 35, while <i>serious</i> moves from 97 to 100, <i>persistent</i> from 82 to 87 and <i>reliable</i> from 84 to 91. Some statistics swing enormously and others scarcely move, and Asch attributes the difference to the content of the terms rather than to their position. Yet eighteen marginals separate full adoption from partial adoption no better than one. Separating them needs the rating of one trait before and after the second cue and against what that cue delivers (Proposition PROP:ADJ), and the check-list asks whether a trait fits and never how strongly two traits are believed to go together beforehand, so the prior association cannot be formed from such data at all. What the paper adds is therefore not that more ratings are needed but that a second object must be elicited. Seeing the sequence effect requires only a marginal read in both sequences, which Asch has; locating the mechanism behind it requires the prior association as well, which he does not. |
| <p><b>Why.</b> Plan 3.B with the audit applied. A11, the check-list is a forced choice between opposite pairs. A15, Asch does give an account of the uneven effects, by content. P13, what is never elicited is the prior association, not the joint. The ADJ clause now states what the recovery formula uses, three ratings of one marginal, not a marginal across values of <i>c</i>. “amnesic updating” becomes “full adoption”. Colon removed. All eight percentages are correct and in the right columns.</p> <p><b>Evidence.</b> Asch.lean (quoted_all, d_neg_exactly, stimulus_disjoint_checklist); Anchoring.lean (dampedB_deviation, the identification).</p> <p><b>Grounds.</b></p> <p><b>Quote.</b> Asch 1946, p. 270 (<i>Experiment VI</i>)</p> <p>“The following series are read, each to a different group: A. intelligent—industrious—impulsive—critical—stubborn—envious B. envious—stubborn—critical—impulsive—industrious—intelligent There were 34 subjects in Group A, 24 in Group B. The two series are identical with regard to their members, differing only in the order of succession of the latter.”</p> <p><i>Six stimulus terms read in either order; the eighteen traits of Table 7 are response items, none of which is a stimulus term (Asch.lean, stimulus_disjoint_checklist).</i></p> <p><b>Quote.</b> Asch 1946, p. 262 (<i>Check List I</i>) and p. 271</p> <p>“To this end we constructed a check list consisting of pairs of traits, mostly opposites. From each pair of terms in this list, which the reader will find reproduced in Table 1, the subject was instructed to select the one that was most in accordance with the view he had formed.” (p. 262) “The check-list data appearing in Table 7 furnish quantitative support for the conclusions drawn from the written sketches.” (p. 271)</p> <p><i>The check-list is a forced choice within a pair of opposites (audit A11); Table 7 reports, for each pair, the percentage choosing the listed member.</i></p> <p><b>Computation.</b> Asch.lean, quoted_all; literature/asch1946/sympy/check_table7.py</p> <p>Table 7 (p. 271), Experiment VI, columns intelligent→envious (<math>N = 34</math>) and envious→intelligent (<math>N = 24</math>): restrained 64 / 9, good-looking 74 / 35, serious 97 / 100, persistent 82 / 87, reliable 84 / 91; also humorous 52 / 21, good-natured 18 / 0, important 85 / 90. Every quoted number is in the right column (proved by decide; script 16/16 PASS).</p> <p><i>The four small moves are all of 7 points or less and all in the opposite direction to the large swings (Asch.lean, d_neg_exactly).</i></p> <p><b>Quote.</b> Asch 1946, p. 264 (<i>Experiment I</i>)</p> <p>“If we assume that the process of mutual influence took place in terms of the actual character of the qualities in question, it is not surprising that some will, by virtue of their content, remain unchanged.”</p> <p><i>Asch’s account of uneven effects is by content; it is given for Experiment I, and he offers no trait-by-trait account for Experiment VI (audit A15).</i></p> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |

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|            | <p><b>Quote.</b> <i>Asch 1946, pp. 271-272</i><br/> “the first terms set up in most subjects a direction which then exerts a continuous effect on the latter terms.” (p. 271) “It is not the sheer temporal position of the item which is important as much as the functional relation of its content to the content of the items following it.” (p. 272)<br/> <i>Content rather than position: the entry’s “rather than to their position”.</i></p> <p><b>Theorem.</b> <i>Anchoring.lean, dampedB_deviation</i><br/> With <math>\text{dampedB}(Q, r_0, \delta)</math> the Jeffrey step on <math>B</math> to <math>(1 - \delta)Q(B=1) + \delta(1 - r_0)</math> and <math>Q(B=1) \neq 0</math>: <math>\text{dampedB}(Q, r_0, \delta)(B=1) - (1 - r_0) = (1 - \delta)(Q(B=1) - (1 - r_0))</math>. Hence <math>\omega = \delta</math> is fixed by three readings of one marginal, before the cue, after it, and the credence it delivers.<br/> <i>Marginals in two sequences, however many, do not fix omega; one marginal read before and after the second cue and against what it delivers does.</i></p> <p><b>Quote.</b> <i>Asch 1946, p. 262 (Technique) and p. 271 (design of the within-subject group)</i><br/> “Following the reading, each subject wrote a brief sketch.” (p. 262) “A new group (<math>N=24</math>) heard Series B, wrote the free sketch, and immediately thereafter wrote the sketch in response to Series A.” (p. 271)<br/> <i>Nothing is elicited before the series is read, and neither the sketch nor the forced-choice list asks how strongly two traits are believed to go together, so no prior association is collected (audit P13).</i></p> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>E.7</b> | Related literature, Bohren-Imas-Rosenberg, after the identification paragraph (was 3.D)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|            | <p><i>New paragraph(s) after the paragraph containing “contributes only a second-order loss in aggregate (Theorem THM:LOS).”</i></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <p>A current formulation of the same problem is given by Bohren et al. (2019), who distinguish discrimination arising from correct beliefs, from biased beliefs, and from preferences, and identify the source from how discrimination evolves along a history of evaluations. Their map from partiality to behaviour is Bayesian. A belief gap between groups is transmitted to evaluations, attenuated by the precision of the signal and along the history, and it vanishes as judgement becomes perfectly objective. Under the reading adopted here that map does not hold. A Jeffrey step sets the marginal of the attribute it addresses to the delivered credence, so an evaluator who takes the impression of quality last evaluates two workers alike whatever her prior beliefs about their groups, and the belief gap is silenced without any gain in objectivity. Under partial adoption with weight <math>\omega</math> on the impression, a fraction <math>1 - \omega</math> of the belief gap survives (Proposition PROP:ADJ). What credence-input updating does is therefore not to add a further source of discrimination to their three, but to relocate where partiality must sit in order to act. Lodged in the prior it is silenced, lodged in the impression it passes through untouched, and their framework has no parameter for the latter.</p> |
|            | <p><b>Why.</b> Plan 3.D with the audit applied. P17b, the belief gap is also attenuated along the history (their Proposition 2), so “vanishes only as” becomes “vanishes as”. Colon removed. Goes directly after C.15’s paragraph, which carries the Heckman precedent that plan 3.C proposed (3.C is superseded by C.15). Presupposes E.10 and the Bohren2019 entry.</p> <p><b>Evidence.</b> <code>check_pinning_kills_partiality.py</code> (7/7, <math>D = 0</math> exactly for arbitrary group priors and covariances); <i>Anchoring.lean</i> <i>dampedB_deviation</i>; <i>BohrenImasRosenberg.lean</i> (<code>prop1_decreasing</code>, <code>prop2_decreasing</code>, <code>endo_gap</code>).</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|            | <p><b>Grounds.</b><br/> <b>Quote.</b> <i>Bohren, Imas and Rosenberg 2019, working paper p. 11 (Belief-Updating)</i><br/> “The evaluator learns about the worker’s ability from the evaluation history. Her posterior belief about ability is derived using Bayes rule, given her model of inference. She combines this updated belief about ability with the signal to learn about the quality of the current task, also using Bayes rule to form her posterior belief about quality.”<br/> <i>Their map from partiality to behaviour is Bayesian by explicit assumption.</i></p> <p><b>Quote.</b> <i>Bohren, Imas and Rosenberg 2019, working paper pp. 11-12, eqs. (2) and (3)</i><br/> “Then her optimal evaluation in period <math>t</math> is <math>v_i(h_t, s_t, g) = \hat{E}_i[q_t \mid h_t, s_t, g] - c_g^i</math>. (2)” (p. 11) “Let <math>D_i(h, s) \equiv v_i(h, s, M) - v_i(h, s, F)</math> (3) denote the difference between type <math>\theta_i</math>’s evaluation of a male and female worker conditional on observing history <math>h</math> and signal <math>s</math>” (p. 12).<br/> <i>With no taste (<math>c_g^i = 0</math>) discrimination is the gap in posterior expected quality at the same history and signal.</i></p>                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |

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|                                                                                                                                                                                                                                                                                                                                                                                                  | <p><b>Quote.</b> <i>Bohren, Imas and Rosenberg 2019, working paper p. 16, Proposition 1</i></p> <p>“Proposition 1 (Subjectivity of Judgement). If the evaluator has belief-based partiality, initial discrimination is decreasing in the precision of the signal <math>\tau_\eta</math> and otherwise, initial discrimination is constant with respect to <math>\tau_\eta</math>. As the signal becomes perfectly objective, <math>\tau_\eta \rightarrow \infty</math>, there is initial discrimination if and only if the evaluator has preference-based partiality.”</p> <p><i>The belief gap is attenuated by the precision of the signal and vanishes as judgement becomes perfectly objective.</i></p> <p><b>Quote.</b> <i>Bohren, Imas and Rosenberg 2019, working paper p. 18, Proposition 2</i></p> <p>“Proposition 2 (Impossibility of Reversal). Suppose there is a single type of evaluator with belief-based partiality and no preference-based partiality. Then fixing an evaluation history, discrimination decreases across periods but never reverses.”</p> <p><i>The belief gap is also attenuated along the history, so the entry says “vanishes as” rather than “vanishes only as” (audit P17b).</i></p> <p><b>Computation.</b> <i>literature/bohren_imas_rosenberg2019/sympy/check_pinning_kills_partiality.py, checks (3)-(5)</i></p> <p>Group <math>g</math> has the <math>2 \times 2</math> joint with <math>P_g(A=1) = a_g</math>, <math>P_g(B=1) = b_g</math> and covariance <math>c_g</math>. A Jeffrey step on <math>A</math> to <math>1 - q_0</math> followed by a Jeffrey step on <math>B</math> to <math>1 - r_0</math> gives <math>P_M(B=1) - P_F(B=1) = 0</math> identically in <math>(a_M, b_M, c_M, a_F, b_F, c_F, q_0, r_0)</math>, so <math>D = 0</math> exactly with <math>c_g^i = 0</math>. With the <math>B</math>-cue read first and the <math>A</math>-cue last, <math>D \neq 0</math>. All checks pass.</p> <p><i>An evaluator who takes the impression of quality last evaluates two workers alike whatever her group priors; the silencing is exact and route-dependent.</i></p> <p><b>Theorem.</b> <i>Anchoring.lean, dampedB_deviation</i></p> <p>For <math>Q</math> with <math>Q(B=1) \neq 0</math>: <math>\text{dampedB}(Q, r_0, \delta)(B=1) - (1 - r_0) = (1 - \delta) (Q(B=1) - (1 - r_0))</math>. Applied to two groups meeting the same delivered credence <math>1 - r_0</math>, the difference of their <math>B</math>-marginals after the cue is <math>(1 - \delta)</math> times the difference before it.</p> <p><i>Under partial adoption with weight <math>\omega = \delta</math> on the impression, a fraction <math>1 - \omega</math> of the belief gap survives.</i></p> <p><b>Theorem.</b> <i>BohrenImasRosenberg.lean, prop1_decreasing, prop2_decreasing</i></p> <p>With <math>\text{eval}(\tau_q, \tau_\eta, \mu, c, s) = (\tau_q \mu + \tau_\eta s) / (\tau_q + \tau_\eta) - c</math> and <math>D(\tau_\eta) = \text{eval}(\tau_q, \tau_\eta, \hat{\mu}_M, 0, s) - \text{eval}(\tau_q, \tau_\eta, \hat{\mu}_F, c_F, s)</math>: if <math>\tau_q &gt; 0</math>, <math>0 &lt; \tau_{\eta,1} &lt; \tau_{\eta,2}</math> and <math>\hat{\mu}_F &lt; \hat{\mu}_M</math> then <math>D(\tau_{\eta,2}) &lt; D(\tau_{\eta,1})</math>. Along any history <math>v : \mathbb{N} \rightarrow \mathbb{R}</math>, with <math>\tau_a, \tau_\varepsilon, \tau_\eta &gt; 0</math>, <math>\hat{\mu}_F &lt; \hat{\mu}_M</math> and <math>c = 0</math>: <math>D_{n+1}(s) &lt; D_n(s)</math> for every <math>n</math>.</p> <p><i>Their map transmits the belief gap attenuated by the signal precision and along the history; the paper’s mechanism sits outside this map.</i></p> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>E.8</b> Section 4, title and opening (was 4.A)                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 1/2                                                                                                                                                                                                                                                                                                                                                                                              | <i>Section title: Sequence effects at the individual level</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <i>Section title: Which statistics register the sequence</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 2/2                                                                                                                                                                                                                                                                                                                                                                                              | Since Bayesian conditioning is sequence-independent by definition, we measure sequence effects with respect the Bayesian benchmark using two <b>primitive</b> statistics – gap and score. As discussed in Section SEC:JEFFREY, the <i>gap</i> is a distance from a Bayes-factor benchmark $P^B$ for the evaluator (see Definition DEF:INDGAP). The second primitive, <i>score-gap</i> $s(P) = \langle v, P \rangle$ defined against the sequence-free reference $P^B$ alongside with a threshold $\tau$ . As we show in Section SEC:INDIVIDUAL_PRIMITIVES, both <i>gap</i> and <i>score-gap</i> are first-order to the evaluator. The belief-statistics and decision-statistics follow from the two primitive statistics. The two <b>belief-statistics</b> that we are interested in are the <i>marginal probability</i> and <i>believed cross-attribute association</i> . The <b>decision-statistics</b> that we discuss for the individual evaluator and the population aggregate are <i>surplus-weighted loss</i> $L(c)$ and a <i>decision-flip</i> . The decision-flip is counted in the population to determine the <i>share</i> of those evaluators whose decision is changed by the sequence. The loss is then simply an aggregation of $ u $ over this share. How these quantities are carried over in aggregate are detailed in Section SEC:AGGREGATION.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | This section asks, of each statistic of a single evaluator’s belief, whether the two reading sequences move it at first order in the prior covariance $c$ or only at second order. Two primitive statistics carry the comparison with the sequence-free benchmark $P^B$ of Section SEC:JEFFREY, the <i>gap</i> , the distance of the evaluator’s belief from $P^B$ (Definition DEF:INDGAP), and the <i>score-gap</i> $s(P) = \langle v, P \rangle$ , the same distance read through the decision weights and set against a threshold $\tau$ . Section SEC:INDIVIDUAL_PRIMITIVES shows that both are first order.                                              |
|                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | The belief statistics and the decision statistics are read from the same belief. The belief statistics are the <i>marginal probability</i> of each attribute and the <i>believed cross-attribute association</i> ; the decision statistics are the surplus-weighted loss $L(c)$ and the decision <i>flip</i> , the indicator that the sequence changes the evaluator’s decision. Whether a statistic registers the sequence is settled here, for one evaluator; Section SEC:AGGREGATION shows that averaging over evaluators who met the cues in either sequence leaves the classification unchanged across an open set of priors and every interior mixture. |
| <p><b>Why.</b> Plan 4.A, colon removed. The section is organised on the statistic rather than on the individual-versus-aggregate axis, which ORD and PRO show to be inert. The third paragraph (why these statistics; FGT, corrected by C.13) is unchanged.</p> <p><b>Evidence.</b> DIV and SCR (both primitives first order); ORD and PRO (the classification is settled before averaging).</p> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |

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| <b>Grounds.</b><br><b>Theorem.</b> <i>PropDIV.lean, propDIV_gap_AB_a00..a11, propDIV_gap_BA_a00..a11, propDIV_seq_a00..a11</i><br>For $\alpha, \beta \notin \{0, 1\}$ and each cell $(i, j)$ , as derivatives at $c = 0$ : $\frac{d}{dc}[P_{AB}^J - P^B]_{ij} = \kappa(R_1)_{ij}$ , $\frac{d}{dc}[P_{BA}^J - P^B]_{ij} = \kappa'(R_2)_{ij}$ , $\frac{d}{dc}[P_{AB}^J - P_{BA}^J]_{ij} = \kappa(R_1)_{ij} - \kappa'(R_2)_{ij}$ , with $R_1 = q \otimes (1, -1)$ , $R_2 = (1, -1) \otimes r$ , $\kappa = (\alpha - q_0)r_0(1 - r_0)/Z$ , $\kappa' = (\beta - r_0)q_0(1 - q_0)/Z$ . Every posterior entry is a ratio of functions affine in $c$ , so this is the exact first-order term.<br><i>Proposition DIV: the gap from the benchmark is first order in <math>c</math> on either sequence, with nonzero coefficient off the locus of <math>\kappa_{eq\_zero}</math> iff.</i><br><b>Theorem.</b> <i>Aggregate.lean, lemmaSCR_gap_AB, lemmaSCR_gap_BA, lemmaSCR_protected_weight</i><br>For any weight $v \in \mathbb{R}^{2 \times 2}$ and $\alpha, \beta \notin \{0, 1\}$ : $\frac{d}{dc}\langle v, P_{AB}^J - P^B \rangle _{c=0} = \kappa\langle v, R_1 \rangle$ and $\frac{d}{dc}\langle v, P_{BA}^J - P^B \rangle _{c=0} = \kappa'\langle v, R_2 \rangle$ ; for $v = xJ + y\nabla\text{assoc}(q_0, r_0)$ both coefficients are 0.<br><i>Lemma SCR: the score-gap <math>s(P) = \langle v, P \rangle</math> is first order for a generic desirability vector and second order only for weights in the protected span.</i><br><b>Theorem.</b> <i>PropORD.lean, lemmaORD_gap, propORD_Amarg, propORD_Bmarg, propORD_const</i><br>For any $v$ and $\alpha, \beta \notin \{0, 1\}$ : $\frac{d}{dc}\langle v, P_{AB}^J - P_{BA}^J \rangle _{c=0} = \kappa\langle v, R_1 \rangle - \kappa'\langle v, R_2 \rangle$ ; in particular $\frac{d}{dc}[P_{AB}^J(A=1) - P_{BA}^J(A=1)]_{c=0} = \kappa'$ , $\frac{d}{dc}[P_{AB}^J(B=1) - P_{BA}^J(B=1)]_{c=0} = -\kappa$ , and the constant weight gives 0. The right-hand side mentions neither $P^B$ nor $\lambda$ .<br><i>Proposition ORD: whether a statistic separates the two sequences at first order is fixed by its differential against <math>R_1</math> and <math>R_2</math>, so the classification is settled for one evaluator before any averaging.</i><br><b>Computation.</b> <i>sympy/verify_ORD.py, steps (5) and (8)</i><br>$\text{assoc}(P_{AB}^J) - \text{assoc}(P_{BA}^J) = \Theta(c^2)$ at a generic point and $\langle \nabla\text{assoc}, \kappa R_1 - \kappa' R_2 \rangle = 0$ . Requiring $\kappa\langle G, R_1 \rangle - \kappa'\langle G, R_2 \rangle = 0$ at four priors $(\alpha, \beta)$ leaves a two-parameter solution space for $G$ , every member of which lies in $\text{span}\{J, \nabla\text{assoc}\}$ and annihilates $R_1$ and $R_2$ ; $\det \partial(\kappa, \kappa')/\partial(\alpha, \beta) \neq 0$ .<br><i>Marginals and generic scores are first order, the association second order, and the dividing line is annihilation of <math>\text{span}\{R_1, R_2\}</math>, the same line Proposition PRO draws for the aggregate.</i> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>E.9 Section 5, title and opening (was 5.A)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 1/2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <i>Section title: Aggregated Effects of Cue-Sequence</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <i>Section title: Averaging over sequences does not change the classification</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 2/2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | We now describe how the effect of arrival order on statistics discussed in the previous section fares in the population aggregate. Throughout this section we consider a population mixing the two encounter sequences: a fraction $\lambda \in [0, 1]$ of evaluators meet the $A$ -first (the credential cue in the example) and the rest $B$ -first (the letter), so the mean belief under Jeffrey conditioning is $\bar{P}_\lambda = \lambda P_{AB}^J + (1 - \lambda) P_{BA}^J$ . As it turns out, what makes a statistic second order or first order is largely the nature of the statistic itself rather than the mechanism of aggregation. | The previous section classified statistics for one evaluator. This section shows that the classification survives averaging over a population in which a fraction $\lambda \in [0, 1]$ of evaluators meet the $A$ -cue first (the credential in the example) and the rest the $B$ -cue first (the letter), so that the mean belief under Jeffrey conditioning is $\bar{P}_\lambda = \lambda P_{AB}^J + (1 - \lambda) P_{BA}^J$ . Across an open set of priors, what makes a statistic first order or second order is the statistic itself and not the mixture $\lambda$ , since the mean belief lies in the plane through $P^B$ spanned by the leading directions of the two sequences, and a statistic's order is fixed by its differential on that plane (Propositions PROP:ORD and PROP:PRO). |
| <b>Why.</b> Plan 5.A, colon removed. “Largely the nature of the statistic itself” hedged an exact result. Presupposes E.10.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Evidence.</b> PropPRO.lean (annihilator_eq_span, propPRO_uniqueness); PropORD.lean.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Grounds.</b><br><b>Theorem.</b> <i>PropPRO.lean, annihilator_eq_span (with propPRO_protection_each, R1_R2_indep)</i><br>If $1 - q_0 \neq 0$ and $\langle V, R_1 \rangle = \langle V, R_2 \rangle = 0$ then $V = (V_{01} + r_0(V_{00} - V_{01}))J + \frac{V_{00} - V_{01}}{1 - q_0} \nabla\text{assoc}(q_0, r_0)$ , where $J$ is the all-ones matrix and $\nabla\text{assoc}(q_0, r_0) = \begin{pmatrix} (1-q_0)(1-r_0) & -(1-q_0)r_0 \\ -q_0(1-r_0) & q_0r_0 \end{pmatrix}$ ;<br>conversely every $xJ + y\nabla\text{assoc}$ annihilates $R_1$ and $R_2$ . For $q_0 \neq 0$ , $R_1$ and $R_2$ are linearly independent.<br><i>The annihilator of the plane spanned by the two leading directions is exactly <math>\text{span}\{J, \nabla\text{assoc}\}</math>.</i><br><b>Theorem.</b> <i>PropPRO.lean, propPRO_protection</i><br>With $M_\lambda = \lambda\kappa R_1 + (1 - \lambda)\kappa' R_2$ the leading direction of $\bar{P}_\lambda - P^B$ : for all $x, y$ and all $\alpha, \beta, \lambda$ , $\langle xJ + y\nabla\text{assoc}(q_0, r_0), M_\lambda \rangle = 0$ .<br><i>Protection is annihilation of each route direction, identically in the mixture, not a cancellation that depends on <math>\lambda</math>.</i><br><b>Theorem.</b> <i>PropPRO.lean, propPRO_uniqueness and coeff_times_Z (with margA_not_in_span)</i><br>Let $x = \langle \nabla F, R_1 \rangle$ , $y = \langle \nabla F, R_2 \rangle$ ; the aggregate first-order coefficient times $Z$ is $\lambda(\alpha - q_0)r_0(1 - r_0)x + (1 - \lambda)(\beta - r_0)q_0(1 - q_0)y$ . If $0 < \lambda < 1$ , $0 < q_0 < 1$ , $0 < r_0 < 1$ , $\beta \neq r_0$ and this vanishes at two priors $\alpha_1 \neq \alpha_2$ , then $x = y = 0$ . At $\lambda = 1$ the $A$ -marginal read-out annihilates $R_1$ but not $R_2$ , so an interior $\lambda$ is needed.<br><i>Across an open set of priors a statistic is second order for some interior mixture iff its differential annihilates <math>\text{span}\{R_1, R_2\}</math>, whatever that mixture is.</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |

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| <p><b>Theorem.</b> <i>PropORD.lean, lemmaORD_gap</i><br/> For any <math>v</math> and <math>\alpha, \beta \notin \{0, 1\}</math>: <math>\frac{d}{dc} \langle v, P_{AB}^J - P_{BA}^J \rangle _{c=0} = \kappa \langle v, R_1 \rangle - \kappa' \langle v, R_2 \rangle</math>, so the contrast between the two sequences is read from the same two directions as the gap from <math>P^B</math>, with no reference to <math>P^B</math> or to <math>\lambda</math>.<br/> Combined with Proposition DIV, <math>\bar{P}_\lambda - P^B = c M_\lambda + \mathcal{O}(c^2)</math> for every <math>\lambda</math>, which is the plane through <math>P^B</math> the entry names.</p> |                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>E.10</b> Section 5, Proposition ORD after the definition of protection (was 5.B, ADJ moved to E.15)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 1/2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | New paragraph(s) after the paragraph containing “unprotected if that distortion is first order, $\mathcal{O}(c)$ .”                                        | <p>The distortion compares the population average with the benchmark. The same classification can be reached without either ingredient, by reading the sequence effect of Definition DEF:SEQEFFECT through the statistic itself. For a smooth statistic <math>F</math> the sequence effect read through it is <math>F(P_{AB}^J) - F(P_{BA}^J)</math>, and it involves neither the benchmark <math>P^B</math> nor the mixing weight <math>\lambda</math>.</p> <p><b>Proposition ORD</b> (between-sequence contrast). <i>For any weight <math>v</math>,</i></p> $\langle v, P_{AB}^J - P_{BA}^J \rangle = c (\kappa \langle v, R_1 \rangle - \kappa' \langle v, R_2 \rangle) + \mathcal{O}(c^2),$ <p><i>with <math>\kappa, \kappa', R_1, R_2</math> as in Proposition PROP:DIV. The sequence effect on the A-marginal is <math>c\kappa' + \mathcal{O}(c^2)</math> and on the B-marginal <math>-c\kappa + \mathcal{O}(c^2)</math>; each is <math>\Theta(c)</math> generically. The sequence effect on the believed association is <math>\mathcal{O}(c^2)</math> exactly. Across an open set of priors, a smooth statistic has a second-order sequence effect if and only if it is protected (Definition DEF:PROTECTION); both conditions hold exactly when the differential of the statistic at independence annihilates <math>\text{span}\{R_1, R_2\}</math> (Proposition PROP:PRO below). Nothing in the display involves <math>P^B</math> or <math>\lambda</math>. The classification is decided by the two reading sequences, and every mixture <math>\bar{P}_\lambda</math> inherits it.</i></p> <p><i>Proof.</i> By Proposition PROP:DIV(ii), <math>P_{AB}^J - P_{BA}^J = c(\kappa R_1 - \kappa' R_2) + \mathcal{O}(c^2)</math>; applying <math>\langle v, \cdot \rangle</math> gives the display. With <math>R_1 = q \otimes (1, -1)</math> and <math>R_2 = (1, -1) \otimes r</math>,</p> $\langle \mathbf{1}_{A=1}, R_1 \rangle = 0, \quad \langle \mathbf{1}_{A=1}, R_2 \rangle = -1, \quad \langle \mathbf{1}_{B=1}, R_1 \rangle = -1, \quad \langle \mathbf{1}_{B=1}, R_2 \rangle = 0,$ <p>giving <math>c\kappa'</math> and <math>-c\kappa</math>. For the association, <math>\langle \nabla \text{assoc}(q \otimes r), R_1 \rangle = \langle \nabla \text{assoc}(q \otimes r), R_2 \rangle = 0</math> (proof of Proposition PROP:PRO), so its sequence effect is <math>\mathcal{O}(c^2)</math>. For the equivalence, with <math>M_\lambda = \lambda \kappa R_1 + (1 - \lambda) \kappa' R_2</math> from Proposition PROP:DIV(i) and <math>\nabla F := \nabla F(q \otimes r)</math>,</p> $\begin{aligned} \langle \nabla F, \Delta_{\text{seq}} \rangle &= \kappa \langle \nabla F, R_1 \rangle - \kappa' \langle \nabla F, R_2 \rangle, \\ \langle \nabla F, M_\lambda \rangle &= \lambda \kappa \langle \nabla F, R_1 \rangle + (1 - \lambda) \kappa' \langle \nabla F, R_2 \rangle; \end{aligned}$ <p>across an open set of priors <math>\kappa</math> and <math>\kappa'</math> vary independently, so either vanishes identically exactly when <math>\langle \nabla F, R_1 \rangle = \langle \nabla F, R_2 \rangle = 0</math>. <math>\square</math></p> |
| 2/2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | A particularly surprising finding is that for a smooth statistic of the belief, the answer is settled before any averaging over the aggregate takes place. | A particularly surprising finding is that for a smooth statistic of the belief, the answer is settled before any averaging over the aggregate takes place (Proposition PROP:ORD).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <p><b>Why.</b> Plan 5.B, mathematics unchanged, prose corrected. Only Proposition ORD stays in Section 5, where it belongs with the classification at full adoption. The adoption-weight material, Proposition ADJ with its worked example and Proposition LAD, moves to the robustness subsection of Section 6 (E.15, 2026-10-01), so that Sections 4 and 5 state the results under the two premises and Section 6 relaxes them. Colons before displays are kept; colons in prose are replaced. ORD must go in before E.1, E.5, E.6, E.7, E.9, E.11 and E.15, which cite it.</p>                                                                                      |                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |

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| <b>Evidence.</b> | PropORD.lean (lemmaORD_gap, propORD_Amarg, propORD_Bmarg, propORD_const); verify_ORD.py (25/25).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    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| <b>Grounds.</b>  | <p><b>Theorem.</b> <i>PropORD.lean, lemmaORD_gap</i><br/> For <math>\alpha \neq 0, 1 - \alpha \neq 0, \beta \neq 0, 1 - \beta \neq 0</math> and any weight <math>v \in \mathbb{R}^{2 \times 2}</math>,</p> $\frac{d}{dc} \langle v, P_{AB}^J(c) - P_{BA}^J(c) \rangle \Big _{c=0} = \kappa \langle v, R_1 \rangle - \kappa' \langle v, R_2 \rangle,$ <p>with <math>\kappa = (\alpha - q_0) r_0 (1 - r_0) / Z, \kappa' = (\beta - r_0) q_0 (1 - q_0) / Z, Z = \alpha \beta (1 - \alpha) (1 - \beta), R_1 = q \otimes (1, -1), R_2 = (1, -1) \otimes r</math>. Here <math>\alpha = P(A=0), \beta = P(B=0)</math> and <math>q_0, r_0</math> are the delivered credences on <math>A=0, B=0</math>.<br/> <i>The display of Proposition ORD, assembled cell by cell from Proposition DIV's route derivatives.</i></p> <p><b>Theorem.</b> <i>PropORD.lean, propORD_Amarg, propORD_Bmarg, propORD_const, propORD_Amarg_eq_negKdrift</i><br/> Same hypotheses. <math>\frac{d}{dc} [P_{AB}^J(A=1) - P_{BA}^J(A=1)]_{c=0} = \kappa', \frac{d}{dc} [P_{AB}^J(B=1) - P_{BA}^J(B=1)]_{c=0} = -\kappa</math>, and for the constant weight <b>1</b> the coefficient is 0. With <math>Z \neq 0, \kappa' = -K</math> for <math>K = q_0(1 - q_0)(r_0 - \beta)/Z</math> of Proposition DRF.<br/> <i>The marginal rows of ORD; no statement mentions <math>P^B</math> or <math>\lambda</math>.</i></p> <p><b>Computation.</b> <i>sympy/verify_ORD.py, steps 5 and 8 (25/25)</i><br/> Step 8: <math>\det \partial(\kappa, \kappa') / \partial(\alpha, \beta) \neq 0</math> at the generic point, so <math>\kappa, \kappa'</math> vary independently across priors; requiring <math>\kappa \langle G, R_1 \rangle - \kappa' \langle G, R_2 \rangle = 0</math> at four priors leaves a two-parameter solution space for <math>G</math>, every member a combination of <b>1</b> and <math>\nabla</math> assoc, and each annihilates <math>R_1</math> and <math>R_2</math>. Step 5: the between-sequence gap of assoc is <math>\Theta(c^2)</math>, and <math>\langle \nabla \text{assoc}(q \otimes r), \kappa R_1 - \kappa' R_2 \rangle = 0</math>.<br/> <i>Grounds for the "if and only if" clause and for the <math>\mathcal{O}(c^2)</math> claim on the association.</i></p> <p><b>Theorem.</b> <i>Anchoring.lean, dampedB_deviation, routeDamped_mA1_deviation</i><br/> For any <math>Q</math> with <math>Q(B=1) \neq 0</math>: <math>(\text{dampedB } Q \text{ } r_0 \delta)(B=1) - (1 - r_0) = (1 - \delta)[Q(B=1) - (1 - r_0)]</math>. For <math>Q_1 = \text{jeffreyA}(\text{prior}(\alpha, \beta, c), q_0)</math> with <math>Q_1(B=0) \neq 0, Q_1(B=1) \neq 0, Q_1(B=0) + Q_1(B=1) = 1</math> and <math>\text{prior}(A=1) \neq 0</math>: <math>P_{AB}^\delta(A=1) - (1 - q_0) = \delta[P_{AB}^J(A=1) - (1 - q_0)]</math>. Identities in <math>c</math>; <math>\delta</math> is the code's name for <math>\omega</math>.<br/> <i>The first two lines of Proposition ADJ, exact in <math>c</math>.</i></p> <p><b>Theorem.</b> <i>Anchoring.lean, orderEffect_damped_mA1, orderEffect_damped_at_indep</i><br/> Under the hypotheses above and <math>Q_B(A=1) \neq 0</math>, where <math>Q_B = \text{jeffreyB}(\text{prior}, r_0)</math> is the belief after the <math>B</math>-cue alone (not the benchmark <math>P^B</math>):</p> $P_{AB}^\delta(A=1) - P_{BA}^\delta(A=1) = \delta[P_{AB}^J(A=1) - (1 - q_0)] + (1 - \delta)[(1 - q_0) - Q_B(A=1)].$ <p>At <math>c = 0</math>, for <math>\alpha, 1 - \alpha, \beta, 1 - \beta \neq 0</math>: <math>P_{AB}^\delta(A=1) - P_{BA}^\delta(A=1) = (1 - \delta)(\alpha - q_0)</math>.<br/> <i>The third line of ADJ and its value at independence.</i></p> <p><b>Theorem.</b> <i>Anchoring.lean, dampedB_at_one, dampedB_at_zero, routeDamped_at_zero_pins_A; sympy/check_zero_slope_identification.py, case 3</i><br/> <math>Q(B=1) \neq 0 \Rightarrow (\text{dampedB } Q \text{ } r_0 \text{ } 1)(B=1) = 1 - r_0. Q(B=0) \neq 0, Q(B=1) \neq 0, Q(B=0) + Q(B=1) = 1 \Rightarrow \text{dampedB } Q \text{ } r_0 \text{ } 0 = Q</math>. Hence, with those hypotheses on <math>\text{jeffreyA}(\text{prior}, q_0)</math> and <math>\text{prior}(A=1) \neq 0</math>: <math>P_{AB}^0(A=1) = 1 - q_0</math> for every <math>c</math>. Case 3 of the script: the last-read <math>c</math>-slope carries the factor <math>(1 - \delta)</math> with a <math>\delta</math>-free quotient, and the first-read slope vanishes only at <math>\delta = 0</math>.<br/> <i>The "exactly when" at the endpoints <math>\omega = 1</math> and <math>\omega = 0</math>.</i></p> <p><b>Computation.</b> <i>sympy/check_zero_slope_identification.py, case 7 (25/25); sympy/verify_example.py</i><br/> <math>\delta = \frac{P^A(B=1) - P_{AB}^\delta(B=1)}{P^A(B=1) - r_1}</math> holds identically in <math>c</math> (7a); the denominator equals <math>r_0 - \beta</math> at <math>c = 0</math> and is nonzero at the generic point (7b). With the numbers of Section 2 and <math>\omega = \frac{1}{2}</math>:<br/> <math>P^A(B=1) = 14/25 = .56, P_{AB}^{1/2}(B=1) = 43/100, r_1 = 3/10</math>, and <math>(.56 - .43)/(.56 - .30) = \frac{1}{2}</math>.<br/> <i>The recovery formula and the closing example sentence.</i></p> <p><b>Quote.</b> <i>Hogarth and Einhorn 1992, T-p.7 (transcription), Eqs. 3-4</i><br/> "In this case Eq. (1) can be written <math>S_k = S_{k-1} + w_k[s(x_k) - S_{k-1}]</math>, (3) which, after rearranging terms, leads to the averaging form <math>S_k = (1 - w_k)S_{k-1} + w_k s(x_k)</math>. (4)"<br/> <i>Eq. 4 is the estimation-mode (<math>R = S_{k-1}</math>) form; the source is a LaTeX transcription and pages are its own.</i></p> <p><b>Quote.</b> <i>Hogarth and Einhorn 1992, T-p.12 (transcription), Eq. 8</i><br/> "Thus, to illustrate by denoting the first piece of evidence as the anchor, Eq. (5) can be rewritten as <math>S_k = s(x_1) + w_k[s(x_2, \dots, x_k) - R]</math>, (8) where for a wide range of functions <math>s(x_2, \dots, x_k)</math>, it follows that the effective weight accorded to <math>s(x_1)</math> must be greater than the weight attached to any of the other pieces of evidence."<br/> <i>Their End-of-Sequence form with the first item as anchor.</i></p> |                  |

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|             | <p><b>Theorem.</b> <i>HogarthEinhorn.lean, oneSided_eq_eq8, eos2_est_orderEffect</i></p> <p>For all <math>w, S_0, a, b</math>: <math>\text{oneSided}(w, S_0, a, b) := \text{estStep}_w(\text{estStep}_1(S_0, a), b) = (1 - w)a + wb = a + w[b - a] = \text{eq8}(w, R=a, s_1=a; b)</math>, Eq. 8 with <math>k = 2</math> and <math>R = s(x_1)</math>, where <math>\text{estStep}_w(S, s) = (1 - w)S + ws</math> is Eq. 4. The two-item sequence effect is <math>(2w - 1)(b - a)</math>.</p> <p><i>The one-sided rule is their Eq. 8 for two items with a constant weight; what is new is the two-attribute Jeffrey embedding.</i></p> <p><b>Theorem.</b> <i>Epstein.lean, compromise_eq_damped, omegaEff_eq</i></p> <p>For <math>\text{marg}_1 p(s_1) \neq 0</math>, <math>\sum p = 1</math> and <math>\alpha(s_1) \geq 0</math>, the conditional of Epstein’s compromise prior (12) is <math>(1 - \omega)p_2(s_2) + \omega p(s_2   s_1)</math> with <math>\omega = 1 - \frac{\alpha(s_1)\lambda(s_1)}{1 + \alpha(s_1)} = \frac{1 + \alpha(s_1)(1 - \lambda(s_1))}{1 + \alpha(s_1)}</math>.</p> <p><i>A published axiomatic model whose choice posterior is the damped target, with the Bayesian conditional as delivered credence.</i></p> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>E.11</b> | Scope, rival mechanisms (was 6.A)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|             | <p><i>New paragraph(s) after the paragraph containing “is outside the scope of stated results.”</i></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <p>A separate scope question concerns rival mechanisms rather than broken assumptions. Sequence effects have an explanation that competes with full adoption. In the belief-adjustment model of Hogarth and Einhorn (1992), when the first cue is adopted in full and later cues only in part the first impression is protected, and at zero adoption of later cues it is never moved, whereas when every cue is adopted only in part from a prior anchor the same model predicts recency. That first account and full adoption point in opposite directions, since the protected impression is the first one there and the last one here. An observable separates them. Under partial adoption of the second cue with weight <math>\omega</math> (Proposition PROP:ADJ), the marginal that ignores the prior association is the last-read one at <math>\omega = 1</math>, the first-read one at <math>\omega = 0</math>, and neither in between. A reading on which neither marginal ignores the association is consistent with an interior weight and with the benchmark alike; the two are separated by the sequence comparison, under which the benchmark shows no sequence effect at any <math>c</math> and an interior weight shows one already at <math>c = 0</math> (Proposition PROP:ADJ). Since every rule here is a separable reweighting, all of them carry the prior’s odds ratio (Lemma LEM:SEP), so the odds ratio separates none of them and the two marginals are the only statistics that do. Three cautions bound the claim. The classification covers this one-parameter family, not every conceivable mechanism. Asch (1946) is evidence that sequence moves marginals, six stimulus terms read in two sequences moving the proportions on eighteen response traits, and not evidence for either endpoint, since his own account is that early terms set a direction for the reading of later ones, so that what a later cue delivers depends on its position, a mechanism outside the family here, in which each cue delivers the same credence in either position. And full adoption is a substantive commitment rather than a consequence of the level reading. Hawthorne (2004) objects that it “seems implausible that the most recent experience or non-propositional state should completely dictate belief strengths for basis sentences, with no regard for the import of previous experiences or states”. The objection applies here, and his own two-basis example shows the overwriting explicitly, so that the credential’s implication for trustworthiness is erased once the letter fixes that attribute. The adoption weight is his objection stated as a parameter, and the reply offered here is not that the objection misses but that the weight is recovered from three ratings of one marginal rather than assumed.</p> |

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|      | <p><b>Why.</b> Plan 6.A with the audit applied. P10, the belief-adjustment model protects the first impression only when the first cue is the anchor and later cues are partly adopted; damping every cue gives recency, so the rival is stated as that case. P14, Experiment VI has six stimulus terms and eighteen response traits. P9, Hawthorne’s objection follows his two-basis medical example, so the claim that his illustration misses a one-cue-per-attribute model is dropped. The recovery uses three ratings of one marginal (plan’s own correction). Colon removed. Goes after the scope paragraph, inside the Scope subsection that E.15 opens, and before E.15.</p> <p><b>Evidence.</b> HogarthEinhorn.lean (eq8_estimation_first_dominates, appB_recency, twoAttr_oneSided_orderEffect); Anchoring.lean; Asch.lean (seriesA_length, checkListI_length); Hawthorne.lean (med_overwrite; quotation pp. 98-99 verbatim).</p> <p><b>Grounds.</b></p> <p><b>Theorem.</b> <i>HogarthEinhorn.lean, appB_recency, appB_recency_pos</i><br/> For all <math>S_0, a, b, w_a, w_b</math>: <math>\text{estStep}_{w_b}(\text{estStep}_{w_a}(S_0, a), b) - \text{estStep}_{w_a}(\text{estStep}_{w_b}(S_0, b), a) = w_a w_b (b - a)</math>, and it is <math>&gt; 0</math> for <math>w_a, w_b &gt; 0, a &lt; b</math> (their B.5). Here <math>\text{estStep}_w(S, s) = (1 - w)S + ws</math> is Eq. 4 applied from a prior anchor <math>S_0</math> to every item.<br/> <i>Every cue damped from a prior anchor gives recency, so that case is the rival stated in the entry.</i></p> <p><b>Quote.</b> <i>Hogarth and Einhorn 1992, T-p.35 (Appendix B) and T-p.11 (transcription)</i><br/> “Anderson and Hovland (1957) assume that <math>w_a = w_{ba}</math> and <math>w_b = w_{ab}</math> in which case Eq. (B.3) can be reexpressed as <math>\dots D = w_a w_b [s(x_b) - s(x_a)]</math>. (B.5) Because <math>s(x_b) &gt; s(x_a)</math>, <math>D &gt; 0</math> and recency always obtains.” And T-p.11: “when <math>R = S_{k-1}</math>, the SbS process always predicts recency (for <math>\alpha, \beta \neq 0</math>). . .”<br/> <i>Audit P10 and H15: partial adjustment of every item protects the last impression, not the first.</i></p> <p><b>Theorem.</b> <i>HogarthEinhorn.lean, eq8_estimation_first_dominates, oneSided_primacy_iff, twoAttr_oneSided_orderEffect</i><br/> For <math>0 \leq w &lt; \frac{1}{2}, c_j \geq 0, \sum_j c_j = 1</math>: <math>w c_j &lt; 1 - w</math> for every <math>j</math>, the first item’s effective weight in Eq. 8 (estimation mode) exceeds every other’s. For <math>a &lt; b</math>: <math>\text{oneSided}(w, S_0, a, b) - \text{oneSided}(w, S_0, b, a) &lt; 0 \iff w &lt; \frac{1}{2}</math>. With two attributes each moved only by its own cue: <math>\text{estStep}_1(S_A, s_A) - \text{estStep}_w(S_A, s_A) = (1 - w)(s_A - S_A)</math>.<br/> <i>With the first cue as anchor the first impression is protected, and never moved at <math>w = 0</math>.</i></p> <p><b>Quote.</b> <i>Hogarth and Einhorn 1992, T-p.12 (transcription)</i><br/> “In these circumstances, the structure of the EoS strategy contains within it a force toward primacy. Moreover, this result holds whether the evidence is all positive, all negative, or mixed. This occurs because, in the absence of an explicit starting point, we assume that the first piece of evidence, or an amalgamation of the first few pieces, serves as the anchor.”<br/> <i>The first account in the entry: the first cue adopted in full as anchor, later cues in part.</i></p> <p><b>Theorem.</b> <i>Anchoring.lean, dampedB_at_one, routeDamped_at_zero_pins_A, orderEffect_damped_at_indep; PropIMM.lean, propIMM_indep</i><br/> <math>Q(B=1) \neq 0 \Rightarrow (\text{dampedB } Q r_0 1)(B=1) = 1 - r_0</math>: at <math>\omega = 1</math> the last-read marginal is its delivered credence for every <math>c</math>. With <math>Q_1 = \text{jeffreyA}(\text{prior}, q_0)</math>, <math>Q_1(B=0), Q_1(B=1) \neq 0</math>, <math>Q_1(B=0) + Q_1(B=1) = 1</math>, <math>\text{prior}(A=1) \neq 0</math>: <math>P_{AB}^0(A=1) = 1 - q_0</math> for every <math>c</math>, the first-read one at <math>\omega = 0</math>. For <math>\alpha, 1 - \alpha, \beta, 1 - \beta \neq 0</math>: <math>P_{AB}^\delta(A=1) - P_{BA}^\delta(A=1) _{c=0} = (1 - \delta)(\alpha - q_0)</math>, while <math>P_{AB}^1(0) = P_{BA}^1(0) = P^B(0)</math>.<br/> <i>Which marginal ignores the association at each endpoint, and the sequence effect at <math>c = 0</math> that separates an interior weight from the benchmark.</i></p> <p><b>Theorem.</b> <i>LemmaSEP.lean, isSeparable_applySteps; PropDEC.lean, oddsRatio_gap_eq_zero, oddsRatio_PJba_eq_prior</i><br/> For any <math>N</math>, any <math>P</math> on <math>\{0, 1\}^N</math> and any finite list <math>l</math> of attribute-local steps, <math>\text{applySteps } P l = P \cdot \prod_a g_a(x_a)</math> for some <math>g_a</math>. With <math>\alpha, 1 - \alpha, \beta, 1 - \beta, q_0, 1 - q_0, r_0, 1 - r_0 \neq 0</math>, the first-step <math>B</math>-marginals, the prior’s off-diagonal cells, and the reweighted table’s total and off-diagonal cells nonzero: <math>\text{OR}(P_{AB}^1) - \text{OR}(P^B) = 0</math>; likewise <math>\text{OR}(P_{BA}^1) = \text{OR}(\text{prior})</math> with the first-step <math>A</math>-marginals nonzero.<br/> <i>Every rule here is a separable reweighting carrying the prior’s odds ratio, so the odds ratio separates none of them; the damped routes are checked in verify_interior_omega.py, row 2.</i></p> <p><b>Quote.</b> <i>Asch 1946, pp. 271-272 and p. 272</i><br/> “the first terms set up in most subjects a direction which then exerts a continuous effect on the latter terms. When the subject hears the first term, a broad, uncrystallized but directed impression is born. The next characteristic comes not as a separate item, but is related to the established direction.” And p. 272: “It is not the sheer temporal position of the item which is important as much as the functional relation of its content to the content of the items following it.”<br/> <i>His account is a direction set by early terms for the reading of later ones, not a weight on position (audit A18, A19, A26).</i></p> <p><b>Theorem.</b> <i>Asch.lean, seriesA_length, seriesB_eq_reverse, checkListI_length, stimulus_disjoint_checklist</i><br/> <math> \text{seriesA}  = 6</math> (Experiment VI, p. 270), <math>\text{seriesB} = \text{seriesA reversed}</math>, <math> \text{checkListI}  = 18</math> (Table 1, p. 262), and no stimulus term occurs in either member of any check-list pair.<br/> <i>Six stimulus terms read in two sequences, eighteen response traits (audit P14).</i></p> |                  |

| Part             | BEFORE (current manuscript, or the anchor for an insert)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | AFTER (proposed) |
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| <b>Quote.</b>    | <p><i>Hawthorne 2004, pp. 98-99 and p. 98</i></p> <p>“The resulting Amnestic Model raises two related concerns. First, it seems implausible that the most recent experience or non-propositional state should completely dictate belief strengths for basis sentences, with no regard for the import of previous experiences or states.” And p. 98, in the two-basis example: “When the x-ray report comes in the physician adopts the radiologist’s degree of confidence that an image of a mass is present, <math>Q_e[E] = Q_{fe}[E] = .90</math>.”</p> <p><i>The sentence begins on p. 98 and ends on p. 99 (audit H7); the objection follows his two-basis medical example (audit H33, P9).</i></p> <p><b>Theorem.</b> <i>Hawthorne.lean, med_rewrite, med_crossBasis</i></p> <p>With his prior (likelihoods <math>19/20</math>, <math>Q[C] = \frac{1}{2}</math>): <math>Q_{fe}[E] = \frac{9}{10}</math> and <math>Q_e[E] = \frac{9}{10}</math>, so the <math>E</math>-marginal after <math>f</math> then <math>e</math> equals the one after <math>e</math> alone; the <math>F</math>-update alone had moved <math>Q[E]</math> from <math>\frac{1}{2}</math> to <math>\frac{22}{125}</math>, which the <math>e</math>-update erases.</p> <p><i>The overwriting on two distinct bases, explicit in his own example.</i></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                  |
| <b>E.12</b>      | <p>Folded into E.15 (was 6.B)</p> <p><i>No manuscript text; see Why.</i></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                  |
| <b>Why.</b>      | <p>The two-channel paragraph, the rubric prediction and the settings table of this entry now form part of the robustness subsection in E.15 (2026-10-01), with the corrections recorded there. Kept as a number so that earlier references to E.12 resolve.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                  |
| <b>Evidence.</b> | <p>See E.15.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                  |
| <b>Grounds.</b>  | <p><b>Computation.</b> <i>sympy/verify_interior_omega.py, row 1 (23/23)</i></p> <p>At <math>c = 0</math>, symbolic in <math>\alpha, \beta, q_0, r_0, \delta</math>: <math>P_{AB}^\delta(A=1) - P_{BA}^\delta(A=1) = (1 - \delta)(\alpha - q_0)</math> and <math>P_{AB}^\delta(B=1) - P_{BA}^\delta(B=1) = -(1 - \delta)(\beta - r_0)</math>.</p> <p><i>The position channel on both marginals at independence; <math>\delta</math> is the script’s name for <math>\omega</math>.</i></p> <p><b>Computation.</b> <i>sympy/verify_interior_omega.py, rows 2-3</i></p> <p>Row 2: each damped step multiplies the table it meets by a column (row) factor, so both routes are separable reweightings of the prior; their odds ratios equal the prior’s at the generic point with <math>\delta = \frac{1}{3}</math>, <math>c = \frac{1}{40}</math>. Row 3: the between-sequence effect on assoc is 0 at <math>c = 0</math> for every <math>\delta</math>; its <math>c^1</math> coefficient is <math>(1 - \delta)G</math> with <math>G</math> affine in <math>\delta</math> and nonzero at the generic prior for <math>\delta = \frac{1}{2}</math> and <math>\delta = 0</math>; at <math>\delta = 1</math> the effect is second order.</p> <p><i>Odds ratio identical across sequences; association zero between sequences at <math>c = 0</math> and first order in <math>c</math>.</i></p> <p><b>Computation.</b> <i>sympy/verify_interior_omega.py, row 4</i></p> <p>At <math>c = 0</math> and interior <math>\lambda</math>, the mean belief <math>\lambda P_{AB}^\delta + (1 - \lambda)P_{BA}^\delta</math> differs from <math>P^B</math> by <math>(1 - \lambda)(1 - \delta)(q_0 - \alpha)</math> on the <math>A</math>-marginal and by <math>-\lambda(1 - \lambda)(1 - \delta)^2(\alpha - q_0)(\beta - r_0)</math> on assoc; nonzero at the generic point for <math>\lambda = \frac{2}{5}</math>, <math>\delta = \frac{1}{2}</math>, and zero at <math>\delta = 1</math>.</p> <p><i>The pooled cross-product association that no member holds.</i></p> <p><b>Computation.</b> <i>sympy/verify_interior_omega.py, rows 5-6</i></p> <p>Scale the second cue’s Bayes factor by <math>\delta</math>: <math>W_{ij} = P_{ij} \ell_i^A (\ell_j^B)^\delta</math> when <math>A</math> is read first and <math>W_{ij} = P_{ij} (\ell_i^A)^\delta \ell_j^B</math> when <math>B</math> is, then normalise. At <math>c = 0</math> the <math>A</math>-marginals differ between sequences (generic prior, <math>\delta = \frac{1}{2}</math>) and agree at <math>\delta = 1</math>. At <math>\delta = 0</math> both readings equal the single Jeffrey step on the first cue, for every <math>c</math>.</p> <p><i>The position channel is not a feature of updating on delivered credences.</i></p> <p><b>Theorem.</b> <i>Anchoring.lean, orderEffect_damped_at_indep</i></p> <p>For <math>\alpha \neq 0</math>, <math>1 - \alpha \neq 0</math>, <math>\beta \neq 0</math>, <math>1 - \beta \neq 0</math>: <math>P_{AB}^\delta(A=1) - P_{BA}^\delta(A=1) _{c=0} = (1 - \delta)(\alpha - q_0)</math>; the amnestic term vanishes there by Proposition IMM.</p> <p><i>Row 1’s first identity, machine-checked.</i></p> <p><b>Quote.</b> <i>Wagner 2002, preprint p. 4, Theorem 3.1</i></p> <p>“Theorem 3.1. Given the probability revision schema (3.1), if the Bayes factor identities (3.2) <math>\beta_{r',q'}(E_{i_1} : E_{i_2}) = \beta_{q,p}(E_{i_1} : E_{i_2})</math>, for all <math>i_1, i_2</math>, and (3.3) <math>\beta_{q',p}(F_{j_1} : F_{j_2}) = \beta_{r,q}(F_{j_1} : F_{j_2})</math>, for all <math>j_1, j_2</math>, hold, then <math>r' = r</math>.” Same page: “where the sequence <math>(r'(E_i))</math> may differ from <math>(q(E_i))</math>, and the sequence <math>(q'(F_j))</math> from <math>(r(F_j))</math>”.</p> <p><i>Commutation requires each cue’s Bayes factor to be the same in either position; read on the rendered page, since the text layer drops the primes.</i></p> <p><b>Theorem.</b> <i>Wagner2002.lean, thm31</i></p> <p>For <math>p</math> a probability, <math>q</math> from <math>p</math> by kinematics on <math>E</math>, <math>r</math> from <math>q</math> on <math>F</math>, <math>q'</math> from <math>p</math> on <math>F</math>, <math>r'</math> from <math>q'</math> on <math>E</math>, and <math>\beta_{r',q'}(E_{i_1} : E_{i_2}) = \beta_{q,p}(E_{i_1} : E_{i_2})</math>, <math>\beta_{q',p}(F_{j_1} : F_{j_2}) = \beta_{r,q}(F_{j_1} : F_{j_2})</math> for all indices: <math>r' = r</math>.</p> <p><i>Field’s theorem as Wagner states it, finite partitions.</i></p> |                  |

| Part        | BEFORE (current manuscript, or the anchor for an insert)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | AFTER (proposed)                                                                                                                                                                                                              |
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|             | <p><b>Theorem.</b> <i>BenjaminBodohCreedRabin.lean, example_marginal_gap, margOddsA_orders_ne</i><br/> Uniform prior on <math>\{0, 1\}^2</math>, <math>\alpha = \frac{1}{2}</math>, likelihood vectors <math>a = b = (49/50, 1/50)</math>: the believed <math>P(A=0)</math> is <math>7/8</math> when <math>A</math> is read first and <math>49/50</math> when read last (the Bayes value is <math>49/50</math>). In general, at independence with <math>\alpha \neq 1</math> and <math>a_0 \neq a_1</math>, the <math>A</math>-marginal odds differ between the two sequences.<br/> <i>A published rule with likelihood inputs and posterior-becomes-prior dynamics shows the position channel at independence.</i></p> <p><b>Theorem.</b> <i>Aggregate.lean, PJab_mB1, PJba_mA1, propDRF_route_AB, propDRF_route_BA; PropIMM.lean, propIMM_indep</i><br/> With the first-step marginal nonzero: <math>P_{AB}^J(B=1) = 1 - r_0</math> and <math>P_{BA}^J(A=1) = 1 - q_0</math> for every <math>c</math> (last-read exact). For <math>\alpha, 1 - \alpha, \beta, 1 - \beta \neq 0</math>: <math>\frac{d}{dc} [P_{AB}^J(A=1) - P^B(A=1)]_0 = 0</math> and, with <math>Z \neq 0</math>, <math>\frac{d}{dc} [P_{BA}^J(A=1) - P^B(A=1)]_0 = K</math>, <math>K = q_0(1 - q_0)(r_0 - \beta)/Z</math>; hence <math>P_{AB}^J(A=1) - (1 - q_0) = -cK + \mathcal{O}(c^2)</math>, the first-read marginal at <math>cK</math> from its delivered credence. At <math>c = 0</math>: <math>P_{AB}^J = P_{BA}^J = P^B</math>.<br/> <i>The rubric prediction under full adoption: the sequence effect lives in the first-read belief and vanishes at independence.</i></p> |                                                                                                                                                                                                                               |
| <b>E.15</b> | Section 6, the partial-adoption subsection (new; ADJ and LAD from E.10, E.12 folded in, Proposition FAC added)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                               |
| 1/9         | We now revisit how Assumptions AS:LOCALC–AS:SURPLUS in Section SEC:ASSUMPTIONS define scope for any conclusions made from the model.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | SUBSECTION HEADING<br><b>Scope</b><br>We now revisit how Assumptions AS:LOCALC–AS:SURPLUS in Section SEC:ASSUMPTIONS define the scope of the conclusions drawn from the model.                                                |
| 2/9         | Section SEC:SCOPE states the scope of results and limitations that would break the results.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Section SEC:SCOPE states the scope of the results and the limitations that would break them, shows that the second-order results require full adoption of the later cue, and gives a test of that condition from the ratings. |

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| 3/9  | <i>New paragraph(s) after the paragraph containing “is outside the scope of stated results.”</i> | <p>SUBSECTION HEADING</p> <p><b>Partial adoption and factor inputs</b></p> <p>The results of Sections SEC:INDIVIDUAL and SEC:AGGREGATION rest on two premises about a cue. It delivers a credence on its own attribute rather than a factor, and the credence is adopted in full. Full adoption is the setting in which an evaluator adopts a cue alike whichever position it arrives in, so that the sequence can act only through the believed link between the attributes. A weight on the later cue that depends on its position is a property of the evaluator rather than of the evidence, and it is the subject of the belief-adjustment literature (Hogarth and Einhorn, 1992). Whether a population adopts in full can be read from its ratings, since Proposition PROP:ADJ below recovers the adoption weight from three marginals of one reading group, and Propositions PROP:LAD and PROP:FAC say what to expect when it does not, under either reading of the cues. What partial adoption leaves in place is the odds ratio, which is the same in both sequences at every weight, and a second-order class of statistics that no longer contains the association. Table TAB:ROBUST collects the answers.</p> <p><b>The two channels.</b> Sequence dependence has two channels in this setting, and the results of Sections SEC:INDIVIDUAL and SEC:AGGREGATION concern one of them. One is the association channel. A cue on one attribute moves the belief about the other through the prior association, so the second cue meets a belief the first has already changed. The association channel’s effect is first order in <math>c</math> and vanishes at independence, where each cue sets its own marginal and touches nothing else (Proposition PROP:IMM). The other is the position channel. If the cue read second is adopted only in part, then which cue is weakened depends on the sequence, and the two sequences differ even at <math>c = 0</math>, by <math>(1 - \omega)(\alpha - q_0)</math> on the competence marginal and <math>-(1 - \omega)(\beta - r_0)</math> on the trustworthiness marginal. The position channel is not a feature of updating on delivered credences. A Bayes-factor update that gives the second cue’s factor the weight <math>\omega</math> is sequence-dependent in the same way, since commutation requires the same factor in either position (Wagner, 2002). Under full adoption the position channel is absent and the association channel is the only one, which is the setting of the results. Under partial adoption the position channel is present at <math>c = 0</math> and moves every marginal at order zero, while the believed association still does not differ between sequences at <math>c = 0</math> and differs at first order (Proposition PROP:LAD). The classification by order in <math>c</math> shifts by one order, each statistic keeping its place relative to the others, and the odds ratio is the same in both sequences at every weight. The mean belief of a population mixing the two sequences acquires a cross-product association of <math>-\lambda(1 - \lambda)(1 - \omega)^2(\alpha - q_0)(\beta - r_0)</math> that no member holds, and the separation between the share of decisions changed and the loss is lost. What the paper says about the position channel is confined to Propositions PROP:ADJ and PROP:LAD, the first recovering its strength from the ratings and the second stating what it does to the order in <math>c</math> at which each statistic registers the sequence. The position channel is a temporal bias of the observer, present whether or not the attributes are believed related, while the association channel runs through what one cue implies about the other and exists only when they are. Table TAB:SETTINGS sets the two settings side by side.</p> |

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| 4/9  | <i>Continues the insertion of the previous part.</i>     | <p><b>Partial adoption.</b> The two sequences of Sections SEC:INDIVIDUAL and SEC:AGGREGATION adopt each delivered credence in full. Suppose instead that the second cue is adopted only in part, so that after the first Jeffrey step the response to the second cue is a Jeffrey step on its partition to the target <math>(1 - \omega)m + \omega r_1</math>, where <math>m</math> is the marginal the first step left and <math>0 \leq \omega \leq 1</math> is the adoption weight of Section SEC:SETUP. The rule is the averaging form of the belief-adjustment model, <math>S_k = (1 - w_k)S_{k-1} + w_k s(x_k)</math> (Hogarth and Einhorn, 1992, Eq. 4), applied to the second cue only, which is their end-of-sequence form for two items with a constant weight, and embedded in the joint law by a Jeffrey step. Write <math>P_{AB}^\omega</math> and <math>P_{BA}^\omega</math> for the two sequences under this rule; <math>\omega = 1</math> recovers <math>P_{AB}^J</math> and <math>P_{BA}^J</math>, and <math>\omega = 0</math> ignores the second cue. Let <math>P^A</math> and <math>P^B</math> denote the belief after the <math>A</math>-cue alone and after the <math>B</math>-cue alone.</p> <p><b>Proposition ADJ</b> (adoption weight). <i>For every <math>c</math> and every <math>\omega</math>, in sequence <math>AB</math>,</i></p> $P_{AB}^\omega(B=1) - r_1 = (1 - \omega)[P^A(B=1) - r_1],$ $P_{AB}^\omega(A=1) - q_1 = \omega[P_{AB}^J(A=1) - q_1],$ $P_{AB}^\omega(A=1) - P_{BA}^\omega(A=1) = \omega[P_{AB}^J(A=1) - q_1] + (1 - \omega)[q_1 - P^B(A=1)].$ <p><i>Hence the last-read marginal equals its delivered credence identically in <math>c</math> exactly when <math>\omega = 1</math>, the first-read marginal does so exactly when <math>\omega = 0</math>, and for <math>0 &lt; \omega &lt; 1</math> neither does. The first-order coefficient in <math>c</math> of the first-read marginal is <math>\omega</math> times the benchmark's coefficient for the same marginal. At <math>c = 0</math> the third line equals <math>(1 - \omega)(\alpha - q_0)</math>, so a partially adopting evaluator shows a sequence effect at independence, where the fully adopting evaluator (Proposition PROP:IMM) and the benchmark show none. Whenever <math>P^A(B=1) \neq r_1</math>, the first line gives the weight from three marginals of one reading group, for every <math>c</math>,</i></p> $\omega = \frac{P^A(B=1) - P_{AB}^\omega(B=1)}{P^A(B=1) - r_1}.$ <p><i>Proof.</i> With <math>Q</math> the belief after the first step and <math>t = (t_0, t_1)</math>, <math>t_1 = (1 - \omega)Q(B=1) + \omega r_1</math>, the target of the second, column <math>j</math> is multiplied by <math>t_j/Q(B=j)</math>, so that</p> $P(B=1) = t_1,$ $P(A=1) = \sum_j Q(A=1, B=j) \frac{t_j}{Q(B=j)}$ $= (1 - \omega)Q(A=1) + \omega \sum_j Q(A=1, B=j) \frac{r_j}{Q(B=j)}.$ <p>In sequence <math>AB</math>, <math>Q = P^A</math>, <math>Q(A=1) = q_1</math>, and the last sum is <math>P_{AB}^J(A=1)</math>, giving</p> $P_{AB}^\omega(B=1) - r_1 = (1 - \omega)[P^A(B=1) - r_1], \quad P_{AB}^\omega(A=1) - q_1 = \omega[P_{AB}^J(A=1) - q_1].$ <p>In sequence <math>BA</math>, by symmetry, <math>P_{BA}^\omega(A=1) = (1 - \omega)P^B(A=1) + \omega q_1</math>; subtracting gives the third line. Neither bracket vanishes identically, since</p> $P^A(B=1) - r_1 \Big _{c=0} = r_0 - \beta, \quad P_{AB}^J(A=1) - q_1 = -c \frac{q_0(1 - q_0)(r_0 - \beta)}{\alpha\beta(1 - \alpha)(1 - \beta)} + \mathcal{O}(c^2),$ |

| Part | BEFORE (current manuscript, or the anchor for an insert) | AFTER (proposed)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
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| 5/9  | <i>Continues the insertion of the previous part.</i>     | <p>Proposition PROP:ADJ reads the weight from the ratings. The next result says what the weight does to the order in <math>c</math> at which each statistic first differs between the two sequences. A statistic is <i>protected for every pair of cues</i> if its differential at every independent belief is a multiple of the differential of the association, the condition of Proposition PROP:PRO imposed at every point of the independence surface rather than at <math>q \otimes r</math> alone; the association and the correlation coefficient are examples.</p> <p><b>Proposition LAD</b> (partial adoption). <i>Let <math>0 \leq \omega \leq 1</math>, <math>t_0 = (1 - \omega)\beta + \omega r_0</math>, <math>s_0 = (1 - \omega)\alpha + \omega q_0</math>, <math>t = (t_0, 1 - t_0)</math>, <math>s = (s_0, 1 - s_0)</math> and <math>Z = \alpha\beta(1 - \alpha)(1 - \beta)</math>.</i></p> <p>(i) <i>At <math>c = 0</math> the two sequences end at the independent beliefs <math>q \otimes t</math> and <math>s \otimes r</math>, and</i></p> $P_{AB}^\omega - P_{BA}^\omega _{c=0} = (1 - \omega)(\beta - r_0) R_1 + (1 - \omega)(q_0 - \alpha) R_2.$ <p><i>Every marginal therefore differs between the sequences at order zero in <math>c</math>, the <math>A</math>-marginal by <math>(1 - \omega)(\alpha - q_0)</math>, while a statistic protected for every pair of cues differs between them by <math>\mathcal{O}(c)</math>.</i></p> <p>(ii) <i>The believed association differs between the sequences by</i></p> $\text{assoc}(P_{AB}^\omega) - \text{assoc}(P_{BA}^\omega) = c \frac{(1 - \omega)H}{Z} + \mathcal{O}(c^2),$ $H = q_0 q_1 [\beta(1 - \beta) + \omega(\beta - r_0)^2] - r_0 r_1 [\alpha(1 - \alpha) + \omega(\alpha - q_0)^2],$ <p><i>which is first order whenever <math>\omega &lt; 1</math> and <math>H \neq 0</math>, and second order at <math>\omega = 1</math>.</i></p> <p>(iii) <i>A statistic protected for every pair of cues differs between the sequences by <math>\mathcal{O}(c^2)</math> for every <math>\omega</math> if and only if its differential at every independent belief is a multiple of the differential of the log odds ratio, as for <math>\text{assoc}(P)/(P(A=0)P(A=1)P(B=0)P(B=1))</math>. The odds ratio itself is the same in both sequences for every <math>c</math> and every <math>\omega</math>.</i></p> <p><i>Proof.</i> Every Jeffrey step, damped or not, multiplies the rows or the columns of the table by constants, so after both steps <math>P_\sigma^\omega(i, j) = a_i b_j P(i, j)</math> for some factors <math>a_0, a_1, b_0, b_1</math>. Such a rescaling multiplies the association by <math>a_0 a_1 b_0 b_1</math> and leaves the odds ratio unchanged, and <math>\text{assoc}(P) = c</math>, so <math>\text{assoc}(P_\sigma^\omega) = c k_\sigma</math> with <math>k_\sigma = a_0 a_1 b_0 b_1</math>, and the odds ratio claim of (iii) follows. (i) At <math>c = 0</math> a step on one attribute leaves the other marginal unchanged (Proposition PROP:IMM), so sequence <math>AB</math> reaches <math>q \otimes \beta</math> and then <math>q \otimes t</math>, and sequence <math>BA</math> reaches <math>s \otimes r</math>; subtracting gives the display. A statistic protected for every pair of cues has a differential that annihilates both tangent directions of the independence surface at each of its points (Lemma LEM:ASC), so it is constant on the surface and takes the same value at <math>q \otimes t</math> and <math>s \otimes r</math>. (ii) At <math>c = 0</math> the factors give <math>k_{AB} = q_0 q_1 t_0 t_1 / Z</math> and <math>k_{BA} = s_0 s_1 r_0 r_1 / Z</math>, and <math>k_{AB} - k_{BA} = (1 - \omega)H / Z</math>. (iii) Near the independence surface a statistic protected for every pair of cues is <math>F = F_0 + \text{assoc} \cdot h</math> with <math>h</math> smooth, and writing <math>h = g/m</math> with <math>m = P(A=0)P(A=1)P(B=0)P(B=1)</math>, the first-order coefficient of <math>F(P_{AB}^\omega) - F(P_{BA}^\omega)</math> is <math>k_{AB} h(q \otimes t) - k_{BA} h(s \otimes r) = [g(q \otimes t) - g(s \otimes r)] / Z</math>, since <math>m(q \otimes t) = q_0 q_1 t_0 t_1</math> and <math>m(s \otimes r) = s_0 s_1 r_0 r_1</math>. It vanishes for every pair of cues and every <math>\omega</math> exactly when <math>g</math> is constant on the independence surface, that is when the differential of <math>F</math> there is a multiple of the differential of <math>\text{assoc}/m</math>, which at an independent belief equals the differential of the log odds ratio. <math>\square</math></p> <p>Partial adoption therefore moves the sequence effect on each belief statistic other than</p> |

| Part | BEFORE (current manuscript, or the anchor for an insert) | AFTER (proposed)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
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| 6/9  | <i>Continues the insertion of the previous part.</i>     | <p><b>Factor inputs.</b> Under the benchmark reading each cue supplies a factor on its own attribute, <math>a = (a_0, a_1)</math> for the credential and <math>b = (b_0, b_1)</math> for the letter, and the belief after both cues is the prior with cell <math>(i, j)</math> multiplied by <math>a_i b_j</math> and renormalised. This is the reading of the factor-based variants of Hawthorne (2004), and with both factors applied in full the two sequences give the same belief for every <math>c</math>, since the same cells are multiplied by the same numbers whichever cue comes first (Wagner, 2002). Partial adoption has a counterpart under this reading. Let the factor read second be adopted only in part, so that sequence <math>AB</math> applies <math>a</math> and then <math>b^\omega = (b_0^\omega, b_1^\omega)</math> and sequence <math>BA</math> applies <math>b</math> and then <math>a^\omega</math>, with <math>\omega = 1</math> the benchmark and <math>\omega = 0</math> the second cue ignored.</p> <p><b>Proposition FAC</b> (factor inputs). <i>Let <math>a</math> and <math>b</math> be positive and <math>0 \leq \omega \leq 1</math>. Write <math>K_\sigma</math> for the product of the four factors that sequence <math>\sigma</math> applies and <math>S_\sigma</math> for its normalising sum.</i></p> <ul style="list-style-type: none"> <li>(i) <i>At <math>\omega = 1</math> the two sequences give the same belief for every <math>c</math>.</i></li> <li>(ii) <i>At <math>c = 0</math> each sequence ends at an independent belief. The <math>A</math>-marginals are <math>q_1</math> in sequence <math>AB</math> and <math>(1 - \alpha)a_1^\omega / (\alpha a_0^\omega + (1 - \alpha)a_1^\omega)</math> in sequence <math>BA</math>, and they agree if and only if <math>a_0 = a_1</math> or <math>\omega = 1</math>. Every marginal therefore differs between the sequences at order zero whenever the cue is informative and <math>\omega &lt; 1</math>, while a statistic protected for every pair of cues differs by <math>\mathcal{O}(c)</math>.</i></li> <li>(iii) <i>In sequence <math>\sigma</math> the believed association equals <math>cK_\sigma/S_\sigma^2</math> exactly, so it is zero at <math>c = 0</math> and first order for <math>\omega &lt; 1</math> whenever <math>K_{AB}/S_{AB}^2</math> and <math>K_{BA}/S_{BA}^2</math> differ at <math>c = 0</math>. A statistic whose differential at every independent belief is a multiple of that of the log odds ratio differs between the sequences by <math>\mathcal{O}(c^2)</math> for every <math>\omega</math>, and the odds ratio is the prior's in both sequences for every <math>c</math> and <math>\omega</math>.</i></li> </ul> <p><i>Proof.</i> Applying a factor on <math>A</math> and then one on <math>B</math> multiplies cell <math>(i, j)</math> by <math>a_i b_j</math> whichever is applied first, and renormalisation divides every cell by the same sum, which gives (i). Both steps rescale rows or columns, so the odds ratio is the prior's, and the association of the rescaled table is <math>K_\sigma \text{assoc}(P) = cK_\sigma</math>, divided by <math>S_\sigma^2</math> after renormalisation. At <math>c = 0</math> the prior is <math>\alpha \otimes \beta</math> and the rescaled table is <math>(\alpha_0 a_0, \alpha_1 a_1) \otimes (\beta_0 b_0, \beta_1 b_1)</math> up to the normalising sum, an independent belief whose <math>A</math>-marginal is <math>(1 - \alpha)a_1 / (\alpha a_0 + (1 - \alpha)a_1)</math> for the factor applied on <math>A</math>. With <math>a_i = q_i / \alpha_i</math> this is <math>q_1</math>, and with <math>a^\omega</math> in its place the two values agree exactly when <math>a_1 a_0^\omega = a_0 a_1^\omega</math>, that is when <math>(a_1 / a_0)^{1 - \omega} = 1</math>. The claims about protected statistics follow as in Proposition PROP:LAD, since both sequences end at independent beliefs. For the last claim, at <math>c = 0</math> the product of the four marginals is <math>ZK_\sigma/S_\sigma^2</math>, so the first-order factor of <math>\text{assoc}/m</math> is <math>1/Z</math> in both sequences and the argument of Proposition PROP:LAD(iii) applies. <math>\square</math></p> |

| Part                                                       | BEFORE (current manuscript, or the anchor for an insert) | AFTER (proposed)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                    |                    |  |                  |  |              |              |              |              |           |             |             |   |             |                      |                    |             |   |             |                                                            |                    |                    |   |                    |            |   |   |   |   |                            |             |             |   |             |                       |                    |             |   |             |
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| 7/9                                                        | Continues the insertion of the previous part.            | <table><tr><th rowspan="2">Sequence effect on</th><th colspan="2">Credence delivered</th><th colspan="2">Factor delivered</th></tr><tr><th><math>\omega = 1</math></th><th><math>\omega &lt; 1</math></th><th><math>\omega = 1</math></th><th><math>\omega &lt; 1</math></th></tr><tr><td>Marginals</td><td><math>\Theta(c)</math></td><td><math>\Theta(1)</math></td><td>0</td><td><math>\Theta(1)</math></td></tr><tr><td>Believed association</td><td><math>\mathcal{O}(c^2)</math></td><td><math>\Theta(c)</math></td><td>0</td><td><math>\Theta(c)</math></td></tr><tr><td>Statistics agreeing with the log odds ratio to first order</td><td><math>\mathcal{O}(c^2)</math></td><td><math>\mathcal{O}(c^2)</math></td><td>0</td><td><math>\mathcal{O}(c^2)</math></td></tr><tr><td>Odds ratio</td><td>0</td><td>0</td><td>0</td><td>0</td></tr><tr><td>Share of decisions changed</td><td><math>\Theta(c)</math></td><td><math>\Theta(1)</math></td><td>0</td><td><math>\Theta(1)</math></td></tr><tr><td>Surplus-weighted loss</td><td><math>\mathcal{O}(c^2)</math></td><td><math>\Theta(1)</math></td><td>0</td><td><math>\Theta(1)</math></td></tr></table> <p>Table caption: The between-sequence effect of each statistic, by what a cue delivers and how fully the later cue is adopted. Entries are orders in the prior covariance <math>c</math>, generic in the prior and the cues, and 0 means no sequence effect at any <math>c</math>. The decision rows of the partial-adoption columns read Theorem THM:LOS with a score gap of order zero and are not separately verified. The columns are, in turn, Sections SEC:INDIVIDUAL and SEC:AGGREGATION, Proposition PROP:LAD, the benchmark of Section SEC:SETUP, and Proposition PROP:FAC.</p> | Sequence effect on | Credence delivered |  | Factor delivered |  | $\omega = 1$ | $\omega < 1$ | $\omega = 1$ | $\omega < 1$ | Marginals | $\Theta(c)$ | $\Theta(1)$ | 0 | $\Theta(1)$ | Believed association | $\mathcal{O}(c^2)$ | $\Theta(c)$ | 0 | $\Theta(c)$ | Statistics agreeing with the log odds ratio to first order | $\mathcal{O}(c^2)$ | $\mathcal{O}(c^2)$ | 0 | $\mathcal{O}(c^2)$ | Odds ratio | 0 | 0 | 0 | 0 | Share of decisions changed | $\Theta(c)$ | $\Theta(1)$ | 0 | $\Theta(1)$ | Surplus-weighted loss | $\mathcal{O}(c^2)$ | $\Theta(1)$ | 0 | $\Theta(1)$ |
| Sequence effect on                                         | Credence delivered                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                    | Factor delivered   |  |                  |  |              |              |              |              |           |             |             |   |             |                      |                    |             |   |             |                                                            |                    |                    |   |                    |            |   |   |   |   |                            |             |             |   |             |                       |                    |             |   |             |
|                                                            | $\omega = 1$                                             | $\omega < 1$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | $\omega = 1$       | $\omega < 1$       |  |                  |  |              |              |              |              |           |             |             |   |             |                      |                    |             |   |             |                                                            |                    |                    |   |                    |            |   |   |   |   |                            |             |             |   |             |                       |                    |             |   |             |
| Marginals                                                  | $\Theta(c)$                                              | $\Theta(1)$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 0                  | $\Theta(1)$        |  |                  |  |              |              |              |              |           |             |             |   |             |                      |                    |             |   |             |                                                            |                    |                    |   |                    |            |   |   |   |   |                            |             |             |   |             |                       |                    |             |   |             |
| Believed association                                       | $\mathcal{O}(c^2)$                                       | $\Theta(c)$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 0                  | $\Theta(c)$        |  |                  |  |              |              |              |              |           |             |             |   |             |                      |                    |             |   |             |                                                            |                    |                    |   |                    |            |   |   |   |   |                            |             |             |   |             |                       |                    |             |   |             |
| Statistics agreeing with the log odds ratio to first order | $\mathcal{O}(c^2)$                                       | $\mathcal{O}(c^2)$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 0                  | $\mathcal{O}(c^2)$ |  |                  |  |              |              |              |              |           |             |             |   |             |                      |                    |             |   |             |                                                            |                    |                    |   |                    |            |   |   |   |   |                            |             |             |   |             |                       |                    |             |   |             |
| Odds ratio                                                 | 0                                                        | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 0                  | 0                  |  |                  |  |              |              |              |              |           |             |             |   |             |                      |                    |             |   |             |                                                            |                    |                    |   |                    |            |   |   |   |   |                            |             |             |   |             |                       |                    |             |   |             |
| Share of decisions changed                                 | $\Theta(c)$                                              | $\Theta(1)$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 0                  | $\Theta(1)$        |  |                  |  |              |              |              |              |           |             |             |   |             |                      |                    |             |   |             |                                                            |                    |                    |   |                    |            |   |   |   |   |                            |             |             |   |             |                       |                    |             |   |             |
| Surplus-weighted loss                                      | $\mathcal{O}(c^2)$                                       | $\Theta(1)$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 0                  | $\Theta(1)$        |  |                  |  |              |              |              |              |           |             |             |   |             |                      |                    |             |   |             |                                                            |                    |                    |   |                    |            |   |   |   |   |                            |             |             |   |             |                       |                    |             |   |             |
| 8/9                                                        | Continues the insertion of the previous part.            | <p><b>Opposite predictions about one procedure.</b> Whether a procedure removes the sequence effect depends on which channel it removes. Consider rubric scoring, in which the rating on each attribute is fixed by the document on that attribute and a set scale, so that what each document delivers no longer depends on what was read first. Under full adoption the rubric removes the position channel and leaves the association channel untouched. The recorded scores carry the delivered credences in either sequence, the belief about the attribute read second agrees with its score, and the belief about the attribute read first sits at <math>cK</math> from its score (Proposition PROP:DRF), so the sequence effect lives in that belief and in the decision made from it, vanishes when the attributes are believed unrelated (Proposition PROP:IMM), and is invisible in the scores and in the association between them. Under partial adoption the rubric removes the position channel from the record only. The belief about the attribute read second sits at <math>(1 - \omega)[P^A(B=1) - r_1]</math> from its score (Proposition PROP:ADJ), which at independence equals <math>(1 - \omega)(r_0 - \beta)</math>, so the two reading sequences differ even when the attributes are believed unrelated. The two settings therefore predict opposite things about the same procedure, and the difference is observable. In the rubric setting <math>r_1</math> is on the record, so Proposition PROP:ADJ needs only the belief about trustworthiness after the credential alone and after both documents to say which prediction holds.</p>                                                                                           |                    |                    |  |                  |  |              |              |              |              |           |             |             |   |             |                      |                    |             |   |             |                                                            |                    |                    |   |                    |            |   |   |   |   |                            |             |             |   |             |                       |                    |             |   |             |

| Part | BEFORE (current manuscript, or the anchor for an insert) | AFTER (proposed)                     |                                                                                                                                                                                                                                                                                                  |
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| 9/9  | <i>Continues the insertion of the previous part.</i>     |                                      | <b>Full adoption, <math>\omega = 1</math></b> (association channel only)<br><b>Partial or no adoption, <math>\omega &lt; 1</math></b> (position channel operates)                                                                                                                                |
|      |                                                          | Where the sequence dependence arises | In the rule. The inputs $q, r$ are the same in either position; the sequence matters because two rigid steps on correlated partitions do not commute (Diaconis and Zabell, 1982).<br>In the inputs. What the second cue delivers depends on its position; the rule adds nothing new at $c = 0$ . |
|      |                                                          | Source in the model                  | Association only. Zero at $c = 0$ (Proposition PROP:IMM).<br>Position, present at $c = 0$ , plus association when $c \neq 0$ .                                                                                                                                                                   |
|      |                                                          | Status of the inputs                 | Independent of the sequence. The credential delivers $q$ whether or not the letter was read.<br>Dependent on the sequence. The later cue is adopted only in part, so what it delivers depends on being second.                                                                                   |
|      |                                                          | Size of the effect                   | Small, first order in $c$ , spread unevenly across statistics.<br>Blunt. $(1 - \omega)(\alpha - q_0)$ on a marginal whatever $c$ is (Proposition PROP:ADJ).                                                                                                                                      |
|      |                                                          | Marginals                            | Last-read exact, first-read drifts at first order (Proposition PROP:DRF).<br>Both differ at zeroth order between sequences.                                                                                                                                                                      |
|      |                                                          | Believed association                 | Second order, between sequences and pooled (Propositions PROP:DEC and PROP:PRO).<br>Zero between sequences at $c = 0$ , first order in $c$ . Pooled, nonzero at $c = 0$ , $-\lambda(1 - \lambda)(1 - \omega)^2(\alpha - q_0)(\beta - r_0)$ .                                                     |
|      |                                                          | Odds ratio                           | Identical across sequences, every $c$ (Lemma LEM:SEP).<br>Identical across sequences, every $c$ (Lemma LEM:SEP). The one statistic that does not move.                                                                                                                                           |
|      |                                                          | Classification by order in $c$       | Is the contribution. Which statistics are protected, and why.<br>Shifts by one order, each statistic keeping its place. Marginals at order zero, protected statistics at first order, the odds-ratio shadow at second, the odds ratio never (Proposition PROP:LAD).                              |
|      |                                                          | Share and loss of decisions          | Share first order, loss second (Proposition PROP:SHR, Theorem THM:LOS).<br>Both of order zero; the separation is lost.                                                                                                                                                                           |
|      |                                                          | Audit implication                    | False negative. An association audit passes a sequence-dependent population.<br>False positive as well. A pooled association audit returns a stereotype no member holds, from mixing alone.                                                                                                      |
|      |                                                          | Identification                       | The reading, from which marginal ignores the prior association (Propositions PROP:ORD and PROP:DRF).<br>The weight, from three ratings (Proposition PROP:ADJ). At its strongest here, since the position channel is what Proposition PROP:ADJ reads.                                             |

Table caption: The two settings. Under full adoption only the association channel

| Part                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | BEFORE (current manuscript, or the anchor for an insert)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | AFTER (proposed)                                                                                                                                                                                                                    |
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|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <p><b>Why.</b> Placement (2026-10-01). The author’s Section 6 is already titled “Scope and robustness”, so the section is split into two subsections. Scope keeps the assumptions paragraph and the rival mechanisms of E.11. The second subsection states that the second-order results require full adoption of the later cue, gives the test of that condition (Proposition ADJ, three marginals of one reading group) and says what partial adoption and a factor reading change (Propositions LAD and FAC). It is not a robustness result and is not called one (author, 2026-10-01): under partial adoption the association differs between sequences at first order and the pooled association acquires a term of order zero, so the paper’s second-order conclusions do not survive; what survives is the odds ratio, exact at every weight, and the form of the characterisation, with the second-order class moving to the statistics that agree with the log odds ratio to first order. The author may want to drop “and robustness” from the section title. Part 1 adds the Scope heading and repairs “define scope for any conclusions made from the model”. Part 2 adds the clause to the roadmap. Parts 3 to 9 are the new subsection, one insertion after E.11 shown as consecutive paragraphs so that the plan can break pages between them. What it gathers: the two-channel paragraph of E.12 unchanged; the adoption-weight lead-in, Proposition ADJ, its example and Proposition LAD from E.10, with “so far” replaced by the section references; the rubric prediction and the settings table of E.12 unchanged. What is new: the framing paragraph, Proposition FAC and Table tab:robust. Proposition FAC is the factor-input counterpart of LAD. Its weighted factor <math>a^\omega</math> is a construction of this paper, log-linear damping of the factor, not one of Hawthorne’s variants; his variants are the <math>\omega = 1</math> column, where updates on distinct bases commute. The Lean record proves FAC for an arbitrary damped factor <math>a'</math> and states the marginal agreement as <math>a_1 a'_0 = a_0 a'_1</math>; the power form and its witnesses are in the sympy suite. Table tab:robust puts the four settings side by side; its decision rows for the partial-adoption columns are Theorem LOS read with an order-zero score gap and are the only entries not checked by sympy, as the caption says. Two tables now sit in the subsection; the author may drop tab:settings if tab:robust carries enough. Order of application: after E.10 (ORD) and E.11; before C.4, C.5 and C.9, which cite LAD or sec:robust; E.13 after it.</p> <p><b>Evidence.</b> Ladder.lean: <code>rescale_rescale</code>, <code>rescale_rescale_eq</code>, <code>assoc_normalize</code>, <code>oddsRatio_normalize</code>, <code>oddsRatio_factorRoute</code>, <code>assoc_factorRoute</code>, <code>factorRoute_at_zero</code>, <code>factorRoute_mA1_zero</code>, <code>factor_mA1_gap_iff</code>, <code>mprod_factorRoute_zero</code>, <code>oddsShadow_factor_coeff</code> (FAC); <code>ladder_gap</code>, <code>ladder_assoc_coeff</code>, <code>ladder_oddsShadow_seqEffect</code> (LAD); Anchoring.lean (ADJ). <code>sympy/verify_ladder.py</code> (53/53, section F for the factor routes, including that the full-adoption factor table is the manuscript’s benchmark); <code>sympy/verify_interior_omega.py</code> rows 5 and 6. Hawthorne.lean (<code>factorUpdate_comm</code>, <code>extUpdate_basisCommuting</code>); Wagner2002.lean (<code>thm31</code>); HogarthEinhorn.lean (<code>oneSided_eq_eq8</code>).</p> |                                                                                                                                                                                                                                     |
| <b>E.13</b> Back matter, AI declaration (was B.B)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                     |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Proposition PROP:PRO (uniqueness of the protected statistic), which was then independently re-derived                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Propositions PROP:PRO (uniqueness of the protected statistic), PROP:ORD (between-sequence contrast), PROP:ADJ (adoption weight), PROP:LAD (partial adoption) and PROP:FAC (factor inputs), which were then independently re-derived |
| <p><b>Why.</b> Plan B.B. Propositions ORD, ADJ, LAD and FAC have the same provenance as PRO, so the declaration names them once E.10 and E.15 are applied.</p> <p><b>Evidence.</b> None needed.</p> <p><b>Grounds.</b></p> <p><b>Quote.</b> <i>PAPER_B_MANUSCRIPT.tex, lines 1220-1228 (author’s draft), Declaration of Generative AI</i></p> <p>“During the preparation of this work the author used Anthropic’s Claude (large language model) in order to draft and refine manuscript prose; propose, prove, and cross-verify Proposition PROP:PRO (uniqueness of the protected statistic), which was then independently re-derived and machine-verified by the author; propose bibliographic sources for positioning against neighbouring literatures; and structure the manuscript to conform to the target journal’s house style.”</p> <p><i>Propositions ORD and ADJ were proposed, proved and cross-verified the same way (PropORD.lean, Anchoring.lean, verify_ORD.py, check_zero_slope_identification.py), so the declaration names them with PRO once E.10 is applied.</i></p> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                     |
| <b>E.14</b> Bibliography entries the Section E texts need (was B.A)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                     |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <i>No manuscript text; see Why.</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                     |
| <p><b>Why.</b> Needed by E.1, E.5, E.7 (Bohren2019, Doring1999, Garber1980) and C.15 (Heckman1998, Bohren2019). Hawthorne2004 is already in bibliography.bib. All six proposed entries are in notes/manuscript_corrections_extra.bib and move to bibliography.bib on approval. Zhao-Osherson is cited only in the introduction (author’s draft) and is already in the bibliography.</p> <p><b>Evidence.</b> Bibliographic details verified against the Drive copies where printed (Doring: Phil. Sci. 66 (Proceedings) S379-S389; Garber: 47(1) 142-145; Heckman: JEP 12(2) 101-116); Bohren et al.’s AER pages are from the reference lists of later papers, the Drive copy being the working paper.</p>                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                     |

## C.12 and E.14: bibliography entries the AFTER texts need

Hawthorne2004 and Zhao0sherson2010 are already in `bibliography.bib`. Six more are needed by C.4’s optional sentence, C.6, C.10, C.15, E.1, E.5 and E.7 (Jeffrey 2004, Benjamin et al. 2019, Heckman 1998, Bohren et al. 2019, Döring 1999, Garber 1980); they are in `notes/manuscript_corrections_extra.bib` so that this document renders them, and move to `bibliography.bib` on approval. The Drive copy of Jeffrey (2004) is the November 2002 draft, so C.6 and C.10 cite the chapter only.

## Section D: where each entry of the archived plan (`notes/manuscript_change_plan_asof_2026-09-30.md`) now lives

| Entry              | Subject                                   | Status and what it needs                                                                                                                    |
|--------------------|-------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| 0.A                | Abstract                                  | Superseded by the author’s rewrite and C.1.                                                                                                 |
| 1.A, 1.B, 1.D, 1.E | Introduction                              | Superseded by the author’s rewrite; what survives is corrected in C.2-C.8. The ORD sentence of 1.E is still unapplied and presupposes E.10. |
| 1.A2               | Two-channel prelude                       | Applied by the author; corrected by C.4.                                                                                                    |
| 1.C                | Why sequence matters beyond one judgement | Re-issued as E.1.                                                                                                                           |
| 2.A, 2.B, 2.C      | Impression; worked example; soft cues     | Re-issued as E.2, E.3 (extended with the soft-versus-hard passage), E.4.                                                                    |
| 3.A, 3.B           | Commutativity literature; Asch            | Re-issued as E.5, E.6.                                                                                                                      |
| 3.C                | Heckman                                   | Superseded by C.15, which carries the identification precedent with P17a applied.                                                           |
| 3.D                | Bohren-Imas-Rosenberg                     | Re-issued as E.7.                                                                                                                           |
| 4.A, 5.A           | Section titles and openings               | Re-issued as E.8, E.9.                                                                                                                      |
| 5.B                | Propositions ORD and ADJ                  | Re-issued as E.10 (mathematics unchanged; prose corrected per P10/P11).                                                                     |
| 6.A, 6.B           | Rival mechanisms; two channels            | Re-issued as E.11, E.12 (6.B’s text pulled in from <code>interior_omega.tex</code> ).                                                       |
| B.A, B.B           | Bibliography; AI declaration              | Re-issued as E.14, E.13.                                                                                                                    |
| W.A                | Input/output tables                       | Withdrawn by the author 2026-09-21; kept in <code>notes/empirical_analytics.tex</code> .                                                    |

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